IN THE HIGH COURT OF JUSTICE **Claim No: IL-2021-000019**

BUSINESS AND PROPERTY COURTS OF ENGLAND & WALES

INTELLECTUAL PROPERTY LIST (ChD)

B E T W E E N :

CRYPTO OPEN PATENT ALLIANCE **Claimant**

-and-

DR CRAIG STEVEN WRIGHT

Defendant

Appendix PM3

Bitcoin White Paper

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1. In this Appendix, I will set out the findings of my analysis regarding various documents provided to me which are related to the Bitcoin White Paper.

2. When I refer to the "Bitcoin White Paper" or the abbreviation "BWP" I do so generally to mean documents that are similar in nature to the document known by that name, without trying to use that term to indicate any version as "official". It is a convenient way to describe different, similar-looking documents as a whole based on their face-value content, and using that term does not indicate any conclusion. As I explain in detail below, I have come to the view that some of the documents in the disclosure dataset are authentic versions of the Bitcoin White Paper contemporaneous to 2008-2009, while others are not authentic and/or do not appear to be contemporaneous to that time period (or to their purported time period). I have tried to state my conclusions as to each document clearly at the end of each section.

3. In this Appendix,

    a. I begin by establishing some control copies of the BWP as a basis for further review.

    b. I also assess the authenticity of a further copy provided to me as a further possible benchmark for contextual review.

    c. I then analyse native format PDF versions of the BWP provided in the disclosure dataset.

    d. I then analyse DOC and DOCX files

    e. Finally, I make some relatively minor observations about files such as hard copy scans, the metadata of which does not provide a good basis for forensic review.

4. In the analyses in the other Appendices to my Main Report, I have tried as far as possible to keep to a chronological account of my investigations, to show how my analysis developed. In the case of the various BWP-related documents however, I analysed them as a single contextual set, moving between them as required to help me form a view of the set as a whole. A chronological account of my investigation would therefore have been very difficult to follow. A chronological account of the documents going by their purported timestamps would also have been problematic for reasons that are clear from the analysis in this Appendix. Therefore the approach I take is to address them in a way that I hope will be much more helpful, grouping the documents as far as I can together to aid review and understanding.

Introduction and Overview

5. By using text searches and by visual inspection of the disclosure dataset, I have established that there are 68 documents that are copies of, or exhibit a relationship to, the Bitcoin White Paper published by Satoshi Nakamoto on 24 March 2009 (and I have been instructed by Bird & Bird that I can take the publication date of that version as a working basis for this analysis). Since there are so many, it is helpful to set out some general observations and explanations about the various documents before analysing them.

{ID_000199}
{ID_000546}
{ID_000870}
{ID_003847}
{ID_000217}
{ID_000537}
{ID_003459}
{ID_003857}
{ID_000226}
{ID_000538}
{ID_003702}
{ID_004007}
{ID_000254}
{ID_000540}
{ID_003732}
{ID_004010}
{ID_000510}
{ID_000849}
{ID_003747}
{ID_004011}
{ID_000515}
{ID_000865}
{ID_003794}
{ID_004023}

6. Of these 68 documents in the dataset, 24 are unique by MD5 hash, with the remainder being duplicates. I therefore focus on the unique documents only. The 24 <u>unique</u> documents are:

***ID_000199.DOC**	ID_000536.PDF	ID_000870.PDF	***ID_003847.PDF**
***ID_000217.DOC**	ID_000537.PDF	ID_003459.DOCX	ID_003857.PDF
ID_000226.PDF	ID_000538.PDF	***ID_003702.DOC**	ID_004007.PDF
***ID_000254.ODT**	ID_000540.PDF	ID_003732.PDF	***ID_004010.PDF**
ID_000510.DOC	ID_000849.PDF	ID_003747.MSG	***ID_004011.PDF**
ID_000515.TIF	ID_000865.PDF	ID_003794.DOCX	ID_004023.PDF

7. Of these,

   a. Some are listed as Reliance Documents, which are shown in the above list with an asterisk and bold font, and

   b. the following documents have been addressed separately elsewhere in my report, having come up in the course of my analysis of Reliance Documents and other documents:

{ID_000199}
{ID_000217}
{ID_000254}
{ID_000849}
{ID_003702}

   ID_000199, ID_000217, ID_000254, ID_000849, ID_003702

8. In addition to the documents from the disclosure dataset, I have access to certain additional copies of versions of the Bitcoin White Paper from sources as follows:

a. I have obtained a sample of the Bitcoin Whitepaper by visiting the Wayback machine and searching through early snapshots of the Bitcoin.org website. The earliest snapshots taken by the Wayback Machine did not include the PDF whitepaper within the snapshots, albeit there was a link to it. The links to the BWP on these earlier snapshots redirect to the earliest snapshot of the document which was taken dated 04 July 2010 at: http://web.archive.org/web/20100704213649/http://www.bitcoin.org/bitcoin.pdf . I will call this version "**BWP Archive 1**".

b. Browsing the Archive.org database of documents, it is also possible to obtain three other copies of versions of the Bitcoin White Paper from the following links:

i. https://archive.org/details/bitcoin-whitepaper ("**BWP Archive 2**")

ii. https://archive.org/details/bitcoin-a-peer-to-peer-electronic-cash-system ("**BWP Archive 3**")

iii. https://archive.org/details/bitcoin_201905 ("**BWP Archive 4**")

c. Bird & Bird has also provided me with a copy of a file named "bitcoin.pdf" which I am informed is part of Exhibit NB1 from a different document in these proceedings (which is referred to in this Appendix as "**BWP-NB1**").

{D/91}

d. I have also been provided with a link to a mailing list discussion dating from January 2015 with the subject "2008 revision of Bitcoin whitepaper" at https://www.metzdowd.com/pipermail/cryptography/2015-January/024433.html and I am informed that is a discussion on the same mailing list as the list originally used by Satoshi Nakamoto to publish the BWP. That discussion does not contain a version of the Bitcoin White Paper but does state that a version was known with an internal date of 2 November 2008 having a SHA256 hash of "`427c63b364c6db914cf23072a09ffd53ee078397b7c6ab2d604e12865a982faa`"

("**Metzdowd Discussion Hash**"), which (as I explain below) matches hashes of other documents in the dataset.

General characteristics and identifying control copies

Contact details

9. Most of the versions of the Bitcoin White Paper listed above contain a name and contact details at the top of the first page. I understand the identity of the author to be important in this case. Since the contact details appear at the top of the front page, this also provides a convenient way to group the documents.

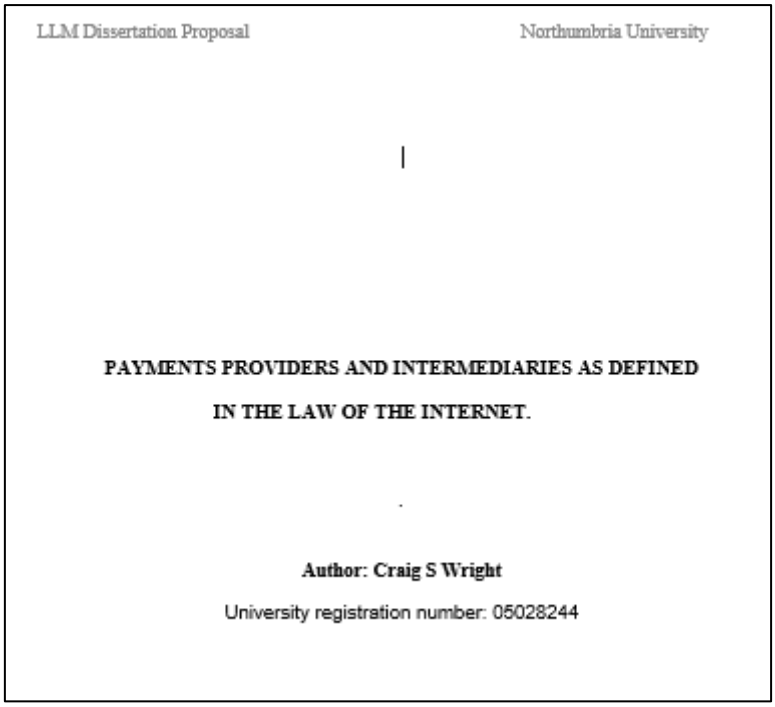
10. The table below is organised by ID_ number according to its first occurrence in the disclosure dataset and is not intended to be chronological. Reliance Documents are shown in the same way as in the list above:

Style No.	Presentation of contact details	Relevant documents
Style 1 {ID_000226} {D/91}	Satoshi Nakamoto satoshi@vistomail.com www.bitcoin.org	ID_000226 BWP Archive 3 BWP-NB1
Style 2 {ID_000254}	Dr Craig S Wright craigswright@acm.org Charles Sturt University	***ID_000254**
Style 3 {ID_000510}	[None]	ID_000510
Style 4 {ID_000515} {ID_000536} {ID_000537} {ID_000538} {ID_000540} {ID_003847}	Dr Craig S Wright craigswright@acm.org Charles Sturt University	ID_000515 ID_000536 ID_000537 ID_000538 ID_000540 ***ID_003847**
Style 5 {ID_000865} {ID_000870} {ID_003794} {ID_003857} {ID_004007} {ID_004010} {ID_004023}	Satoshi Nakamoto satoshin@gmx.com www.bitcoin.org	ID_000865 ID_000870 ID_003794 ID_003857 ID_004007 ***ID_004010** ID_004023 BWP Archive 1 BWP Archive 2 BWP Archive 4

	Style No.	Presentation of contact details	Relevant documents
{ID_003732}	**Style 6**	Dr Craig S Wright csw26@leicester.ac.uk nChain Ltd	ID_003732
{ID_003459}	**Style 7**	Dr Craig S Wright craigswright@acm.org	ID_003459
{ID_004011}	**Style 8**	Satoshi Dr Craig S Wright craigswright@acm.org Charles Sturt University GACS.	***ID_004011**

11. I exclude from this table above the set of three documents which appear quite different, being ID_000199, ID_000217, and ID_003702. For clarity, although addressed separately in my report at Appendix PM25, they present as per the following screenshot:

{ID_000199}
{ID_000217}
{ID_003702}
{H/118}



LLM Dissertation Proposal Northumbria University

PAYMENTS PROVIDERS AND INTERMEDIARIES AS DEFINED IN THE LAW OF THE INTERNET.

Author: Craig S Wright

University registration number: 05028244

Hash assessments of online sources against the disclosure dataset

12. It was helpful to begin by trying to establish a set of control documents against which to test other documents in the disclosure dataset. I begin by an MD5 hash function of the Archive documents mentioned above. The MD5 hashes of the four BWP Archives are as follows:

BWP Archive 1	`d56d71ecadf2137be09d8b1d35c6c042`
BWP Archive 2	`d56d71ecadf2137be09d8b1d35c6c042`

BWP Archive 3	b7026c5be02de23871fc1d80a49e087b
BWP Archive 4	d56d71ecadf2137be09d8b1d35c6c042

13. The SHA256 hashes[1] of the same files are as follows:

BWP Archive 1	b1674191a88ec5cdd733e4240a81803105dc412d6c 6708d53ab94fc248f4f553
BWP Archive 2	b1674191a88ec5cdd733e4240a81803105dc412d6c 6708d53ab94fc248f4f553
BWP Archive 3	427c63b364c6db914cf23072a09ffd53ee078397b7 c6ab2d604e12865a982faa
BWP Archive 4	b1674191a88ec5cdd733e4240a81803105dc412d6c 6708d53ab94fc248f4f553

14. It can be observed that:

    a. BWP Archive 1, BWP Archive 2, and BWP Archive 4 are identical files by MD5 and SHA256 – they are the same file.

    b. BWP Archive 3 is a different file by MD5 and SHA256, but the SHA256 output of that file matches the Metzdowd Discussion Hash, which states:

    "Specifically, it's the 183697 byte version of bitcoin.pdf received here 2008 November 8, sha256 427c63b364c6db914cf23072a09ffd53ee078397b7c6ab2d604e12865a982faa."

{H/21} 15. Turning to the ID_ numbered files (those provided in the disclosure dataset), **Exhibit PM3.1** is a table of hash function outputs, listing hashes by MD5 and SHA256, for the relevant unique documents that I have listed at the beginning of this Appendix. It can be observed that:

{ID_000865}
    a. BWP-Archive 1 is hash-identical to ID_000865, and

{ID_000226}
    b. BWP-Archive 3 is hash-identical to ID_000226.

Scrutiny to establish control copies

{ID_000865} *Scrutiny of ID_000865*

16. I do not take these documents at face value and have subjected them both to an equivalent level of scrutiny as any other document addressed in my report.

[1] SHA256 is a different hashing algorithm to MD5 which for the present purpose provides an equivalent function. However they are different processes and the SHA256 hash of an input will be very different to the MD5 hash of the same input: an MD5 hash cannot be compared to a SHA256 hash.

{ID_000865} 17. Although I have found no reason to doubt them, I list my observations for ID_000865 as follows.

{ID_000865} 18. ID_000865 is hash-identical to the following 22 other documents within the disclosure DATASET:

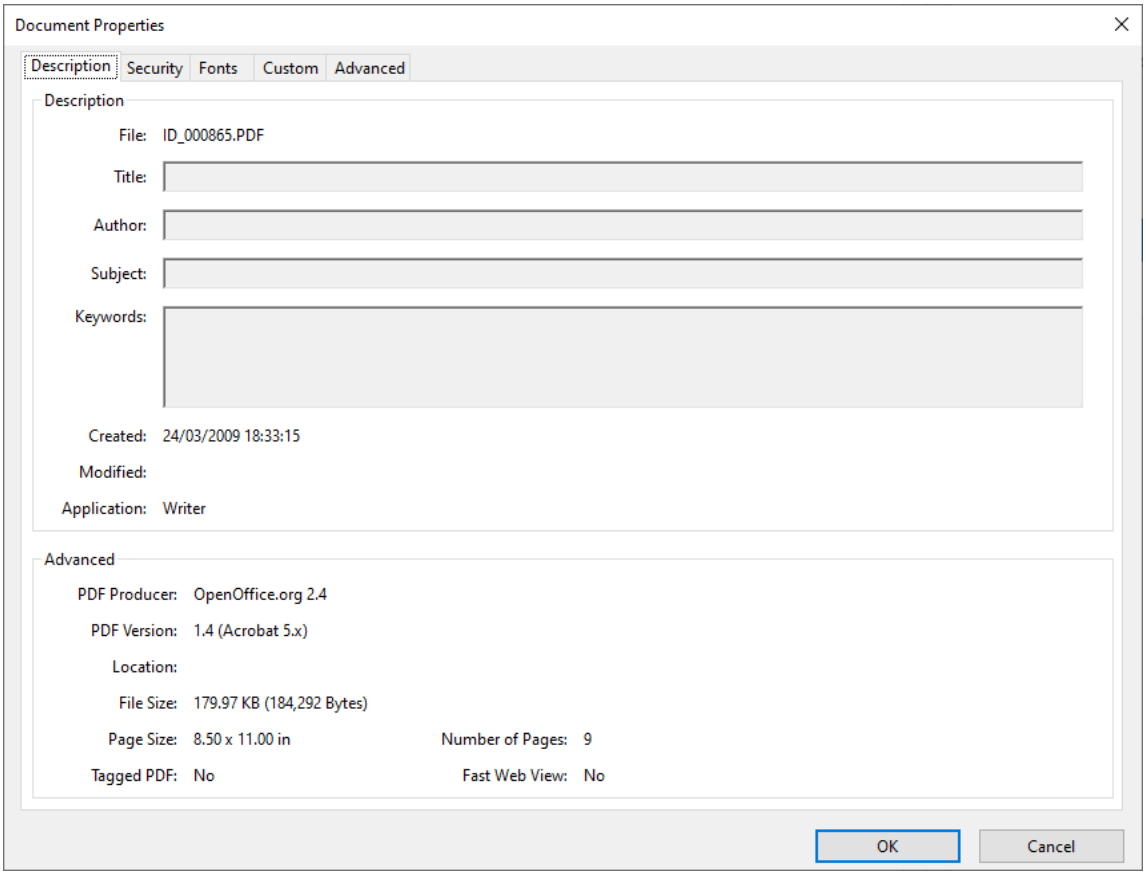
{ID_004040}
{ID_004032}
{ID_003819}
{ID_004039}
{ID_004031}
{ID_000869}
{ID_004038}
{ID_004030}
{ID_000868}
{ID_004037}
{ID_004029}
{ID_000867}
{ID_004036}
{ID_004025}
{ID_000866}
{ID_004035}
{ID_004022}
{ID_000865}
{ID_004034}
{ID_004021}
{ID_004033}
{ID_003820}

ID_004040.PDF	ID_004032.PDF	ID_003819.PDF
ID_004039.PDF	ID_004031.PDF	ID_000869.PDF
ID_004038.PDF	ID_004030.PDF	ID_000868.PDF
ID_004037.PDF	ID_004029.PDF	ID_000867.PDF
ID_004036.PDF	ID_004025.PDF	ID_000866.PDF
ID_004035.PDF	ID_004022.PDF	ID_000865.PDF
ID_004034.PDF	ID_004021.PDF	
ID_004033.PDF	ID_003820.PDF	

19. While the provided OS File date and timestamps for these documents differ between them, the content is identical. The analysis regarding the internal metadata and content of these documents will therefore match identically.

20. There is no date on the printed face of the document with the only dates featured relating to the References on the last page. On the front page of the document, just below the title, the document list authorship to “Satoshi Nakamoto” with an email address “satoshin@gmx.com” and the website “www.bitcoin.org”, akin to the “**style 5**” screenshots shown above.

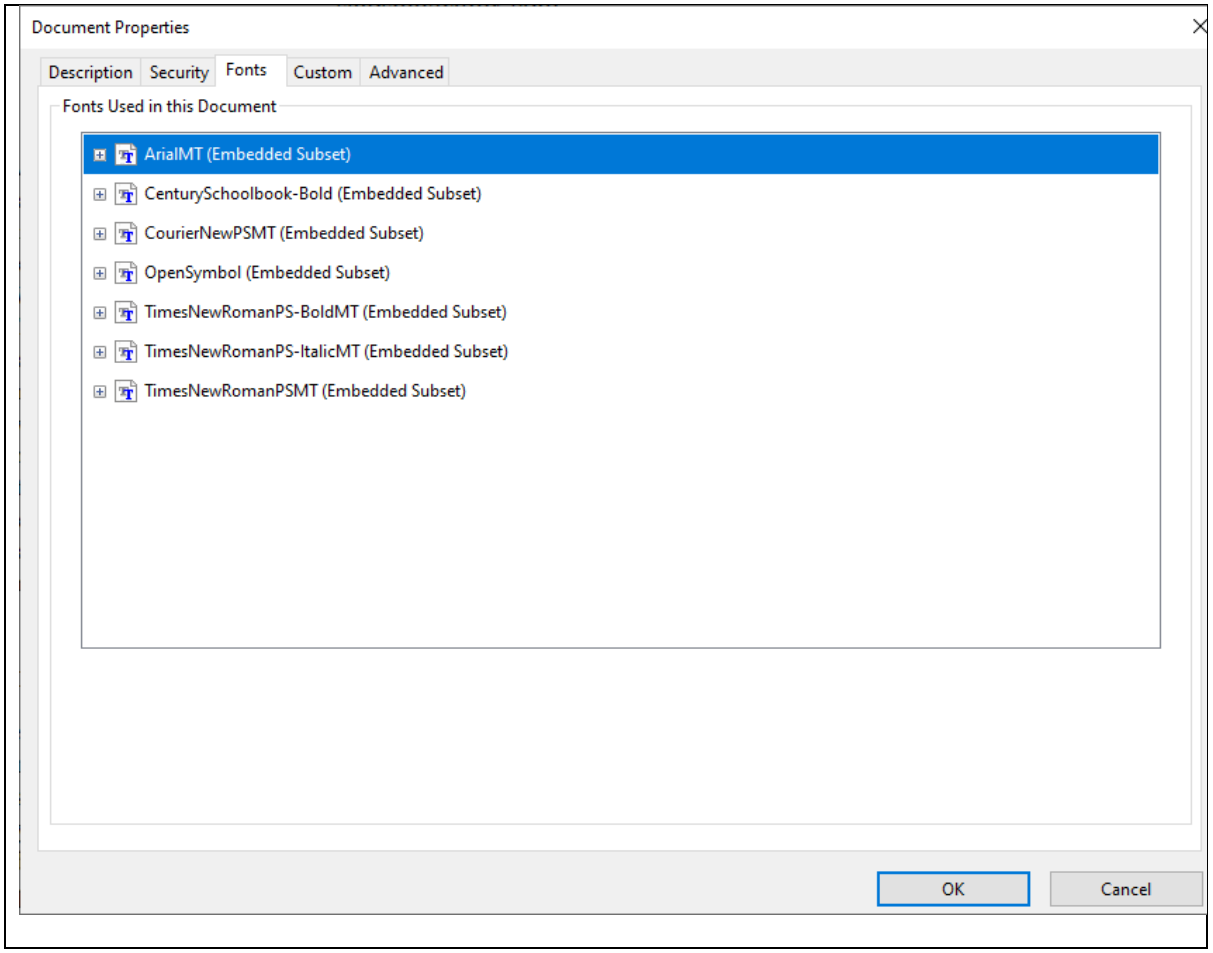
21. The Adobe Acrobat Reader properties dialog for the document indicate a recorded creation date of 24/03/2009 (23 March 2009) at 18:33:15 (expressed in BST, my local time at the time of this analysis) - as shown in the following screenshot:



22. I pause to note that the UTC time zone offset recorded for this document is UTC-6. The UTC time zone offset for ID_000226 is UTC-7. These are also relevant to my review of a third document, and so I have discussed these time zone offsets in greater detail in the next section of this appendix, below. {ID_000226}

23. There is no information recorded in the Title, Author, Subject or Keywords sections of the document properties. The document also does not have a recorded Modified date and timestamp. The metadata has recorded an application of "Writer" and PDF producer of "OpenOffice.org 2.4".

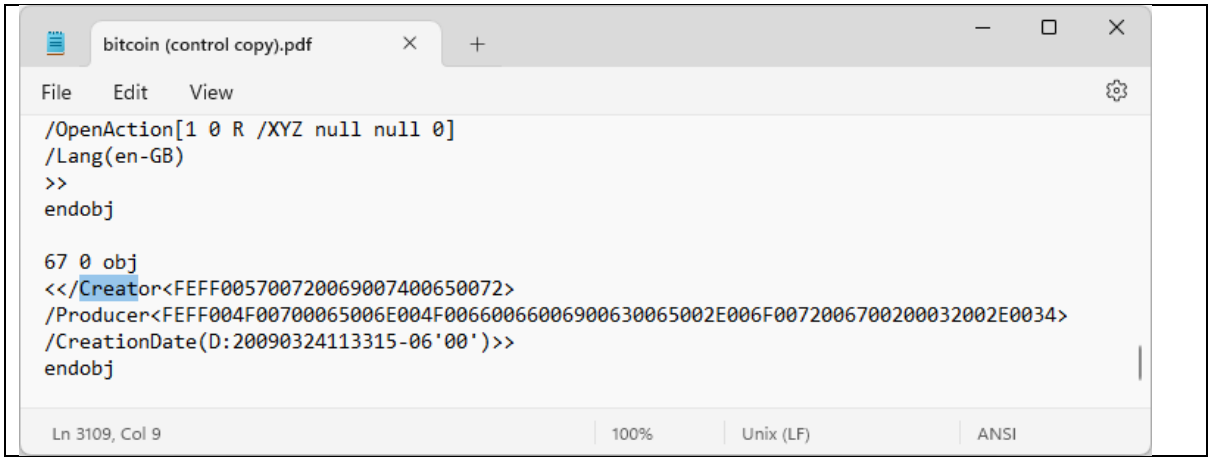
24. The Properties dialog also includes a reference to the 7 "Fonts Used in this Document" as per the screenshot below:



25. An inspection of the metadata within the document reveals that it has little by way of recorded metadata information, as can be seen in the screenshot below. The screenshot also demonstrates how the metadata timestamps can be examined in plain text within the document ID_000865 by viewing it in Notepad. I note however that the majority of the text content of the document is encoded differently, in a manner that is not plain text and requires processing in order to render in human readable form (which is a normal encoding method for PDFs, and which I have explained in considerably more detail in my Appendix PM4 addressing "Touchup_textedit" tags in other documents in the disclosure dataset):

{ID_000865}

{H/29}



26. The two fields For “Creator” and “Producer” contain the text “Writer” and “OpenOffice.org 2.4” respectively encoded in Hexidecimal. It can also be seen that the CreationDate field includes a reference to a Time Zone Offset of -06 hours (minus 6 hours), as mentioned briefly above.

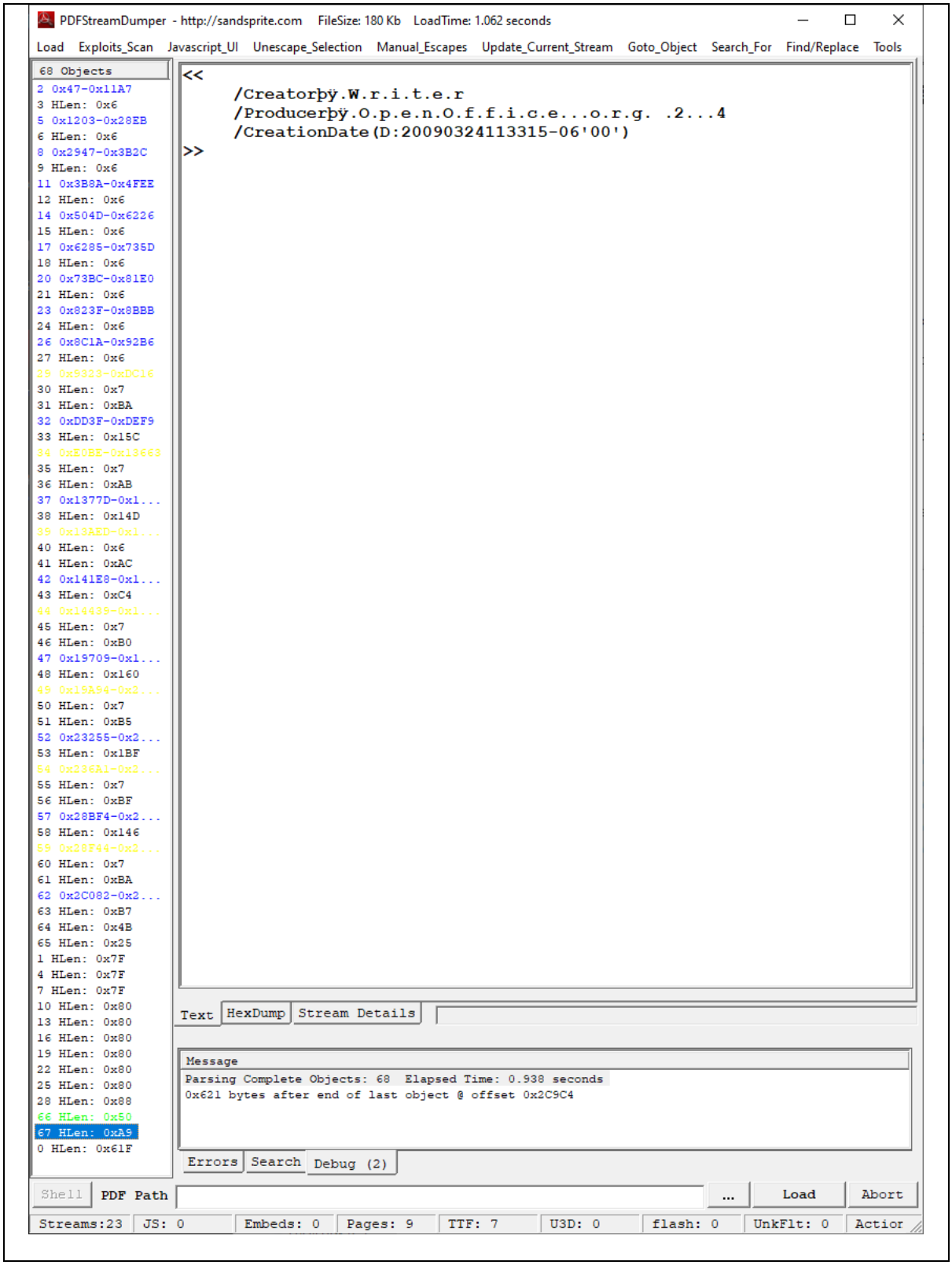
27. However, there is a little additional metadata at the very end of the document, which terminates the file with the following “trailer” information, including a document ID number, a Document Checksum, and an “EOF” (End of file) notation as follows. These appear to be created by hashes of content relating to the document’s creation and content:

```
trailer
<</Size 68/Root 66 0 R
/Info 67 0 R
/ID [ <CA1B0A44BD542453BEF918FFCD46DC04>
<CA1B0A44BD542453BEF918FFCD46DC04> ]
/DocChecksum /6F72EA7514DFAD23FABCC7A550021AF7
>>
startxref
182727
%%EOF
```

28. There are no records indicating the XMP Core or other information regarding the authorship of the file.

29. There are no “Touchup_textedit” tags within the document.

30. The software application PDF Stream Dumper can be used to further explore the structure of the PDF file. A screenshot demonstrating this can be seen below:



31. From this it can be seen that there are 68 Object streams that comprise the document (see the column heading of the field on the left hand side of the screenshot). These relate to the separate parts such as the text content, formatting, font files and formatting instructions for the PDF.

32. The screenshot above displays the only data stream in the file that contains the property information and timestamp.

Each page or component of the document is encoded within a different stream, with each receiving its own collection of formatting and content streams. This is a normal encoding method for PDFs. I have found the structure and layout of ID_000865 to be simple and straight forward. There are no indications of manipulation, irregularities or metadata conflicts.

{ID_000865} *Further corroboration for ID_000865*

33. After explaining the analysis above to Bird & Bird and explaining the sources I have referred to in order to establish control copies, they provided me with a link to the following URL and
{H/22} a copy of the source code of that web page which is at **Exhibit PM3.2**:

https://web.archive.org/web/20091128203045if_/http://sourceforge.net:80/api/file/index/project-id/244765/rss

34. I am informed by Bird & Bird that this is a link to an earlier capture of a page at Sourceforge.net project on which the Bitcoin White Paper and source code were published by Satoshi Nakamoto. The capture is dated 28 November 2009 and relates to an RSS feed (a format of content listing used by web publishing services). Viewing the source information in the file indicates that the RSS feed on 28 November 2009 included MD5 hashes of the content hosted within it.

35. It can be observed that one of the items listed there is titled “bitcoin.pdf” and has a “pubDate”
{ID_000865} tag of 24 Mar 2009 at 17:50:18 UTC, with an MD5 hash identical to that of ID_000865. This appears to further corroborate ID_000865 as a copy of the BWP. This can be seen in the excerpt below (I have added highlighting to aid review):

```
<item>
      <title><![CDATA[/Research Paper/bitcoin.pdf/bitcoin.pdf]]></title>

<link>http://sourceforge.net/projects/bitcoin/files%2FResearch%20Paper%2Fbit
coin.pdf%2Fbitcoin.pdf/download</link>

<guid>http://sourceforge.net/projects/bitcoin/files%2FResearch%20Paper%2Fbit
coin.pdf%2Fbitcoin.pdf/download</guid>
      <description><![CDATA[/Research
Paper/bitcoin.pdf/bitcoin.pdf]]></description>
      <pubDate>Tue, 24 Mar 2009 17:50:18 +0000</pubDate>
      <category>Research Paper</category>
      <files:sf-package-id
xmlns:files="http://sourceforge.net/api/files.rdf#">299212</files:sf-
package-id>
      <files:sf-release-id
xmlns:files="http://sourceforge.net/api/files.rdf#">640697</files:sf-
release-id>
```

```
      <files:version
xmlns:files="http://sourceforge.net/api/files.rdf#">bitcoin.pdf</files:versi
on>
      <files:release-notes-url
xmlns:files="http://sourceforge.net/api/files.rdf#">http://sourceforge.net/p
roject/shownotes.php?release_id=640697</files:release-notes-url>
      <files:sf-file-id
xmlns:files="http://sourceforge.net/api/files.rdf#">1602398</files:sf-file-
id>
      <files:extra-info
xmlns:files="http://sourceforge.net/api/files.rdf#">PDF
document</files:extra-info>
      <media:content xmlns:media="http://video.search.yahoo.com/mrss/"
type="application/pdf"
url="http://sourceforge.net/project/bitcoin/files/Research
Paper/bitcoin.pdf/bitcoin.pdf/download"
filesize="184292"><media:title>Research Paper</media:title><media:hash
algo="md5">d56d71ecadf2137be09d8b1d35c6c042</media:hash></media:content>
      <files:download-count
xmlns:files="http://sourceforge.net/api/files.rdf#">83</files:download-
count>
    </item>
```

{ID_000226} *Scrutiny of ID_000226*

36. I conducted the same level of scrutiny on ID_000226 and found it to be simple and straight forward in exactly the same way. The structure, fonts, and all other aspects were identical. The only notable difference, apart from textual changes which I discuss further below, was the different time zone offset which was UTC-7 as I already mentioned briefly above.

{ID_000865} {ID_000226}

37. I note that as with ID_000865, the end of the document ID_000226 terminates with a set of "trailer" information, this time including a different document ID number, a different Document Checksum, and an "EOF" (End of file) notation, as shown below. The differences are consistent with the document content being slightly different and thus resulting in the ID and Checksum values being different.

```
trailer
<</Size 65/Root 63 0 R
/Info 64 0 R
/ID [ <BBD1C86BA0031ECDEFBD1FEDE45329B2>
<BBD1C86BA0031ECDEFBD1FEDE45329B2> ]
/DocChecksum /94B5A950BE82F71A76361664D9DA141F
>>
startxref
182192
%%EOF
```

{ID_000226} {ID_000865}

38. I therefore do not repeat my analysis for ID_000226 which would be simply repetitive of ID_000865. I found no reason to doubt the authenticity of the document and no indications of manipulation or irregularities.

Establishing controls

{ID_000865} {ID_000226} 39. After significant scrutiny and verification by third party sources, I therefore take **ID_000865 and ID_000226** to be 'control' copies of versions of the Bitcoin White Paper, on the basis that both of these documents are:

a. present in the disclosure dataset of Dr Wright's documents, and therefore provide for a good basis for comparative review of Dr Wright's other documents from the same disclosure dataset,

b. verified as hash identical to other copies from apparently independent sources, including bitcoin.org (as captured in 2010 on the archive.org Wayback Machine), the archive.org document database, and also the 2015 discussion on the cryptography mailing list mentioned above,

c. and it therefore appears to be a reasonable assumption that they are authentic copies of versions of the Bitcoin White Paper produced by Satoshi Nakamoto and therefore that they provide a control point for assessment.

I will refer to them in the rest of this Appendix as the Control Copies (or Copy, as appropriate).

40. The readily-available internal metadata properties of these two files is as follows:

{ID_000226} {ID_000865}

	Created date	Modified date	Creator Tool	PDF Producer	PDF Version
ID_000226	3 October 2008 at 13:49:58 UTC-0700	n/a	Writer	OpenOffice.org 2.4	1.4 (Acrobat 5.x)
ID_000865	24 March 2009 at 11:33:15 UTC-0600	n/a	Writer	OpenOffice.org 2.4	1.4 (Acrobat 5.x)

{D/91} **Exhibit NB1**

41. Turning to the BWP-NB1 PDF which was provided to me by Bird & Bird, I do not have any knowledge of its provenance other than that it was provided by a third party witness. I therefore investigated the document to assess its authenticity.

NB1 hash comparison

{D/91} 42. The relevant hash values for BWP-NB1 are as follows:

BWP-NB1	**MD5:** 3e5e11e1e3208d2829e887fb1c86bd05 **SHA256:**

	e6cc7c952c688b234f9872c3e2f50060ae6556fd27925cba503c6460048e50a9

{H/21} {D/91}

43. By comparison of these hashes with Exhibit PM3.1 and the BWP Archive-sourced hashes presented above, I determined that BWP-NB1 is:

a. not present in the disclosure dataset (does not match any of the ID_ numbers), and

b. not among the BWP Archive-sourced documents.

NB1 external searches

{D/91}

44. Since Exhibit NB1 has not been disclosed with an ID_ number, there was no comparative metadata from any load file for me to assess it against from the disclosure dataset. I therefore investigated whether I could find a third party source which might indicate any data about its provenance at all.

45. First, searching the internet using the hashes as keywords, Google, neither the MD5 hash nor the SHA256 hash produce any results at all (by contrast, equivalent searches for the hashes of the Control Copies produce many results), and therefore did not help.

{D/91}

46. Second, by comparing the text of BWP-NB1 to the Control Copies it is possible to establish that it contains textual differences to each of those control copies.

{D/91}

47. In order to approach this in context I insert the BWP-NB1 metadata into the same table shown above for the Control Copies (with BWP-NB1, the subject of this examination, highlighted in red):

	Created date	Modified date	Creator Tool	PDF Producer	PDF Version
{ID_000226} {D/91} ID_000226	3 October 2008 at 13:49:58 UTC-0700	n/a	Writer	OpenOffice.org 2.4	1.4 (Acrobat 5.x)
BWP-NB1	*11 November 2008 at 08:00:34 UTC-0800*	*n/a*	*Writer*	*OpenOffice.org 2.4*	*1.4 (Acrobat 5.x)*
{ID_000865} ID_000865	24 March 2009 at 11:33:15 UTC-0600	n/a	Writer	OpenOffice.org 2.4	1.4 (Acrobat 5.x)

{D/91}

48. Taking the metadata at face value for present purposes, BWP-NB1 is contextually dated between ID_000226 and ID_000865 as shown above. I therefore take ID_000226 as a point of comparison. {ID_000226}

49. Bird & Bird has prepared the following exhibits to indicate the textual changes between the three documents listed above, which are automatic comparisons of the text content only:

{H/23}
{ID_000226}
{D/91}
{H/24}
{ID_000865}

a. **Exhibit PM3.3 ,** which is a comparison of ID_000226 against BWP-NB1 and,

b. **Exhibit PM3.4**, which is a comparison of BWP-NB1 against ID_000865.

{ID_000226}
{D/91}

50. As demonstrated between these versions, the following text has been edited from ID_000226 to BWP-NB1:

~~The incentive may also~~¶
The incentive is also funded with transaction fees. The output value of every transaction is equal to the input value minus a transaction fee, and the incentive is increased by the total transaction fees in the block. Once a predetermined number of coins have entered circulation, the incentive can transition entirely to transaction fees and be completely inflation free.¶
The incentive may help encourage nodes to stay honest. If a greedy

{D/91}
{ID_000865}

51. And then between BWP-NB1 and ID_000865 the same block of text has been edited again:

The incentive ~~is~~can also be funded with transaction fees. ~~The~~If the output value of ~~every~~a transaction is ~~equal to the~~less than its input value ~~minus~~, the difference is a transaction fee~~, and~~ that is added to the incentive ~~is increased by the total~~value of the block containing the transaction ~~fees in the block~~. Once a predetermined number of coins have entered circulation, the incentive can transition entirely to transaction fees and be completely inflation free.

{D/91}

52. Therefore, among the three documents the following strings are unique to BWP-NB1:

- "The incentive is also funded with transaction fees"
- "The output value of every transaction is equal to the input value minus a transaction fee"
- "and the incentive is increased by the total transaction fees in the block"

53. Searching the disclosure dataset for those three phrases, including various parts of them, {D/91} produced no results at all, indicating that neither a document equivalent to BWP-NB1 nor the text of document BWP-NB1 is included in the disclosure dataset.[2]

54. Searching the internet using Google for those three phrases produced no relevant results, {D/91} indicating that neither the document BWP-NB1 nor the text of document BWP-NB1 is publicly available on a website indexed by Google.[3]

[2] And as a cross check, an equivalent search of the disclosure dataset for a fragment of one of those phrases, "funded with transaction fees" produced more than 50 hits, that text corresponding to text which was retained {ID_000865} in ID_000865.

[3] Out of all the searches, there was just one result for one of the strings but it was not relevant. The first string, which is the shortest, returned one result which was a 2018 publication written in Swedish and English and which does not appear relevant: http://uu.diva-portal.org/smash/get/diva2:1313719/FULLTEXT01.pdf . The second and third strings produced no results at all.

55. This document is not among the documents that I was able to find on archive.org or the Wayback Machine archive of Bitcoin.org.

56. I was not therefore able to find any other instance of the PDF or text corresponding to Exhibit NB1 (BWP-NB1) from any source at all, including within the disclosure dataset, nor any reference to the hashes of that file online.

{D/91}

Comparative analysis of the file itself

57. Since I was not able to find any resource which might indicate the provenance of BWP-NB1, and since nothing equivalent is included in the disclosure dataset, I proceeded to investigate the file itself in detail against the Control copies to establish as far as I could whether it was authentic or whether it might be the product of manipulation.

{D/91}

{ID_000226} 58. Comparing ID_000226, ID_000865 and BWP-NB1 side by side, and subjecting BWP-NB1 to the same level of scrutiny and analysis as the others, my observations were as follows:
{D/91}

{D/91}
{ID_000865}
{ID_000226}

a. Overall BWP-NB1 was structured simply and straightforwardly in exactly the same way as ID_000865 and ID_000226.

b. None of the documents contained any data to indicate their XMP core.

c. The encoding of data within streams was done in the same way as the Control Copies.

{D/91}
{ID_000226}

d. Looking at the raw data, as a starting point, I observed that the first few kilobytes of BWP-NB1 are an identical match for ID_000226, suggesting a close relation.

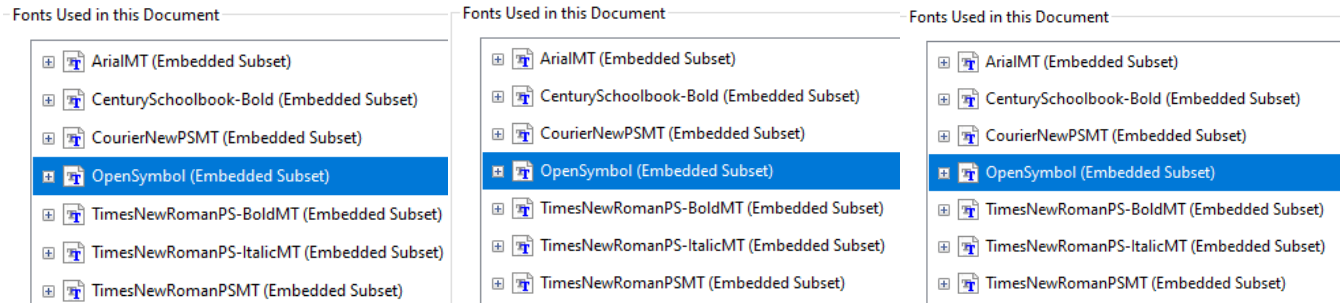
e. I did not observe any characteristics of the metadata having been altered by poor forensic handling.

f. Looking at the available metadata, the metadata characteristics for all three files were a match, with the same information recorded for Creator, and PDF producer. All three included a “Create” date but did not include a “modified” date suggesting that they were not modified in PDF form, but produced by exporting to PDF from OpenOffice.org Writer version 2.4.

g. However, I noted that the time zone offsets encoded in the three files were different (as displayed above, and as discussed further below).

h. I then opened all three files in the forensic examination tools Encase PDF Stream Dumper, and reviewed them for any differences.

i. Overall, the structure and content of the files is very similar. For example, I note that in all three cases, the embedded fonts in the PDFs are the same and they are embedded in the same order:



{ID_000226}
{D/91}
{ID_000865}

Comparison of identical embedded fonts in ID_000226, BWP-NB1, and ID_000865 respectively. In these screenshots, "OpenSymbol" is highlighted for ease of reference, because it will become relevant to a later analysis but is not relevant to the present point.

j. I then expanded the streams in each of the documents using qPDF and examined the expanded data of the files, and again observed that they corresponded very closely.

k. Overall, the three documents all appear to have been generated in the same way using the same configuration.

l. The only other differences I could observe corresponded to the dates of creation and the textual changes shown in Exhibits PM3.3 and PM3.4 and the hashes shown below.

{H/23}
{H/24}

m. The document BWP-NB1 also exhibits the same style of trailer section as found in ID_000865 and ID_000226, which reads as follows. Again, the ID and Checksum values are different, which is consistent with the textual changes to the document.[4]

{D/91}
{ID_000865}
{ID_000226}

```
trailer
<</Size 68/Root 66 0 R
/Info 67 0 R

/ID [ <693BFE527899A65C197B4CD42561A6C4>
<693BFE527899A65C197B4CD42561A6C4> ]
/DocChecksum /34F0F5B5B6D9FFA3BECB3A3AABF39B1D
>>
startxref
182801
%%EOF
```

{ID_000865}
{ID_000223}
{D/91}

[4] I note here that the presence of a "trailer" section in this way is not common in PDF documents, and it is not present in any other BWP version that I have seen, other than ID_000865, ID_000223, and BWP-NB1, save in one case that I describe below in which the Checksum seems to have been copied from ID_000865.

59. Having observed the similarities, I also looked for any additional differences or possible indications of manipulation in BWP-NB1. I could not find any indications that it had been edited or manipulated at all. There were no tags indicative of editing such as "Touchup_textedit" tags; the streams of the PDF did not appear to contain any redundant information, and there did not appear to be any areas where the formatting or encoding changed between sections. There were no artefacts of conversion between formats, altering of metadata, or content having been removed.

{D/91}

60. Finally, taking a step back and looking at the textual changes between the versions, from a forensic document examination perspective, the changes from ID_000226 to BWP-NB1, and then from BWP-NB1 to ID_000865, do appear to be consistent with ordinary user editing behaviour and are not inherently suspicious. I note however that this is not an opinion on the technical content of the changes themselves, which is outside my expertise and I do not form a view about. It is simply a general view based on my experience of typical user editing behaviour.

{ID_000226}
{D/91}
{ID_000865}

Time zone observation

61. The only difference that I could observe between the two files other than the textual changes was that the encoding of the time zone in each case was different:

{ID_000226}
{D/91}
{ID_000865}

	Created date	TZ offset
ID_000226	3 October 2008 at 13:49:58	UTC-0700
BWP-NB1	11 November 2008 at 08:00:34	UTC-0800
ID_000865	24 March 2009 at 11:33:15	UTC-0600

62. I have therefore considered this very carefully.

Daylight saving time and Pacific time zone

63. First, I observe that some differences in time zone offset are to be expected, in view of normal daylight saving changes. To illustrate this, noting that all three time zones are consistent with North America (although not only North America), the dates for the beginning and end of daylight saving time were as follows in the USA in those years: [5]

Year	DST Begin date	DST End date
2008	8 March 2008	2 November 2008

[5] These dates are taken from https://en.wikipedia.org/wiki/History_of_time_in_the_United_States, and confirmed for Canada and the USA according to https://www.timeanddate.com/time/change/canada?year=2008 and https://www.timeanddate.com/time/change/canada?year=2009, as well as the equivalent pages for Canada.

2009	9 March 2009	1 November 2009

Bringing these together:

		Created date	TZ offset	DST?
{ID_000226}	ID_000226	3 October 2008 at 13:49:58	UTC-0700	DST active
{D/91}	BWP-NB1	11 November 2008 at 08:00:34	UTC-0800	DST ended
{ID_000865}	ID_000865	24 March 2009 at 11:33:15	UTC-0600	DST active

{ID_000226} 64. Taking that into account, the time zone change from ID_000226 (i.e., UTC-7 during DST) and BWP-NB1 (i.e., UTC-8 after DST ended) are both consistent with the Western US or Canada, which operate in the Pacific time zone. As I understand it, that corresponds to the states California, Idaho, Nevada, Oregon and Washington (in the USA) or British Colombia (in Canada). See Exhibit PM3.5.

{H/25}

65. I note that Mexico is listed on https://en.wikipedia.org/wiki/UTC%E2%88%9207:00 as using UTC-7 during daylight saving hours, but that according to https://www.timeanddate.com/time/change/mexico?year=2008 and https://www.timeanddate.com/time/change/mexico?year=2009 Daylight Saving Time did not begin in Mexico until April 2009, which is not consistent with the time zones of ID_000226 and BWP-NB1.

{ID_000226}
{D/91}

{D/91} 66. In view of these factors, the stated time zone encoding for BWP-NB1 therefore does appear to be consistent with the first Control Copy, ID_000226.

{ID_000226}

Time zone of Second control copy

{ID_000865} 67. However, the second Control Copy, ID_000865 is encoded in the time zone UTC-6.

68. As indicated by the table above, by 24 March 2009, the US and Canada would have been in their daylight saving periods again: therefore the time zone indicated by ID_000865 is 1 hour different to the time zone indicated by ID_000226 and BWP-NB1.

{ID_000865}
{ID_000226}
{D/91}

69. Considering this carefully, I do not think this is necessarily irregular or suspicious. I note that both the US and Canada have multiple time zones, and that a difference of 1 hour is consistent with ordinary travel around the country.

{ID_000226} 70. I also observe that the period between ID_000226 and BWP-NB1 is around 5 weeks, whereas the period between BWP-NB1 and ID_000865 is longer, being over 4 months.

{D/91}
{ID_000865}

{D/91} 71. Finally, in the context of the current analysis (which is the authenticity of BWP-NB1 as a comparator for other related documents in the disclosure dataset), I observe that the

{D/91}
[ID_000226} differences between the time zones are not indicative of any irregularity with BWP-NB1 (the time zone for which correlates to Control Copy ID_000226, but of differences between the two Control Copies. The Control Copies appear to be independently verified by independent sources, as stated above. Therefore, the apparent difference between the time zones on their face does not indicate BWP-NB1 to be inauthentic, in my opinion.

{D/91} *Conclusion on BWP-NB1*

{D/91} 72. My opinion, in view of these findings, is that the correspondence between BWP-NB1 and the two Control Copies of the BWP indicate BWP-NB1 is an authentic version of the Bitcoin White Paper.

73. I caution that it is not possible for me to determine with certainty that a document is "authentic" or "inauthentic" but only to indicate an opinion based on the factors I observe. I do not form any opinion on the provenance of the document itself (which I am not aware of) or the informational content of the document, and I point out that it is absolutely possible to create a perfect forgery which would evade detection. However, absent any reason to doubt
{D/91} BWP-NB1, the only conclusion I can reach is that it is very likely to be an authentic document from the same source as the original Bitcoin White Paper, and that it appears to be an intermediate draft between the two known versions.

Turning to non-control documents: PDFs

74. I turn below to the various other Bitcoin White Paper documents in the disclosure dataset referred to in paragraph 6 above. As I explain in detail below, I have found significant points
{D/91} of concern with many of these documents (other than the Control Copies and BWP-NB1). I will first address the BWP documents which were disclosed in the PDF format.

{ID_003732} **ID_003732**

75. I report first on this document, as though it has a relatively high ID number, I have come to the opinion that it chronologically precedes some other iterations of the BWP seen within the disclosure dataset. On the face of the document it presents with the text as per the partial screenshot below. It is the only version of the document which bears these contact details (labelled as "Style 6" in the table earlier in this Appendix).

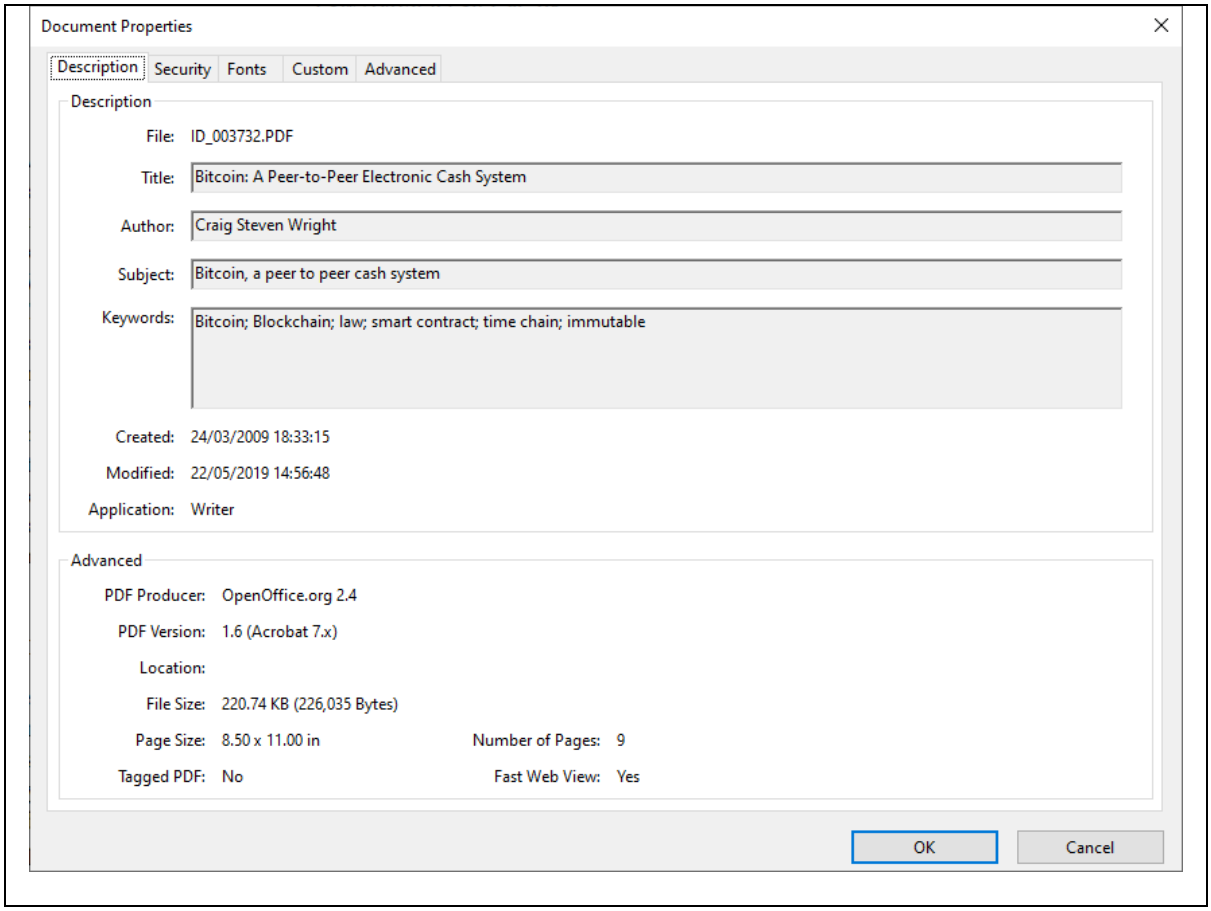
Bitcoin: A Peer-to-Peer Electronic Cash System

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Abstract. A purely peer-to-peer version of electronic cash would allow online payments to be sent directly from one party to another without going through a financial institution. Digital signatures provide part of the solution, but the main benefits are lost if a trusted third party is still required to prevent double-spending. We propose a solution to the double-spending problem using a peer-to-peer network. The network timestamps transactions by hashing them into an ongoing chain of hash-based proof-of-work, forming a record that cannot be changed without redoing the proof-of-work. The longest chain not only serves as proof of the sequence of events witnessed, but proof that it came from the largest pool of CPU power. As long as a majority of CPU power is controlled by nodes that are not cooperating to attack the network, they'll generate the longest chain and outpace attackers. The network itself requires minimal structure. Messages are broadcast on a best effort basis, and nodes can leave and rejoin the network at will, accepting the longest proof-of-work chain as proof of what happened while they were gone.

{ID_003732} *ID_003732 Edited in May 2019*

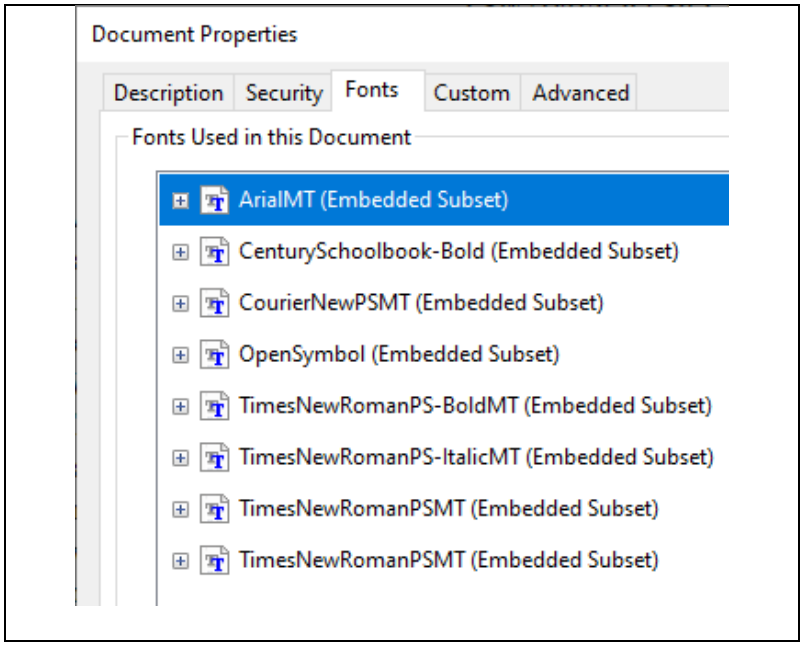
76. The Adobe Reader properties for the document indicate a created date of 24/03/2009 at 17:33:15 UTC (24 March 2009), and a modified date of 22/05/2019 at 13:56:48 UTC (22 May 2019). There are several entries within the Title, Author, Subject and Keywords property fields. A screenshot of these properties is shown below:



77. Comparing to the Control Copies, I note that this puts the document 'Created' time as {ID_000865} identical to ID_000865, with subsequent modifications being made on 22 May 2019.

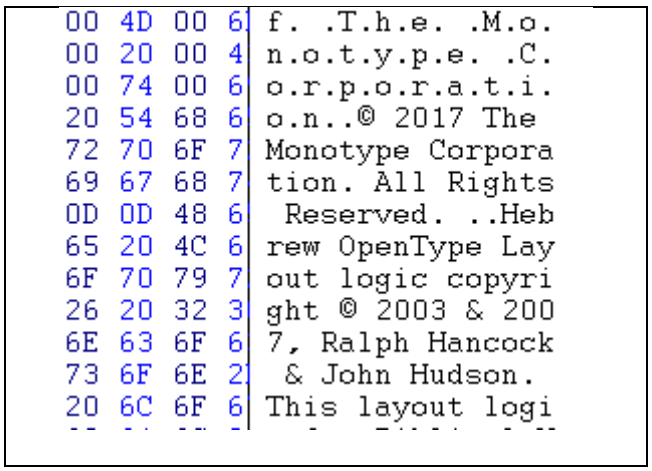
{ID_003732} *ID_003732 - Additional embedded font from 2017*

{ID_003732} 78. It can also be determined from the Adobe Reader properties tab that ID_003732 includes references to several "Fonts Used in this Document" as per the screenshot below



79. By comparison with the equivalent screenshots relating to control copies of the Bitcoin White Paper earlier in this Appendix, it can be observed that this list of fonts contains an additional entry as compared to the Control Copies. Specifically, there is a second copy of "TimesNewRomanPSMT (Embedded Subset)", which appears twice at the bottom of this screenshot. This is consistent with the second copy of the font corresponding to the later modification of the PDF by inserting text, which has been encoded with a different embedded font file (relating to the same typeface) to the one originally included in the document at the point of creation.

80. I proceeded to analyse the content of this PDF file using PDF Stream Dumper. I observed that one of the embedded font files included copyright information for "`© 2017 The Monotype Corporation`" as shown in the partial screenshot below. The other embedded font files did not include such a statement. This is consistent with the document having been edited in 2019.



{ID_003732} *ID_003732 - Embedded Metadata property streams*

{ID_003732} 81. I found that ID_003732 contained two metadata property streams containing the following two sets of metadata properties which are not present in the Control Copies. (I have changed line breaks and added highlighting to aid review):

```
41 0 <</Author(Craig Steven Wright)
/CreationDate(D:20090324113315-06'00')
/Creator(Writer)
/Keywords(Bitcoin; Blockchain; law; smart contract; time chain; immutable)
/ModDate(D:20190522145648+01'00')
/Producer(OpenOffice.org 2.4)
/Subject(Bitcoin, a peer to peer cash system)
/Title(Bitcoin: A Peer-to-Peer Electronic Cash System)>>
```

```
<?xpacket begin="ï»¿" id="W5M0MpCehiHzreSzNTczkc9d"?>
<x:xmpmeta xmlns:x="adobe:ns:meta/" x:xmptk="Adobe XMP Core 5.6-c016 91.163616,
2018/10/29-16:58:49        ">
   <rdf:RDF xmlns:rdf="http://www.w3.org/1999/02/22-rdf-syntax-ns#">
      <rdf:Description rdf:about=""
            xmlns:xmp="http://ns.adobe.com/xap/1.0/"
            xmlns:pdf="http://ns.adobe.com/pdf/1.3/"
            xmlns:xmpRights="http://ns.adobe.com/xap/1.0/rights/"
            xmlns:photoshop="http://ns.adobe.com/photoshop/1.0/"
            xmlns:dc="http://purl.org/dc/elements/1.1/"
            xmlns:xmpMM="http://ns.adobe.com/xap/1.0/mm/">
         <xmp:CreateDate>2009-03-24T11:33:15-06:00</xmp:CreateDate>
         <xmp:CreatorTool>Writer</xmp:CreatorTool>
         <xmp:ModifyDate>2019-05-22T14:56:48+01:00</xmp:ModifyDate>
         <xmp:MetadataDate>2019-05-22T14:56:48+01:00</xmp:MetadataDate>
         <pdf:Producer>OpenOffice.org 2.4</pdf:Producer>
         <pdf:Keywords>Bitcoin; Blockchain; law; smart contract; time chain;
immutable</pdf:Keywords>
         <xmpRights:WebStatement>https://cocatalog.loc.gov/cgi-
bin/Pwebrecon.cgi?v1=8&amp;ti=1,8&amp;Search_Arg=bitcoin&amp;Search_Code=TALL&amp
;CNT=25&amp;PID=smt26T35HVzgQS6nr_ENlOg0IjL7h&amp;SEQ=20190522095558&amp;SID=1</x
mpRights:WebStatement>
         <photoshop:AuthorsPosition>Dr</photoshop:AuthorsPosition>
         <photoshop:CaptionWriter>Satoshi Nakamoto</photoshop:CaptionWriter>
         <dc:format>application/pdf</dc:format>
         <dc:title>
            <rdf:Alt>
               <rdf:li xml:lang="x-default">Bitcoin: A Peer-to-Peer Electronic
Cash System</rdf:li>
            </rdf:Alt>
         </dc:title>
         <dc:description>
            <rdf:Alt>
               <rdf:li xml:lang="x-default">Bitcoin, a peer to peer cash
system</rdf:li>
            </rdf:Alt>
         </dc:description>
         <dc:creator>
            <rdf:Bag>
               <rdf:li>Craig Steven Wright</rdf:li>
            </rdf:Bag>
         </dc:creator>
         <dc:subject>
            <rdf:Bag>
               <rdf:li>Bitcoin</rdf:li>
               <rdf:li>Blockchain</rdf:li>
               <rdf:li>law</rdf:li>
```

```
               <rdf:li>smart contract</rdf:li>
               <rdf:li>time chain</rdf:li>
               <rdf:li>immutable</rdf:li>
            </rdf:Bag>
         </dc:subject>
         <dc:rights>
            <rdf:Alt>
               <rdf:li xml:lang="x-default">(C) Craig Steven
Wright&#xA;2019</rdf:li>
            </rdf:Alt>
         </dc:rights>
         <xmpMM:DocumentID>uuid:3658ec91-de13-42e3-9796-
6369d7172efc</xmpMM:DocumentID>
         <xmpMM:InstanceID>uuid:3f572223-3a06-4323-a01f-
b8c613362b5d</xmpMM:InstanceID>
      </rdf:Description>
   </rdf:RDF>
</x:xmpmeta>

<?xpacket end="w"?>
```

82. The metadata indicates that this document was modified on 22 May 2019 in a UTC+1 time zone. This is consistent with modification on a computer localised to the United Kingdom, during daylight saving time (British Summer Time) or to other countries such as Portugal, and several countries in Africa which were on UTC+1 in May 2019.

83. The Adobe XMP core is recorded as being "Adobe XMP Core 5.6-c016 91.163616, 2018/10/29-16:58:49" which is also consistent with a document modified in 2019.

{ID_003732} 84. Compared to the Control Copies ID_003732 includes a number of additional metadata fields that have been populated with information. I note specifically the following fields:

```
/Keywords(Bitcoin; Blockchain; law; smart contract; time chain; immutable)
/ModDate(D:20190522145648+01'00')
/Producer(OpenOffice.org 2.4)
/Subject(Bitcoin, a peer to peer cash system)
/Title(Bitcoin: A Peer-to-Peer Electronic Cash System)>>
```

85. These metadata fields are blank within the Control Copies of the Bitcoin White Paper, and so are not original to it.

{ID_003732} *ID_003732 - "Touchup_textedit"*

{ID_003732} 86. I observed that document ID_003732 included a "Touchup_textedit" reference typical of an edit being made to the content of the document using the Adobe Acrobat suite of products. I have analysed "Touchup_textedit" references in some detail in Appendix PM4.

{H/29}

87. The text determined to be added using the Touchup process is that shown below:

Dr Craig S Wright
csw26@leicester.ac.uk
nChain Ltd

{ID_003732} *ID_003732 - conclusion*

88. The above analysis indicates that these name and contact details were not part of the original content of the BWP. It has not been possible to extract from ID_003732 what content was removed prior to the insertion of those contact details, but in view of the close association between ID_003732 and ID_000865, it is my view that ID_003732 was created by taking a copy of the 23 March 2009 Control Copy and altering the name and address information as well as adding the metadata fields as listed above, using Adobe Acrobat software to perform that edit. The evidence is consistent with these modifications having been undertaken on 22 May 2019.

{ID_003732}
{ID_003732}
{ID_000865}

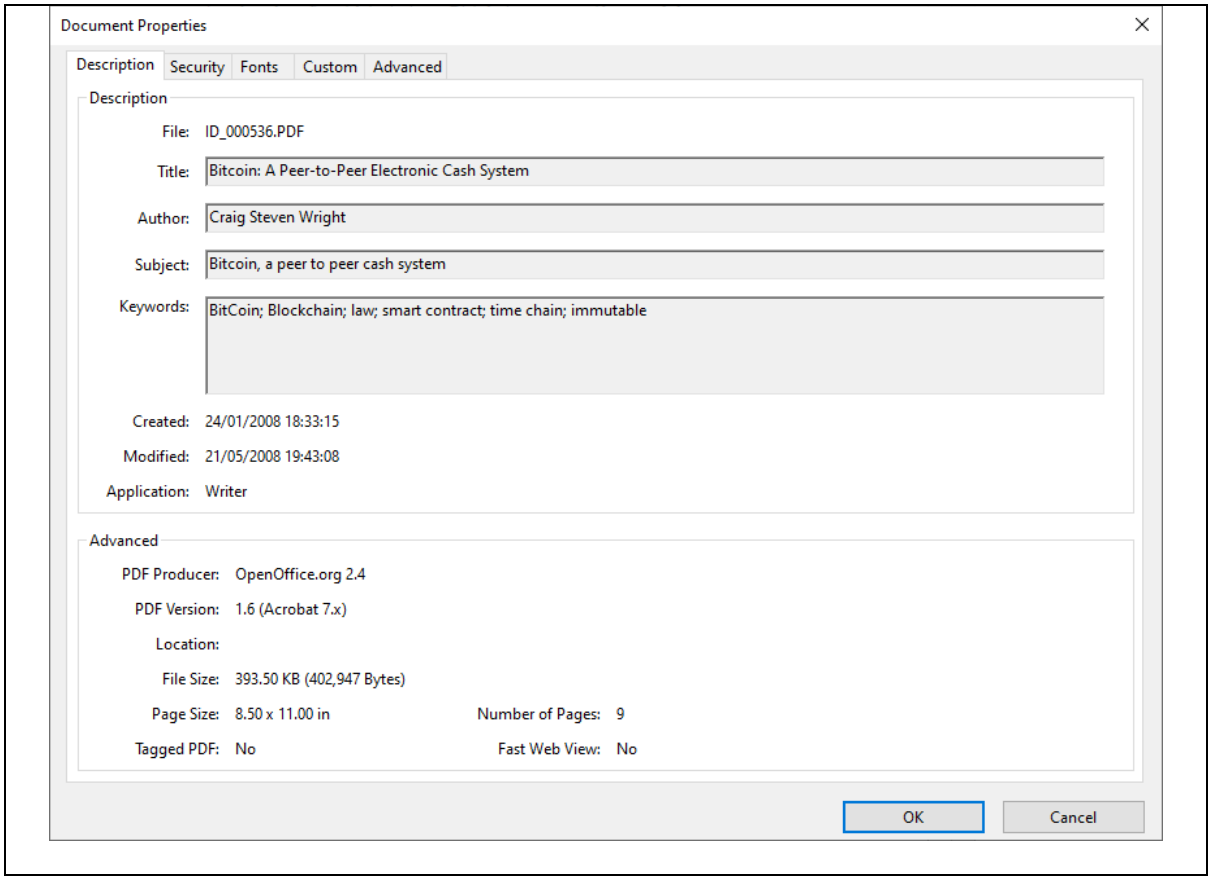
{ID_000536} **ID_000536**

89. The MD5 hash value of ID_000536 is "42fa5efcd463e895c4a1aa7f5612f02f". This correlates with 2 other documents within the disclosure dataset, ID_000539.PDF, and ID_003816.PDF. As before, while the file date and timestamps for these documents may differ, the content is identical. The analysis regarding the internal metadata and content of these documents will therefore match.

{ID_000536}
{ID_000539}
{ID_003816}

90. The Adobe Acrobat reader properties for the document are shown in the screenshot below in BST (with time in BST)[6]. I note that the "Created" and "Last Modified" dates shown here, as well as the OS file property metadata provided in the load file, all indicate the document to be from 2008.

[6] BST has been my consistent localisation setting throughout the preparation of this report other than at times that I have reproduced clock manipulation techniques. Where I express times in BST, I am indicating that the readout of the time was in BST at the point it was displayed to me, which I have not adjusted for time zones unless the analysis required doing so for clarity. Where the time zone offset does matter, as it has done earlier in this Appendix, I have tried to be clear and to express it in terms of UTC offset relating it, so far as possible, to the appropriate time zone.



{ID_000536} *ID_000536 – Metadata irregularities*

91. It can be seen that the document includes a creation timestamp of 24/01/2008 18:33:15 BST. I note that:

{ID_000865}
a. the day, hour, minutes, and seconds portions of this timestamp identically match that of the ID_000865 Control Copy, but

b. The month and year do not, purporting to be from over a year earlier.

{ID_000536}
c. ID_000536 also has a modification date of 21/05/2008 at 19:43:08 (BST), which the two BWP Control Copies do not.

{ID_000865}
d. The PDF version is recorded as “1.6 (Acrobat 7.x)” rather than “1.4 (Acrobat 5.x)” as seen on the ID_000865 Control Copy.

{ID_000536}
{ID_003732}
e. The fields for Title, Author, Subject and Keywords are also populated with text. These fields are empty in both the Control Copies of the BWP. In ID_000536, the fields are similar to those in ID_003732, except for the word Bitcoin in the Keywords section.

{ID_003732}
{ID_000536}

In ID_003732 it is "Bitcoin" whereas ID_000536 lists it as "BitCoin" with a capital "C" character.

{ID_000536}
{ID_000865}
{ID_000226}

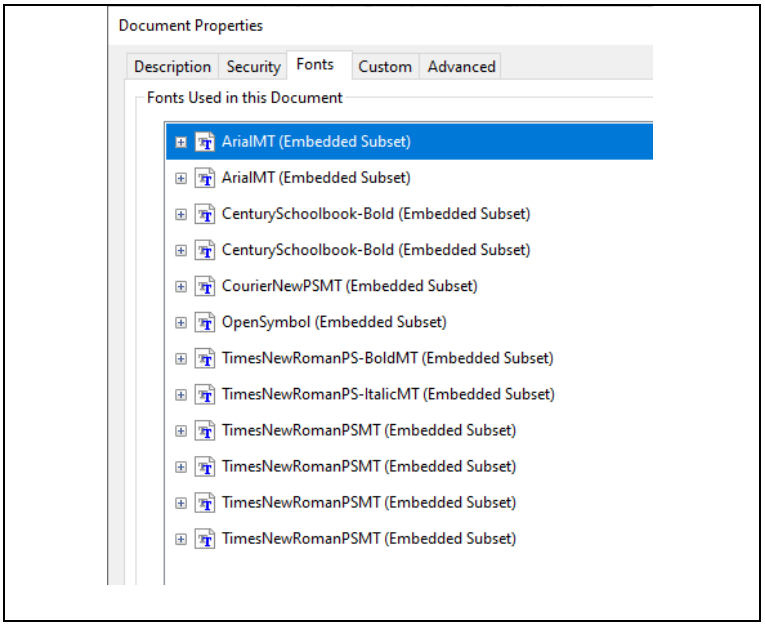
92. In terms of its content, a comparison shows that ID_000536 contains content which matches content in the ID_000865 Control Copy of the BWP, but does not match the ID_000226 Control Copy. Given that ID_000536 purportedly pre-dates ID_000226, which in turn pre-dates ID_000865, this does not appear consistent with normal editing behaviour, as it implies that content would have had to be present in May 2008, in ID_000536, but that it was then deleted by October 2008 (in ID_000226), and then the same content reinstated in identical terms by March 2009 (in ID_000865). The textual differences between ID_000536 and ID_000226 are shown in the comparison document at **Exhibit PM3.6**.

{ID_000536}
{ID_000226}
{ID_000865}

{H/26}

{ID_000536} *ID_000536 – additional duplicate fonts*

93. A screenshot of the Fonts tab can be seen in the screenshot below:



94. As can be seen in the above dialog, this is similar to the phenomenon observed with ID_003732 which had an additional instance of the Times New Roman PSMT font embedded within it. However, in the case of ID_000536, the same font has been embedded two more additional times, for a total of 4 instances of the same font.

{ID_003732}

{ID_000536}

{ID_000536} *ID_000536 – file size and indications of content manipulation*

95. On the face of it, the content of this document closely resembles the Control Copies of the BWP with some of the pages being identical. A prominent difference, however, being the

{ID_000865}
{ID_000536}

attribution of authorship. The screenshots below illustrate the ID_000865 Control Copy BWP on the left compared to ID_000536 BWP version on the right:

ID_000865 Control Copy –	ID_000536
ɾ-to-Peer Electronic Satoshi Nakamoto satoshin@gmx.com www.bitcoin.org	**r-to-Peer Electronic** Dr Craig S Wright craigswright@acm.org Charles Sturt University

96. {ID_000865} {ID_000536} There is also a significant file size difference between the two, with the Control Copy ID_000865 being 180KB and ID_000536 being 394KB in capacity, which is more than double the file size.

{ID_000536} 97. I observed that the document ID_000536 contained several "Touchup_Textedit" metadata flags within the document data streams[7]. I observed that multiple portions of ID_000536 had been altered with content being modified. This is consistent with changes that can be identified through a visual comparison of the two documents. Bird & Bird has prepared an automated PDF comparison between the files ID_000536 and ID_000865, which is at **Exhibit PM3.7** and which corresponds to the areas bearing "Touchup_textedit" tags.

{ID_000536}
{ID_000865}
{H/27}

98. {H/27} {ID_000536} However, one observation that I thought noteworthy and which does not appear in Exhibit PM3.7 is the inclusion of one such "Touchup_textedit" edit that is not present on the face of ID_000536 but can be decoded from examination of the raw data of the file:

a. {ID_003732} The text content of this touchup matched that which was applied to the face of ID_003732 described in paragraph 87 above (i.e., the identification of Dr Craig S Wright at nChain Ltd as the author of the BWP). {ID_000536} However, as set out above, that content is not present on the face of ID_000536, which instead indicates the author as being Dr Craig S Wright at Charles Sturt University. Below I list the three authorship texts for these documents:

{H/29}

[7] The analysis of these is covered in more detail regarding "Touchup_textedit" documents in Appendix PM4 and I do not go into the same level of detail here.

{ID_000865}
{ID_003732}
{ID_000536}

Control Copy – ID_000865	ID_003732	ID_000536
r-to-Peer Electronic	**r-to-Peer Electronic**	**r-to-Peer Electronic**
Satoshi Nakamoto satoshin@gmx.com www.bitcoin.org	Dr Craig S Wright csw26@leicester.ac.uk nChain Ltd	Dr Craig S Wright craigswright@acm.org Charles Sturt University

{ID_000536}
{ID_003732}
{ID_000865}
{ID_000536}

b. The inclusion of this information within the PDF structure of ID_000536 is consistent with edits being applied to ID_003732 (which I have found derives from Control Copy ID_000865) to create document ID_000536.

{ID_003732}

c. In the process of that editing, the edited text stream which had been inserted into ID_003732 was removed from the face value content of the document, but remained embedded within the raw data.

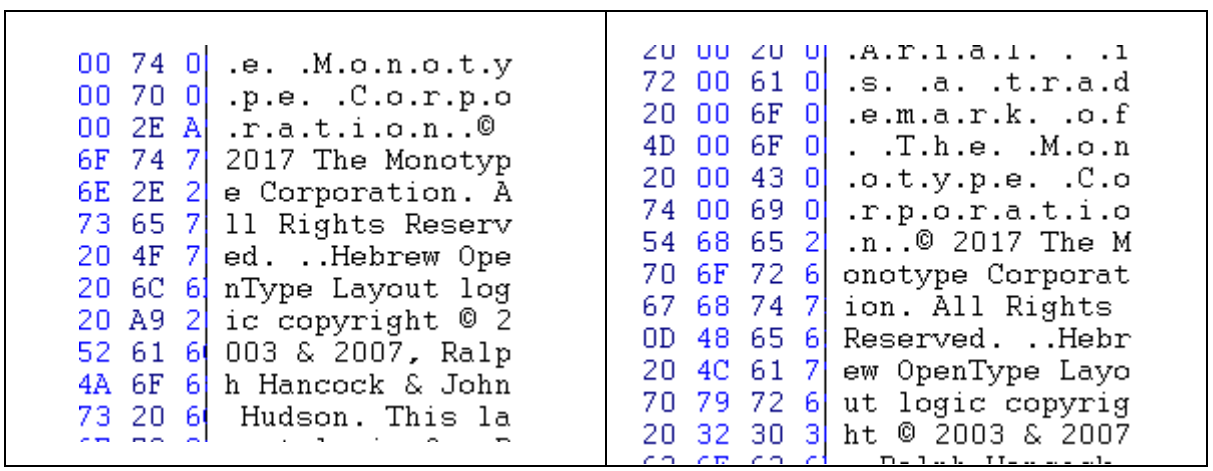
{ID_003832} 99. It should be noted that the internal metadata properties for ID_003732 indicate it as having been Last Modified on 22 May 2019 which is not consistent with the recorded 2008 creation and last modification dates of ID_000536.

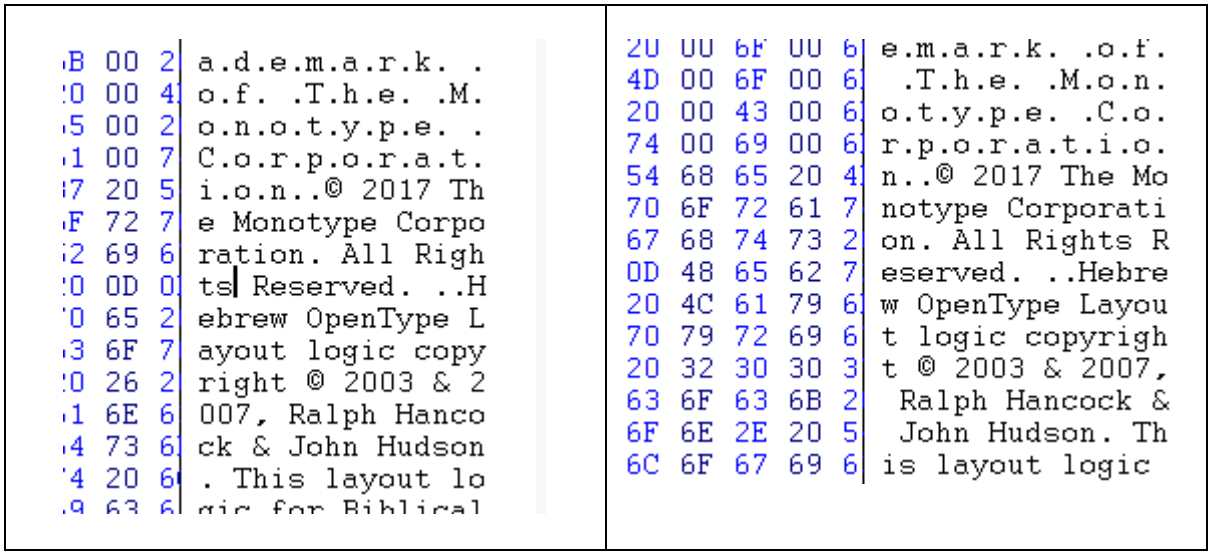
{ID_000536}

{ID_000536} *ID_000536 – post-dating fonts*

100. I next explored the content of ID_000536 with PDF Stream Dumper. Based on that review, I make the following observations.

{ID_000536} 101. Despite ID_000536 being recorded as having been Last Modified in May 2008 it includes embedded within it four Font files which include 2017 copyright information. I demonstrate the copyright information from within these embedded font files in the table below





102. In each example one can see the words "© 2017 The Monotype Corporation…"

{ID_000536} 103. The inclusion of these 2017 font files embedded within ID_000536 is inconsistent with the stated creation and last modification dates for the document, suggesting the possible use of clock manipulation to set the clock back when making changes to the content of the document.

{ID_000536} *ID_000536 – conflicting metadata streams*

104. I next examined the metadata property streams within ID_000536. I observed that it had four different data streams which contained metadata properties as per the table below. I have added highlighting and changed line breaks to aid review of the four streams, which are as follows:

68 0 <</Author(Craig Steven Wright) /CreationDate(D:20090324113315-06'00') /Creator(Writer) /Keywords(Bitcoin; Blockchain; law; smart contract; time chain; immutable) /ModDate(D:20190821193528+01'00') /Producer(OpenOffice.org 2.4) /Subject(Bitcoin, a peer to peer cash system) /Title(Bitcoin: A Peer-to-Peer Electronic Cash System)>>
68 0 <</Author(Craig Steven Wright) /CreationDate(D:20090324113315-06'00') /Creator(Writer) /Keywords(Bitcoin; Blockchain; law; smart contract; time chain; immutable)/ModDate(D:20080521194308+01'00') /Producer(OpenOffice.org 2.4) /Subject(Bitcoin, a peer to peer cash system) /Title(Bitcoin: A Peer-to-Peer Electronic Cash System)>>
<?xpacket begin="ï»¿" id="W5M0MpCehiHzreSzNTczkc9d"?> <x:xmpmeta xmlns:x="adobe:ns:meta/" x:xmptk="Adobe XMP Core 1.1-c016 91.132716, 2008/10/29-16:58:49 "> <rdf:RDF xmlns:rdf="http://www.w3.org/1999/02/22-rdf-syntax-ns#"> <rdf:Description rdf:about="" xmlns:xmp="http://ns.adobe.com/xap/1.0/" xmlns:pdf="http://ns.adobe.com/pdf/1.3/"

```
            xmlns:xmpRights="http://ns.adobe.com/xap/1.0/rights/"
            xmlns:photoshop="http://ns.adobe.com/photoshop/1.0/"
            xmlns:dc="http://purl.org/dc/elements/1.1/"
            xmlns:xmpMM="http://ns.adobe.com/xap/1.0/mm/">
         <xmp:CreateDate>2008-01-24T11:33:15-06:00</xmp:CreateDate>
         <xmp:CreatorTool>Writer</xmp:CreatorTool>
         <xmp:ModifyDate>2008-08-21T19:35:28+01:00</xmp:ModifyDate>
         <xmp:MetadataDate>2008-08-21T19:35:28+01:00</xmp:MetadataDate>
         <pdf:Producer>OpenOffice.org 2.4</pdf:Producer>
         <pdf:Keywords>Bitcoin; Blockchain; law; smart contract; time chain;
immutable</pdf:Keywords>
         <xmpRights:WebStatement>
</xmpRights:WebStatement>
         <photoshop:AuthorsPosition>Dr</photoshop:AuthorsPosition>
         <photoshop:CaptionWriter>Craig S. Wright.</photoshop:CaptionWriter>
         <dc:format>application/pdf</dc:format>
         <dc:title>
            <rdf:Alt>
               <rdf:li xml:lang="x-default">Bitcoin: A Peer-to-Peer Electronic
Cash System</rdf:li>
            </rdf:Alt>
         </dc:title>
         <dc:description>
            <rdf:Alt>
               <rdf:li xml:lang="x-default">Bitcoin, a peer to peer cash
system</rdf:li>
            </rdf:Alt>
         </dc:description>
         <dc:creator>
            <rdf:Bag>
               <rdf:li>Craig Steven Wright</rdf:li>
            </rdf:Bag>
         </dc:creator>
         <dc:subject>
            <rdf:Bag>
               <rdf:li>BitCoin</rdf:li>
               <rdf:li>Time-Chain</rdf:li>
               <rdf:li>law</rdf:li>
               <rdf:li>EDI-E-contract</rdf:li>
               <rdf:li>time chain</rdf:li>
               <rdf:li>immutable</rdf:li>
            </rdf:Bag>
         </dc:subject>
         <dc:rights>
            <rdf:Alt>
               <rdf:li xml:lang="x-default">(C) Craig Steven
Wright&#xA;2008</rdf:li>
            </rdf:Alt>
         </dc:rights>
         <xmpMM:DocumentID>uuid:3658ec91-de13-42e3-9796-
6369d7172efc</xmpMM:DocumentID>
         <xmpMM:InstanceID>uuid:59f6e63e-2497-4516-8521-
93e9410b9263</xmpMM:InstanceID>
      </rdf:Description>
   </rdf:RDF>
</x:xmpmeta>

<?xpacket end="w"?>
```

```
<?xpacket begin="ï»¿" id="W5M0MpCehiHzreSzNTczkc9d"?>
<x:xmpmeta xmlns:x="adobe:ns:meta/" x:xmptk="Adobe XMP Core 1.1-c016 91.132716,
2008/10/29-16:58:49        ">
   <rdf:RDF xmlns:rdf="http://www.w3.org/1999/02/22-rdf-syntax-ns#">
      <rdf:Description rdf:about=""
            xmlns:xmp="http://ns.adobe.com/xap/1.0/"
            xmlns:pdf="http://ns.adobe.com/pdf/1.3/"
            xmlns:xmpRights="http://ns.adobe.com/xap/1.0/rights/"
            xmlns:photoshop="http://ns.adobe.com/photoshop/1.0/"
            xmlns:dc="http://purl.org/dc/elements/1.1/"
```

```
            xmlns:xmpMM="http://ns.adobe.com/xap/1.0/mm/">
         <xmp:CreateDate>2008-01-24T11:33:15-06:00</xmp:CreateDate>
         <xmp:CreatorTool>Writer</xmp:CreatorTool>
         <xmp:ModifyDate>2008-05-21T19:43:08+01:00</xmp:ModifyDate>
         <xmp:MetadataDate>2008-05-21T19:43:08+01:00</xmp:MetadataDate>
         <pdf:Producer>OpenOffice.org 2.4</pdf:Producer>
         <pdf:Keywords>BitCoin; Blockchain; law; smart contract; time chain;
immutable</pdf:Keywords>
         <xmpRights:WebStatement>
</xmpRights:WebStatement>
         <photoshop:AuthorsPosition>Dr</photoshop:AuthorsPosition>
         <photoshop:CaptionWriter>Satoshi Nakamoto</photoshop:CaptionWriter>
         <dc:format>application/pdf</dc:format>
         <dc:title>
            <rdf:Alt>
               <rdf:li xml:lang="x-default">Bitcoin: A Peer-to-Peer Electronic
Cash System</rdf:li>
            </rdf:Alt>
         </dc:title>
         <dc:description>
            <rdf:Alt>
               <rdf:li xml:lang="x-default">Bitcoin, a peer to peer cash
system</rdf:li>
            </rdf:Alt>
         </dc:description>
         <dc:creator>
            <rdf:Bag>
               <rdf:li>Craig Steven Wright</rdf:li>
            </rdf:Bag>
         </dc:creator>
         <dc:subject>
            <rdf:Bag>
               <rdf:li>BitCoin</rdf:li>
               <rdf:li>Blockchain</rdf:li>
               <rdf:li>law</rdf:li>
               <rdf:li>smart contract</rdf:li>
               <rdf:li>time chain</rdf:li>
               <rdf:li>immutable</rdf:li>
            </rdf:Bag>
         </dc:subject>
         <dc:rights>
            <rdf:Alt>
               <rdf:li xml:lang="x-default">(C) Craig Steven
Wright&#xA;2008</rdf:li>
            </rdf:Alt>
         </dc:rights>
         <xmpMM:DocumentID>uuid:3658ec91-de13-42e3-9796-
6369d7172efc</xmpMM:DocumentID>
         <xmpMM:InstanceID>uuid:02a9a63b-da5a-449f-b280-
36f9a69f1962</xmpMM:InstanceID>
      </rdf:Description>
   </rdf:RDF>
</x:xmpmeta>

<?xpacket end="w"?>
```

105. The metadata values of the fourth of the listed streams correlate with the internal metadata properties for the document, but others do not. Overall, the recorded times for some of these metadata fields contradict others. In my opinion, the observable characteristics from the streams above are not achievable without either manipulation of the computer clock to backdate the time, or subsequent manual modification of the timestamps using a hex editor.

106. For example, comparing the Modified date between the 1st and 3rd streams it can be seen that they are inconsistent, and purport to be exactly 11 years apart to the second. I have added spaces and removed the "T" time signifiers to aid review.

1st metadata stream Modified	2019 08 21 19 35 28+01'00'
3rd metadata stream Modified	2008-08-21 19:35:28+01:00

107. Comparing the Created date and timestamps, the 1st and 2nd streams match, but are different from the 3rd and 4th which also match each other. I observe also that the time of day is the same between all four, with only the year and month being different by 14 months.

1st metadata stream Created	2009 03 24 11 33 15-06'00'
2nd metadata stream Created	2009 03 24 11 33 15-06'00'
3rd metadata stream Created	2008-01-24 11:33:15-06:00
4th metadata stream Created	2008-01-24 11:33:15-06:00

{ID_000536} *ID_000536 – invalid XMP Core*

108. {ID_000536} The information relating to the XMP Core (the version of the standardised metadata format used to create ID_000536) also exhibits an irregularity. The 3rd and 4th metadata streams contained the following two XMP Core records:

3rd metadata XMP Core	"Adobe XMP Core 1.1-c016 91.132716, 2008/10/29-16:58:49 "
4th metadata XMP Core	"Adobe XMP Core 1.1-c016 91.132716, 2008/10/29-16:58:49 "

109. This XMP Core version was not familiar to me and so I attempted to research information about it. A Google search for this XMP Core information produces only two search results. The search result listings for both of these results appeared that they might relate to this case, so I did not click on them or read the search result summary listing any further than necessary. I wanted to ensure that I did not take them into account, and so immediately navigated away.

110. Attempting to avoid those results, I shortened my search term to just "Adobe XMP Core 1.1", but that produced the same two results only, and took me no further.

111. Disregarding those search results, there were no search results listing this XMP Core version at all, and I do not recall seeing it in my experience of forensic examination in the past. This leads me to the view that it did not exist.

{ID_000536} {ID_003732} 112. I next approached it another way, comparing the XMP core records for ID_000536 to that of ID_003732 (as listed below):

{ID_003732} {ID_000536}

ID_003732 XMP Core	"Adobe XMP Core 5.6-c016 91.163616, 2018/10/29-16:58:49 "
ID_000536 XMP Core	"Adobe XMP Core 1.1-c016 91.132716, 2008/10/29-16:58:49 "

113. As illustrated above in colour coding, the characters are very similar including the hour, minute and second of their creation but some characters in the version numbering have changed. Even if XMP Core 1.1 did exist, it is irregular and not plausible for different versions of XMP Core to have been created at the same month, day, hour, minute, and second as each other, precisely 10 years apart.

{ID_003732} 114. As a cross-check, a Google search for the XMP Core of ID_003732 returns many search hits in relation to entirely unrelated documents. I consider this comparison sufficient to determine that the XMP Core of ID_003736 is authentic to the editing time (in 2019) and that the XMP Core of ID_000536 is invalid and not a real world XMP Core version.

{ID_003736}

{ID_000536}

{ID_000536} *ID_000536 – purported XMP Core post-dates document creation and editing even if valid*

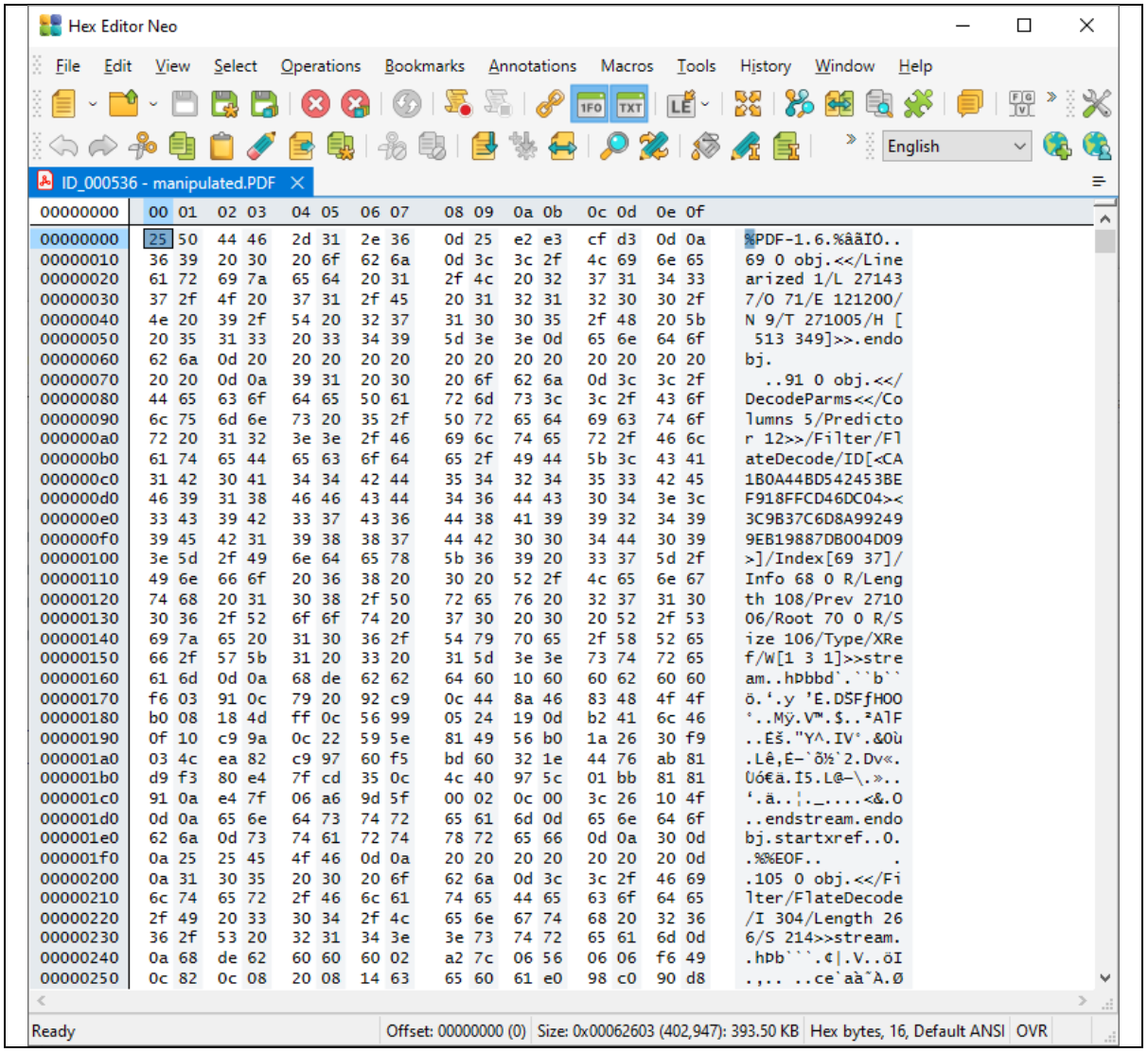
115. Even if it were genuine, I note that the purported date of the XMP Core (October 2008) post-dates the purported authorship (January 2008) and Last Modified (May 2008) times of the document.

{ID_000536} *ID_000536 – demonstration of hex editing*

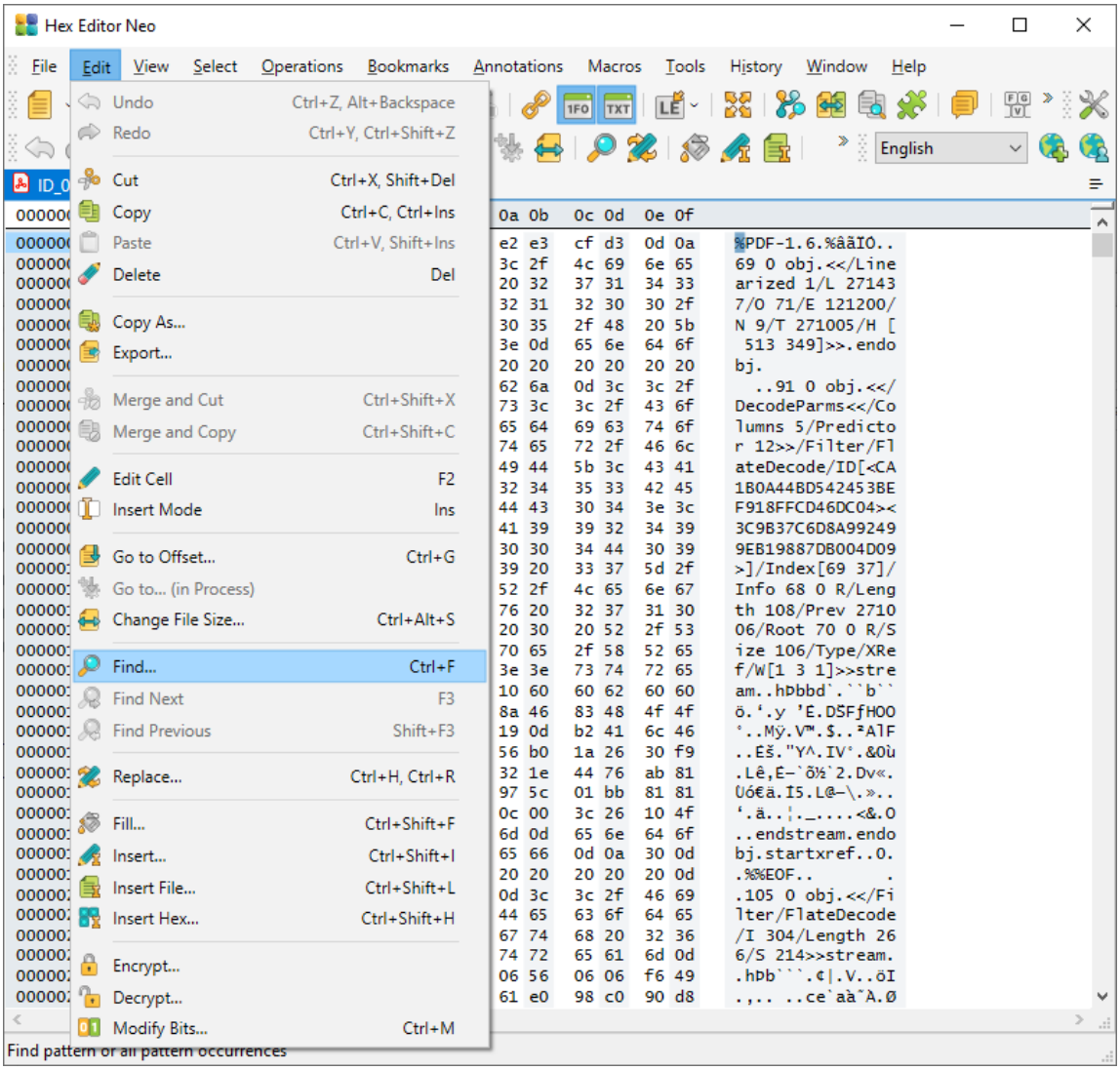
116. Considering the anomalous characteristics of both of the document's timestamps, and the content of the XMP Core, I have formed the opinion that these irregularities are likely the result of someone using a hex editor to manipulate the content of this file at a byte level.

{ID_000536} 117. To demonstrate this process I have used Hex Editor Neo. I have taken a copy of ID_000536 and manipulated the content thereof.

118. Upon opening the document, the user is presented with a screen as follows:

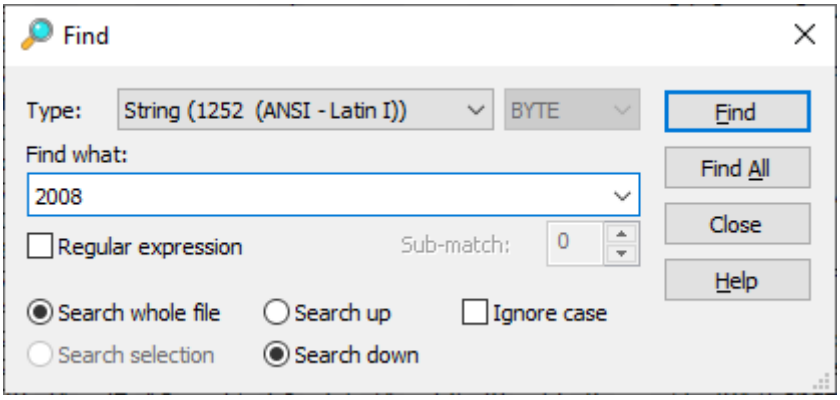


119. I next selected “Find” from the Edit dropdown menu:

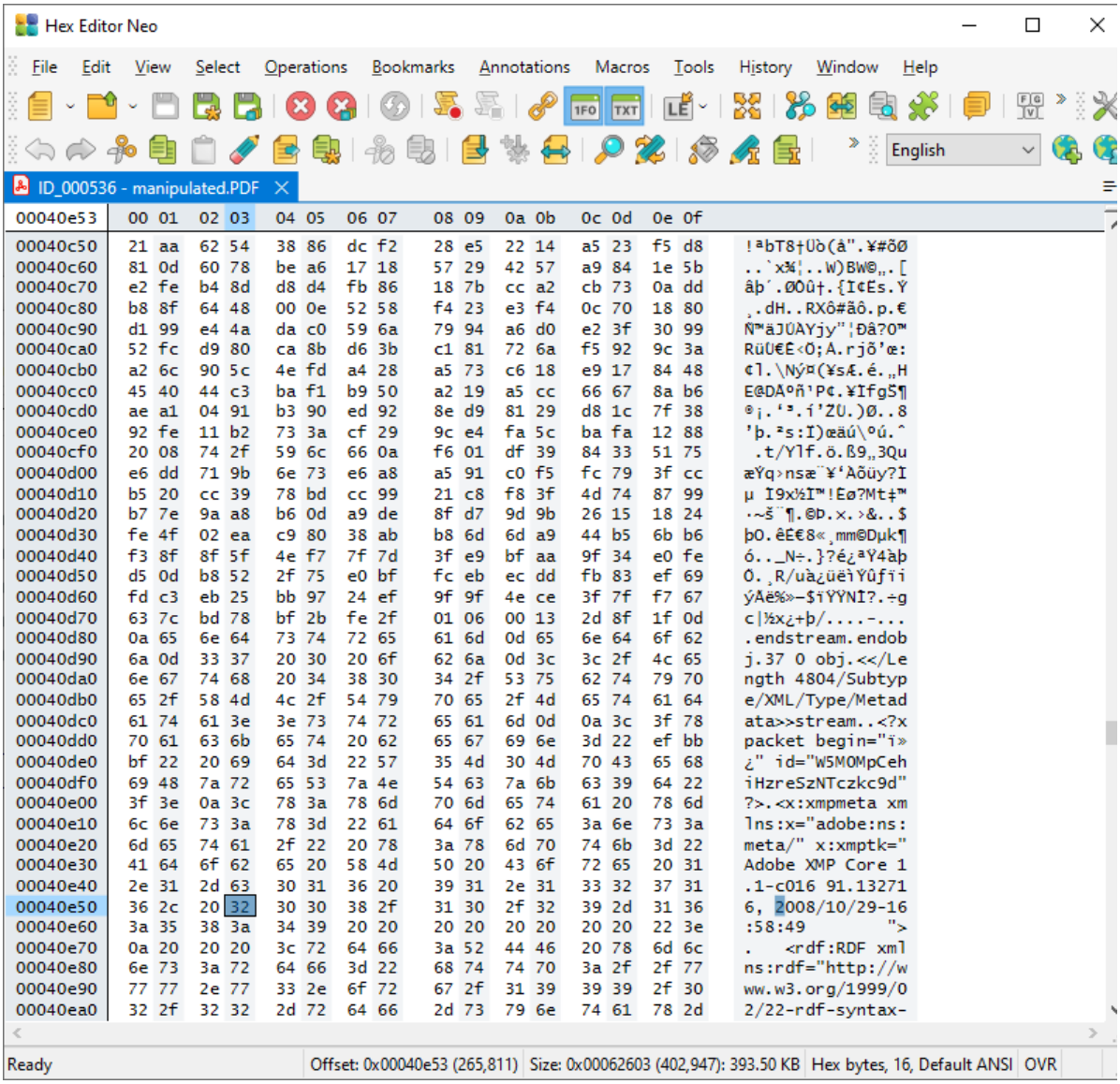


120. A user could choose the Replace option to have all instances of text replaced in one operation. For example all instances of "2008" replaced with "2025". There is however a risk that this could damage the document should "2008" be recorded in a manner that is not indicative of a date.

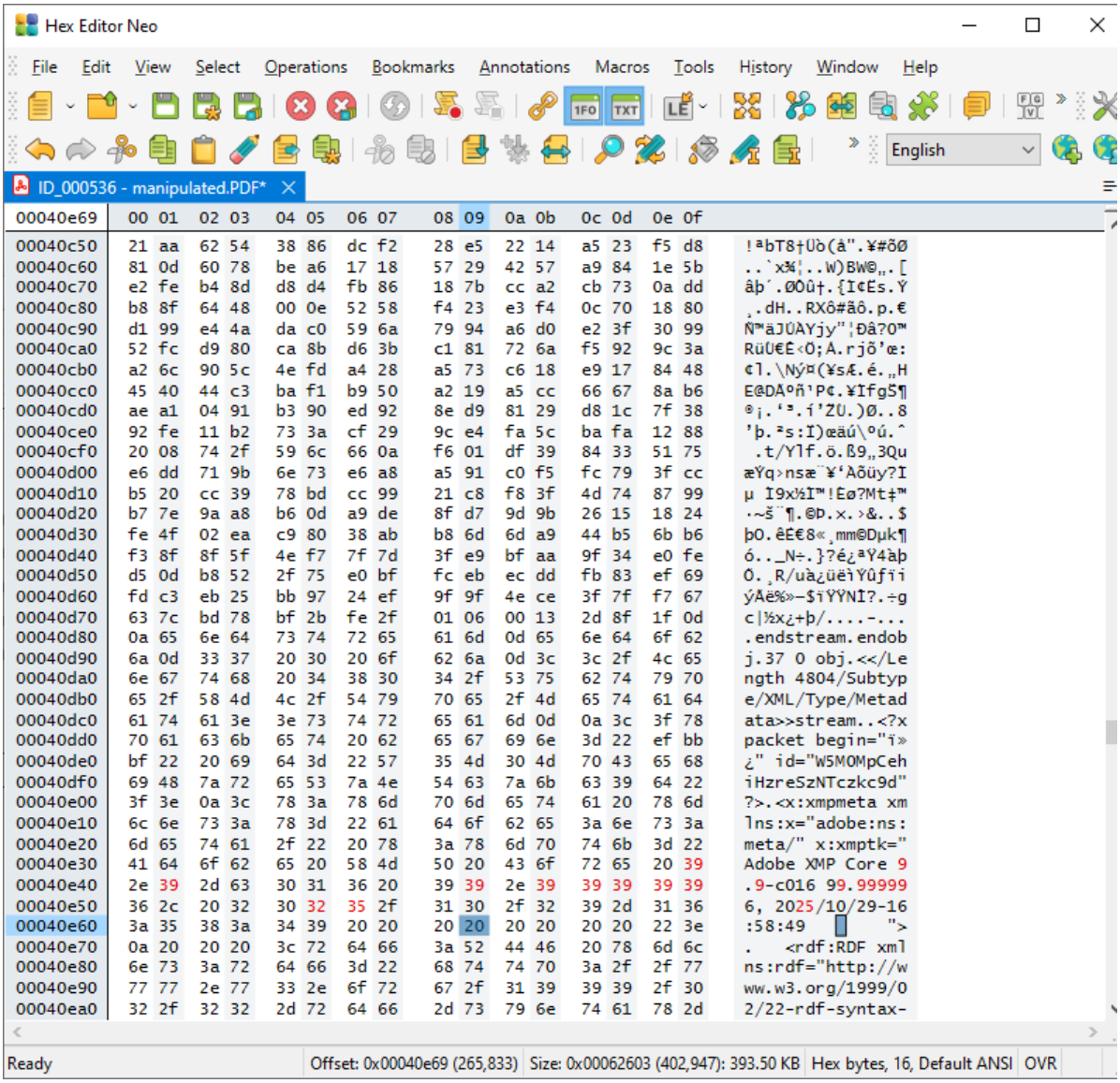
121. In my demonstration, I have kept to using the Find search to locate the various instances of 2008 and replacing them one by one.



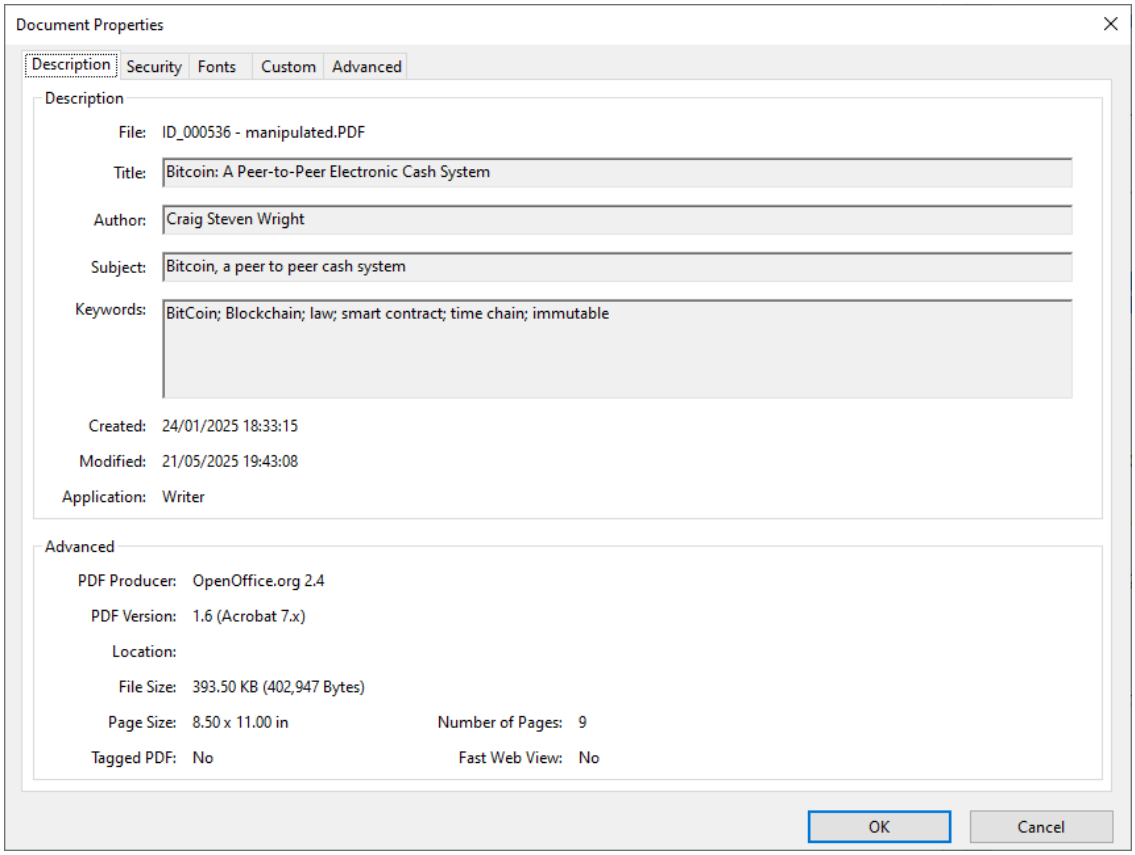
122. The first result is demonstrated below as an XMP Core record:



123. I proceeded to alter the content of the XMP as per the screenshot below, changing the creation date to 2025 and the XMP Core version to 99.99999. While using this method to manipulate the content of a document, it is important to replace bytes 1:1 and not to add or remove bytes from the file which might otherwise leave the data aligned differently in a way that corrupts the file. The alterations are highlighted in red until saved.

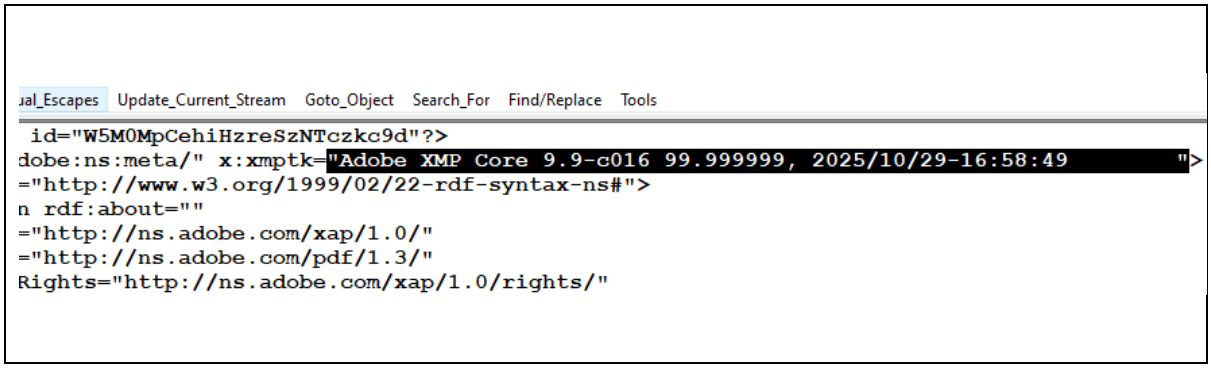


124. I proceeded through the document making similar alterations to the various relevant entries in the document. Upon completion of the task, I saved the changes and opened the document in Adobe Reader. I opened the properties tab as per the screenshot below:



125. It can be seen that the resulting document (modified in 2023) appears to be created in the future (2025) although I could have inputted any date and time of my choosing in this way.

126. Opening the manipulated file within PDF Stream Dumper, it can be observed that the XMP Core modification has also been entered into the metadata stream effectively.



{ID_000536} *ID_000536 - conclusion*

127. It is my opinion that this or a similar method has been used to manually alter the timestamps and XMP Core property information of ID_000536. Based on my findings set out above, in my opinion the timestamps within the document is not authentic to its purported date of Creation or Last Modification.

{ID_000536}

{ID_000536} {ID_000865} 128. I also observe that that ID_000536 retains within it both the Creation timestamp of the ID_000865 Control Copy (24 March 2009 at 17:33:11 GMT) as well as content found in ID_000865 but not the earlier Control Copy. It also contains evidence of the amendments that {ID_003732} had been made to ID_003732 to change the displayed author name. This confirms my view that it is a downstream edit of ID_003732, which itself I have found derives from the {ID_000865} ID_000865 control copy.

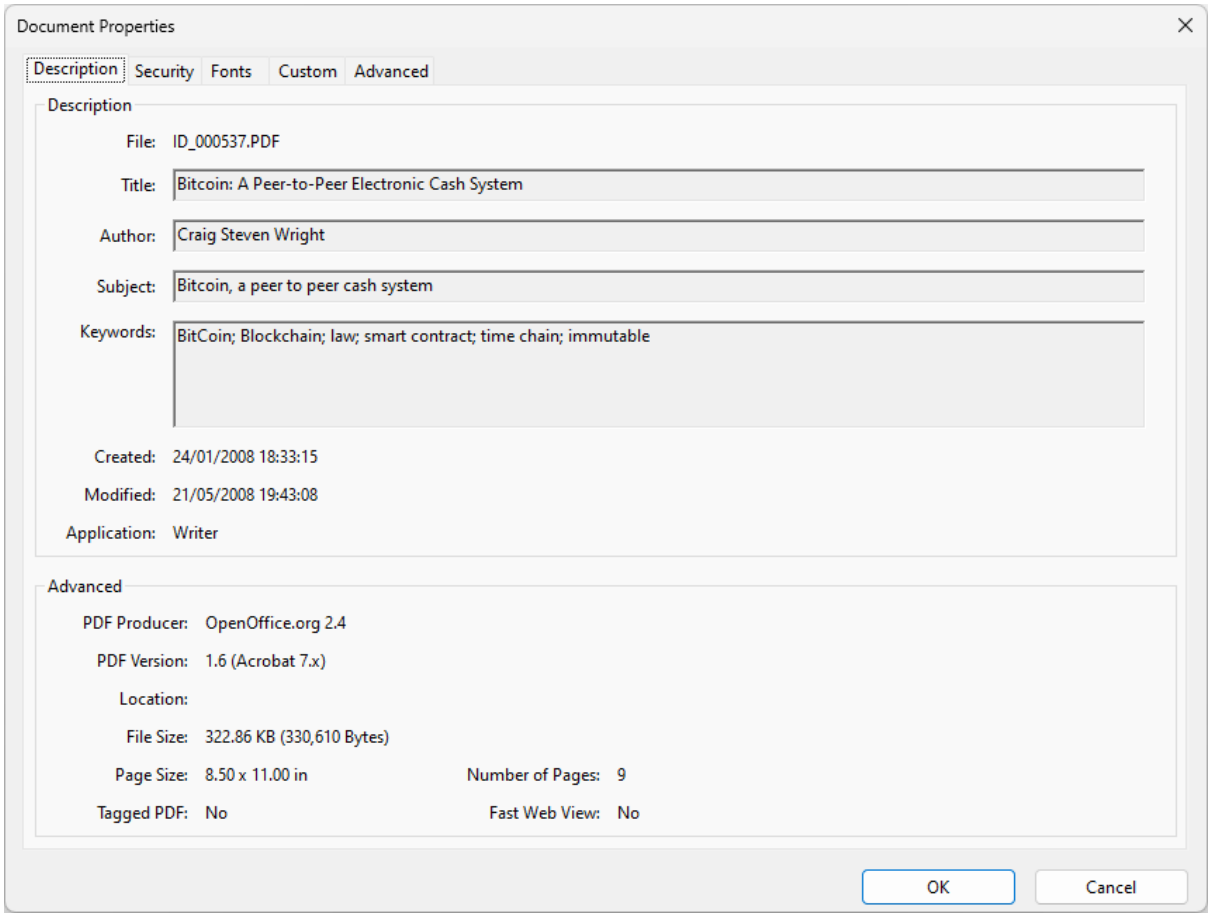
{ID_000537} **ID_000537.PDF "SSRN-id3440802.pdf"**

{ID_000537} 129. The original filename attributed to disclosed document ID_000537.PDF is "SSRN-id3440802.pdf". The MD5 Hash value for it is "5e210ce3003ffaed204b7b2076c2fc92". The external (OS File) metadata provided with the load file indicates that it was created on 24 March 2009 and Last Modified on 21 May 2008.

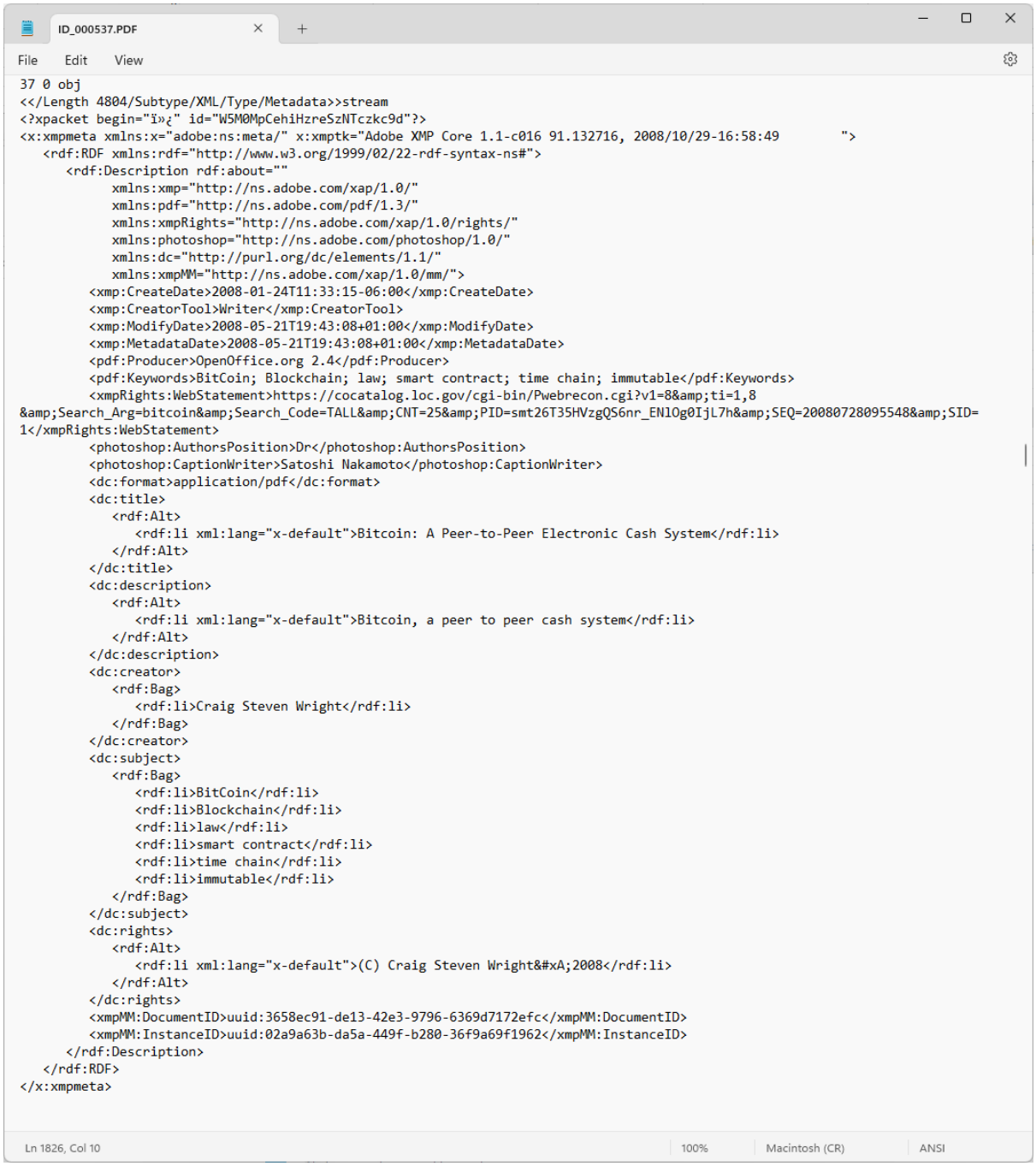
130. Aside from a change to the title, and the addition of a footer banner, the face-value content of {ID_000537} {ID_000536} ID_000537 appears to match that of ID_000536 as compared by eye.

{ID_000537} *ID_000537 – metadata overview*

{ID_000537} {ID_000536} 131. The Adobe Acrobat Reader properties for ID_000537 mirror those of ID_000536 as shown in the screenshot below:



{ID_000537}132. On a plaintext analysis, I found one metadata property data stream within ID_000537, as per the Notepad plain-text view in the screenshot below. From this, it can be seen that ID_000537 {ID_000536} does not include the August 2008 dates or the 2019 dates as found in ID_000536, or the 24 March 2009 timestamp as found within ID_000536, ID_003732 and the Control Copy {ID_003732} {ID_000865} ID_000865:



{D/91} 133. As with other documents within the dataset (though not the Control Copies or BWP-NB1), expanding the PDF file using qPDF makes more streams of metadata available. As with ID_000536, ID_000537 also includes a redundant metadata field "/CreationDate (D:20090324113315-06'00')", which is consistent with the ID_000865 Control Copy.

{ID_000536}
{ID_000537}
{ID_000865}

{ID_000537} 134. The timestamps in ID_000537, as can be observed above, are encoded directly in the file in the UTC+1 timezone, which is different to the Control Copies of the BWP, but which I do not think is not anomalous on its own.

135. I note that included within this stream, the metadata field for "photoshop:CaptionWriter" is recorded as "Satoshi Nakamoto" although I make no conclusion directly therefrom.

{ID_000537} *ID_000537 - Invalid XMP Core*

136. Notably the metadata stream makes reference to "Adobe XMP Core 1.1-c016 91.132716, 2008/10/29-16:58:49" which I consider is an invalid and manipulated XMP Core version stamp for the reasons set out above.

{ID_000537} *ID_000537 – "Touchup_textedit"*

{ID_003732} {ID_000536} 137. It has also carried the "Touchup_textedit" history from ID_003732 that was no longer present on the face of ID_000536.

{ID_000537} *ID_000537 – textual changes – missing title hyphen*

{ID_000536} 138. On the face of the document, ID_000537.PDF appears almost identical to ID_000536.PDF apart from two features.

{ID_000537} 139. Firstly, in the first line of ID_000537.PDF, in the title, there is a missing hyphen "-" comparing "**Bitcoin: A Peer-to-Peer Electronic Cash System**" from

{ID_000536} ID_000536.PDF to "**Bitcoin: A Peer-toPeer Electronic Cash System**" from

{ID_000537} ID_000537.PDF (colouring added for ease of reference). Although this text has been copied and pasted to demonstrate how it renders in a text editor, when viewed in the documents themselves the alignment is different, exhibiting a wider space between "toPeer" and

{ID_000537} {ID_000536} "Electronic" in ID_000537. Below, I compare the titles of ID_000536 and ID_000537 side by side:

Bitcoin: A Peer-to-Peer Electronic Cash System
Bitcoin: A Peer-toPeer Electronic Cash System

{ID_000537} *ID_000537 – textual changes – SSRN ID*

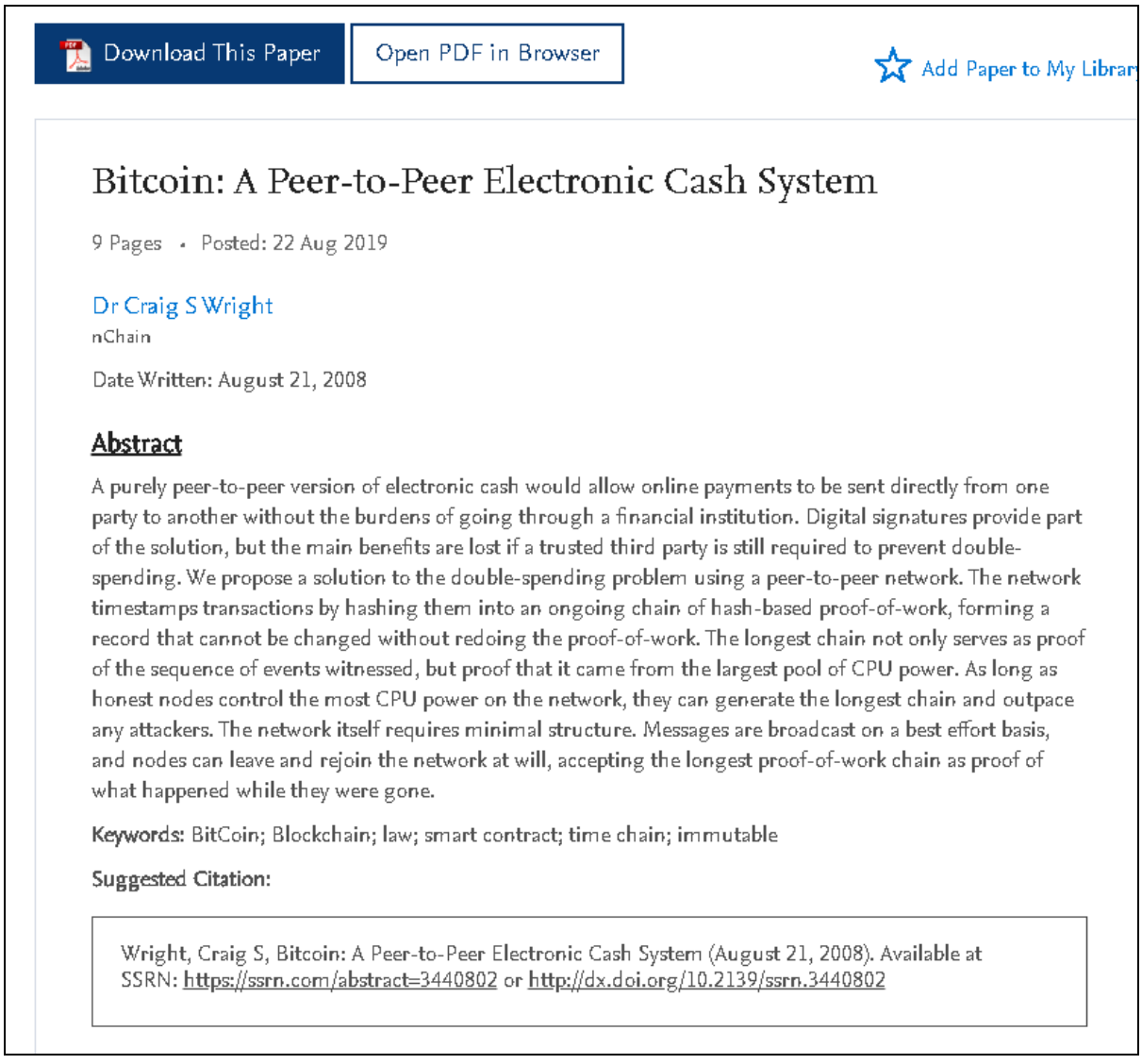
{ID_000537} 140. The second visual difference I observed is that ID_000537.PDF includes the footer text "*Electronic copy available at: https://ssrn.com/abstract=3440802"* on each page. Investigating this:

a. The website SSRN.com describes itself as follows: *"SSRN is devoted to the rapid worldwide dissemination of research and is composed of a number of specialized research networks."*, and I take it to be a repository of academic research papers.

b. The filename attributed to this document "SSRN-id3440802.pdf" is consistent with the naming convention for other files that are available on that website,

c. In the course of preparing this Appendix, I was able to visit the above link to the SSRN website and downloaded a copy of the PDF file hosted there.

d. The website indicates that the document was posted on 22 August 2019 and lists "Date Written" as "August 21, 2008". My understanding is that the "Date Written" field is a user-editable field specified by the person submitting the document. The screenshot below shows a partial screenshot of this section of the web page:



{ID_000537} *ID_000537 – SSRN download not present in disclosure dataset*

141. While the file available to download shared the same filename "SSRN-id3440802.pdf", the MD5 hash value for the file is different, being "ef765a25bfc98358acfe0d3b82d9b9a0". However, no document by that MD5 hash is included in the disclosure dataset (and none is recorded in Exhibit PM3.1).

{H/21}

{ID_000537} 142. The document I downloaded (on 16 May 2023) also does not visually match ID_000537, in that it did does not exhibit the missing hyphen in "toPeer" as described above: the title better matches that of ID_000536 rather than ID_000537. Aside from this, the downloaded SSRN document visually matched ID_000537.

{ID_000536}
{ID_000537}

{ID_000537} *ID_000537 – redundant embedded metadata streams*

143. A metadata property data stream for the SSRN downloaded document also includes a redundant "/CreationDate (D:20090324113315-06'00')" entry. This is consistent with the Creation Date of the ID_000865 Control Copy, but not with the data displayed in the Adobe Properties tab for ID_000537 (which states a creation date in January 2008).

{ID_000865}
{ID_000537}

144. I observe that even though the Adobe properties for the SSRN document list it as being modified on 21 May 2008, the SSRN listing for the document states that it was written on 22 August 2008.

{ID_000537} *ID_000537 – conclusions*

145. Considering these points, it is my opinion that the stated Creation and Last Modification dates for ID_000537 (in 2008) cannot be relied on as authentic. In my opinion, it is most likely that ID_000537 is a downstream edit from ID_000536. It includes many of same redundant metadata characteristics (which can be inherited from one document to the next). I consider it most likely that ID_000536 came before ID_000537 as ID_000536 has retained more redundant content than ID_000537. Considering the overlaps and differences, I consider ID_000537 to be part of the consecutive chain of editing beginning with ID_003732, and therefore to post-date May 2019.

{ID_000537}
{ID_000536}

{ID_000536}
{ID_000537}

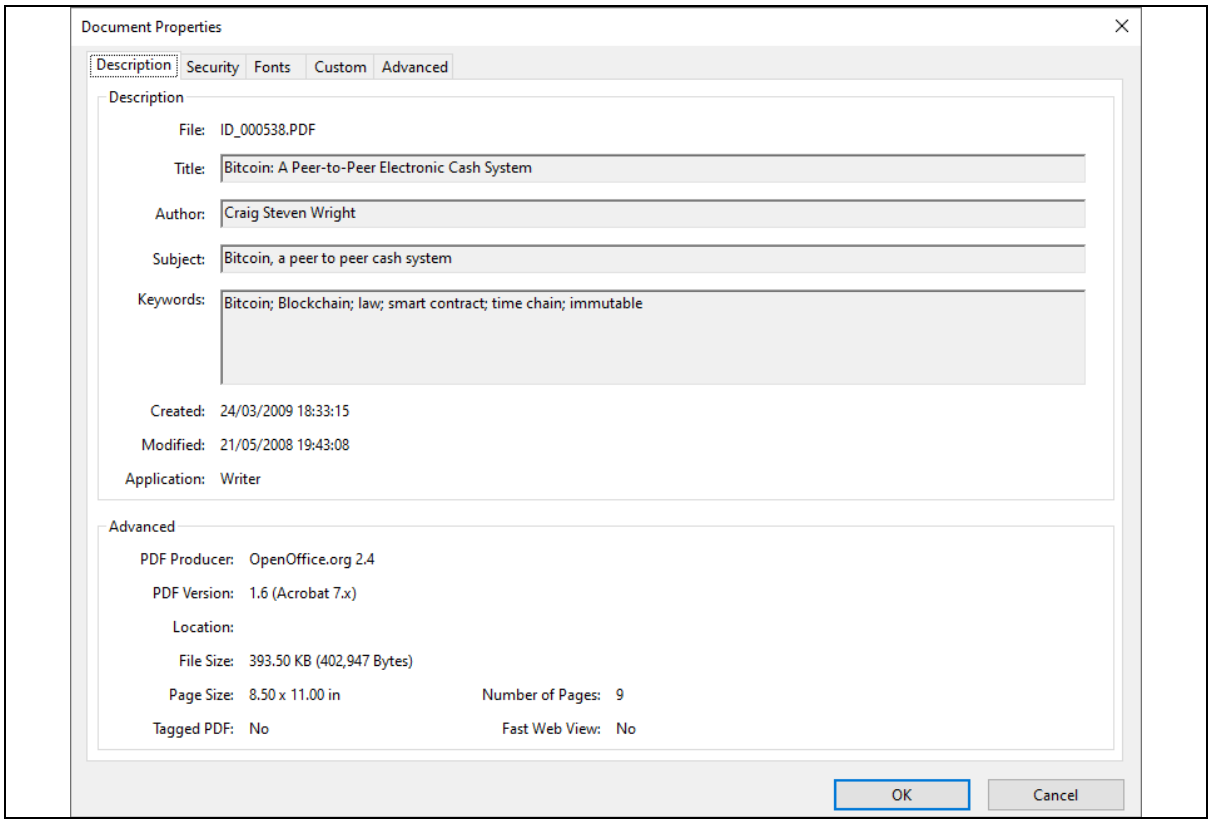
{ID_003732}

{ID_000538} **ID_000538 (bitcoin.pdf)**

{ID_000536} 146. The document ID_000538 is a PDF document that by visual comparison matches ID_000536. There are however differences between the two documents in other areas of the electronic files.

147. Both documents are exactly the same capacity at 402,947 bytes.

{ID_000538} {ID_000536} 148. The Adobe Reader Properties tab, in the screenshot below, indicates a later Created date in ID_000538 (March 2009) than that listed for ID_000536 (January 2008).



{ID_000538} *ID_00538 – contradictory metadata*

{ID_000538} {ID_000865} 149. As can be seen, ID_000538 lists a recorded Created date of 24/03/2009 (24 March 2009) at 18:33:15 (expressed in BST). This is identical to the Created date of the ID_000865 Control Copy of the BWP. However, it is in apparent in conflict with the recorded Modified date which is 21/05/2008 at 19:43:08 (21 May 2008). Logically speaking, the PDF metadata modified timestamp should not precede the metadata created timestamp.[8]

150. In my view, the conflict between the timestamps is characteristic of the use of clock manipulation techniques, whereby the file in question has been created after 24 March 2009, but at a time when the computer clock had been set to a back-dated date in May 2008. This conflict could also be the result of the manual manipulation of the content of the file using a

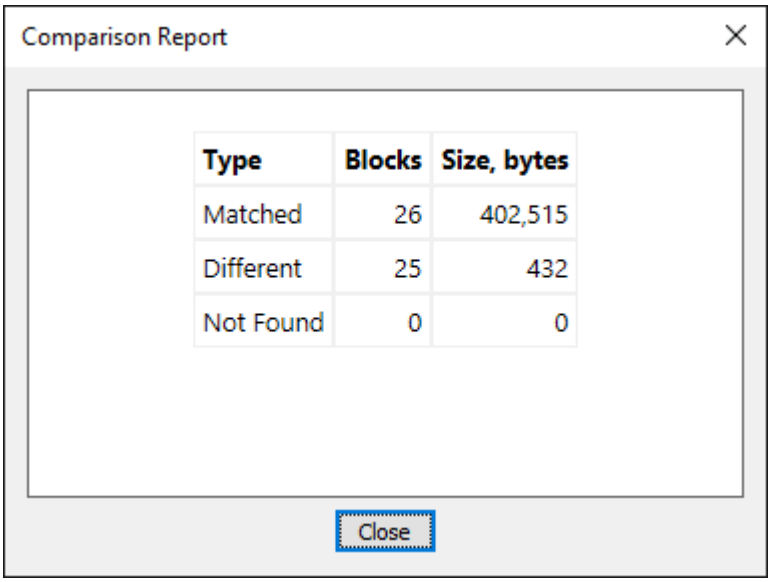
[8] While this can be explained in the context of file property timestamps, it is illogical in regard to internal metadata timestamps

hex editor as demonstrated above. It indicates to me that the timestamps are not reliable sources of information.

{ID_000538} *ID_00538 – hex-editing comparison*

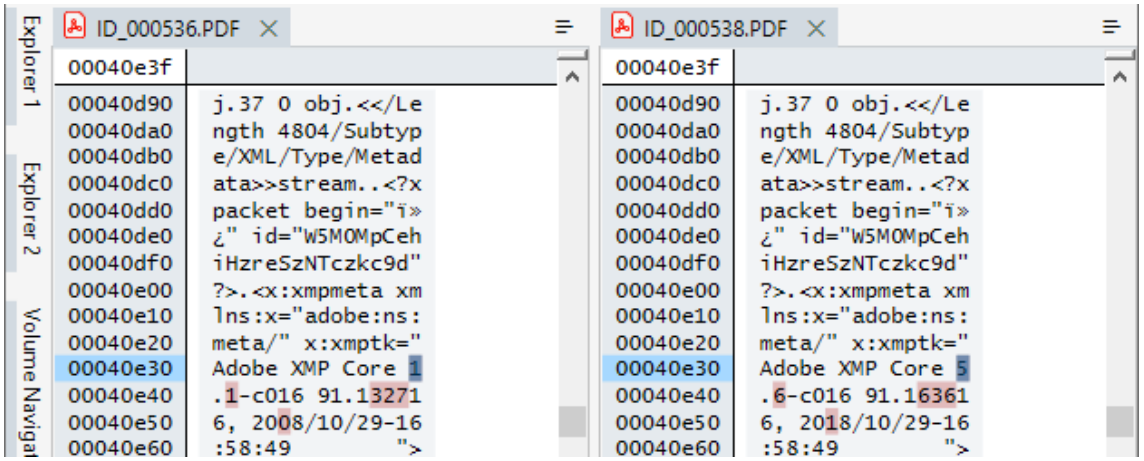
{ID_000538} {ID_003732} {ID_000536} {ID_000538} 151. I noted that the keywords field of ID_000538 lists Bitcoin as "Bitcoin" as it is seen in ID_003732, which is different to that observed in ID_000536 which lists "BitCoin" with a capital "C". Considering that ID_000536 and ID_000538 are the exact same file size, and an initial review between the two files in the Encase forensic software indicated an obvious electronic matchup between the two files, I next conducted a side by side file comparison between those two files using Hex Editor Neo.

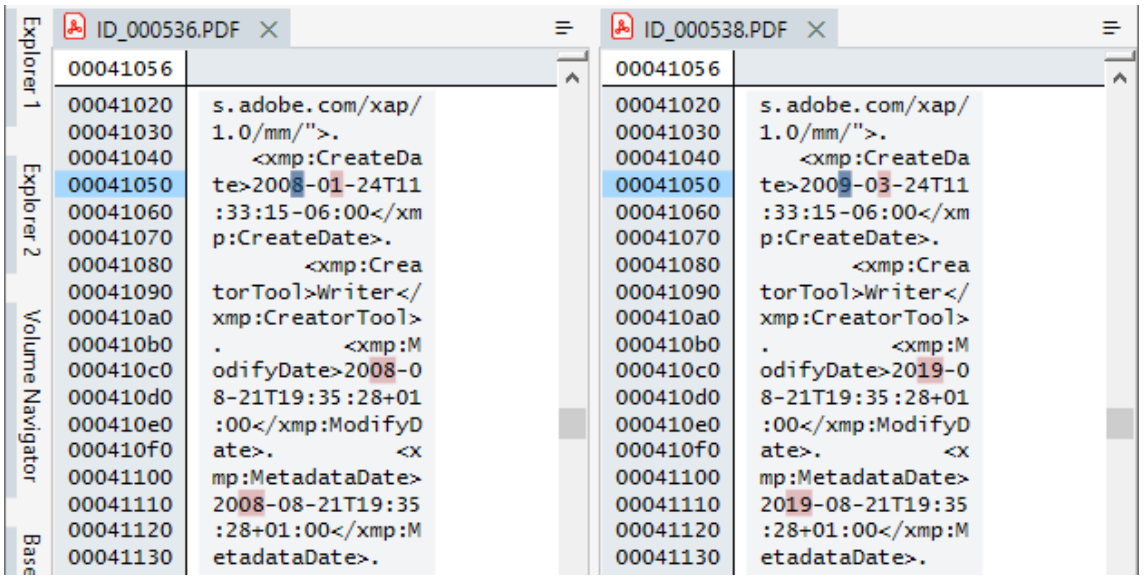
152. The comparison of the files reported that there were 25 differences between the two documents as per the screenshot below:

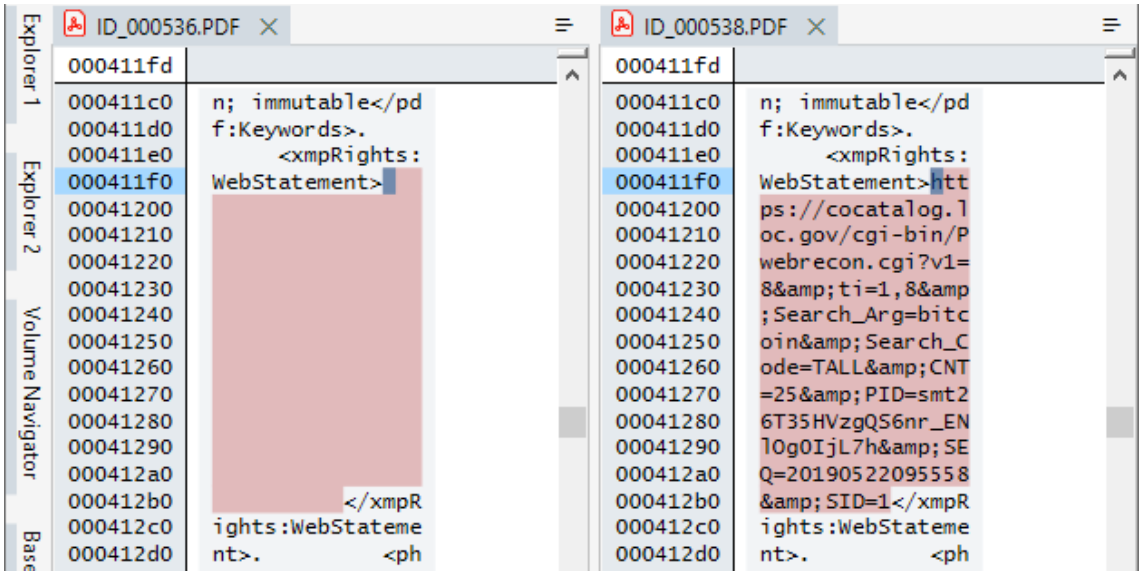


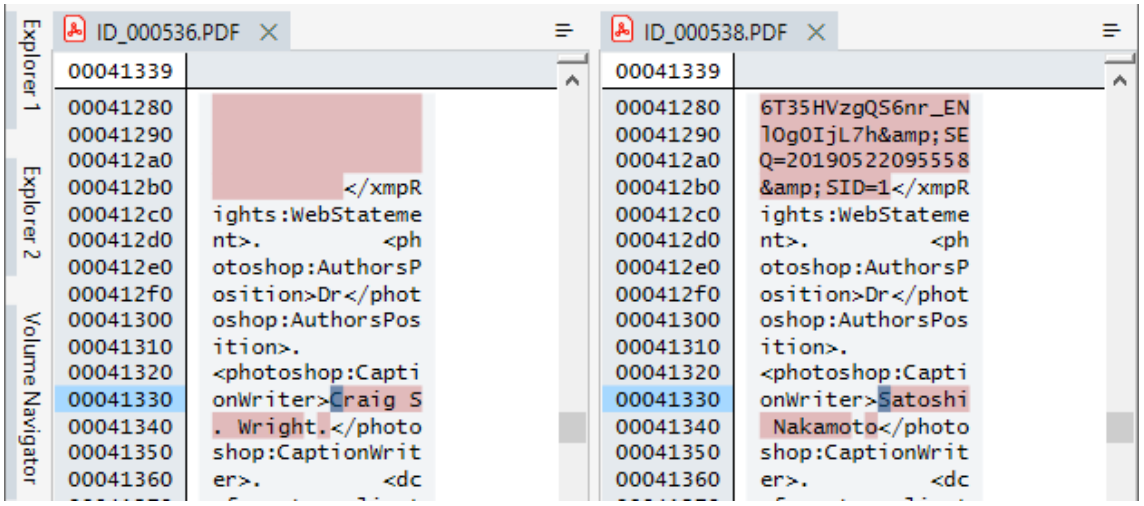
153. This indicates that there are 432 bytes (characters) of difference between the two documents, and that these were found in 25 locations within the documents.

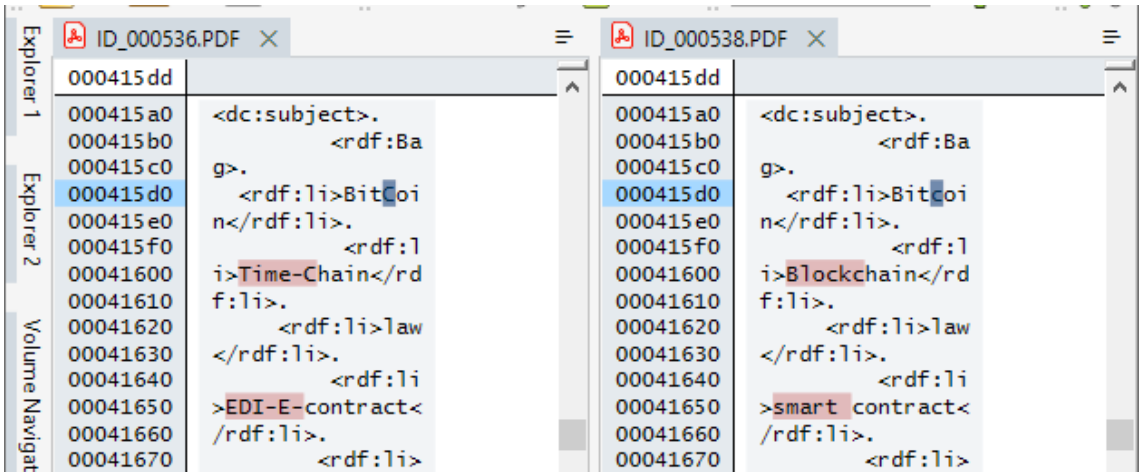
154. Below are a series of screenshots of Hex Editor Neo that depict these differences. Some of the screenshots account for multiple differences at a time.

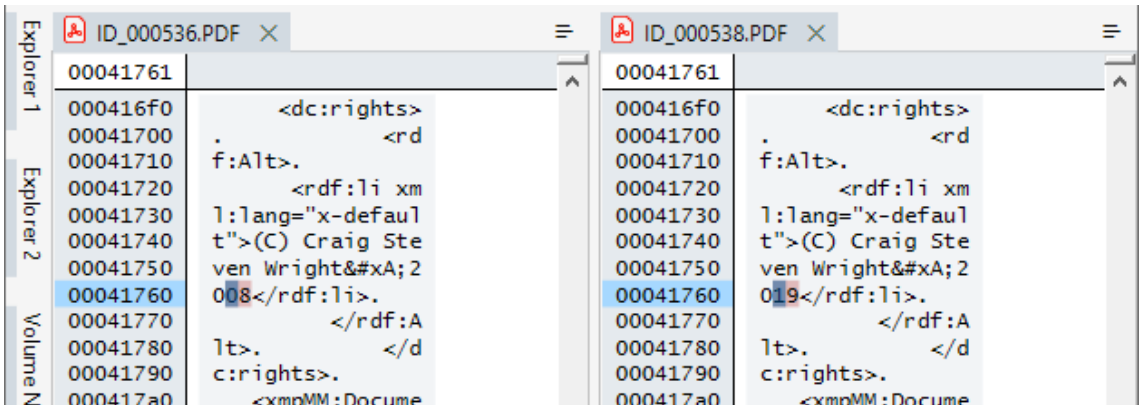

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00040de0 ¿" id="W5M0MpCeh
00040df0 iHzreSzNTczkc9d"
00040e00 ?>.<x:xmpmeta xm
00040e10 lns:x="adobe:ns:
00040e20 meta/" x:xmptk="
00040e30 Adobe XMP Core 1
00040e40 .1-c016 91.13271
00040e50 6, 2008/10/29-16
00040e60 :58:49 ">
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00040db0 e/XML/Type/Metad
00040dc0 ata>>stream..<?x
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00040de0 ¿" id="W5M0MpCeh
00040df0 iHzreSzNTczkc9d"
00040e00 ?>.<x:xmpmeta xm
00040e10 lns:x="adobe:ns:
00040e20 meta/" x:xmptk="
00040e30 Adobe XMP Core 5
00040e40 .6-c016 91.16361
00040e50 6, 2018/10/29-16
00040e60 :58:49 ">
Explorer 1
Explorer 2
Volume Navigat

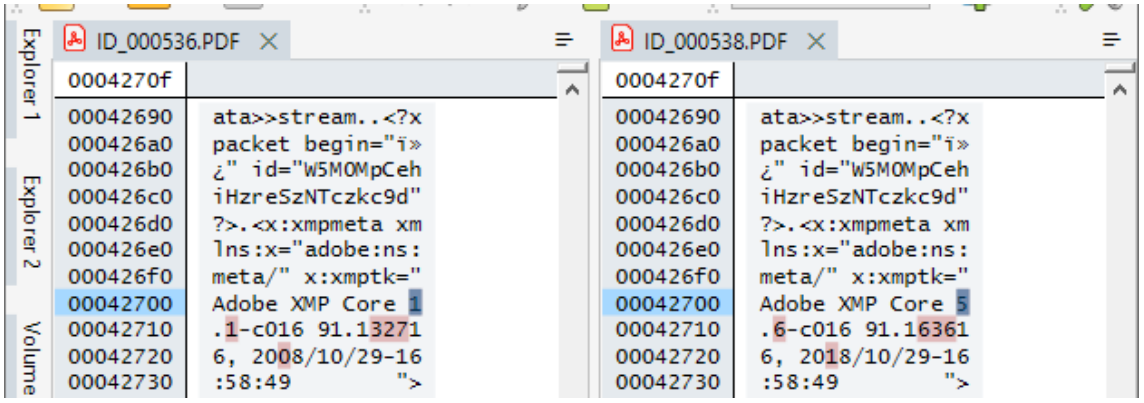

ID_000536.PDF
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00041030 1.0/mm/">.
00041040 <xmp:CreateDa
00041050 te>2008-01-24T11
00041060 :33:15-06:00</xm
00041070 p:CreateDate>.
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000410a0 xmp:CreatorTool>
000410b0 . <xmp:M
000410c0 odifyDate>2008-0
000410d0 8-21T19:35:28+01
000410e0 :00</xmp:ModifyD
000410f0 ate>. <x
00041100 mp:MetadataDate>
00041110 2008-08-21T19:35
00041120 :28+01:00</xmp:M
00041130 etadataDate>.
ID_000538.PDF
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00041020 s.adobe.com/xap/
00041030 1.0/mm/">.
00041040 <xmp:CreateDa
00041050 te>2009-03-24T11
00041060 :33:15-06:00</xm
00041070 p:CreateDate>.
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000410a0 xmp:CreatorTool>
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00041100 mp:MetadataDate>
00041110 2019-08-21T19:35
00041120 :28+01:00</xmp:M
00041130 etadataDate>.
Explorer 1
Explorer 2
Volume Navigator
Base

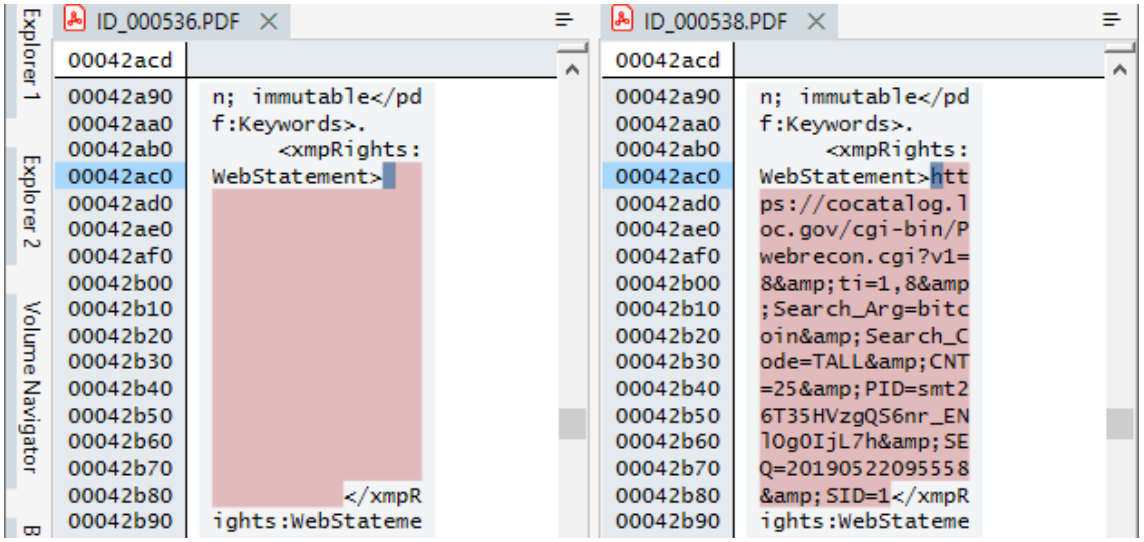

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000411fd
000411c0 n; immutable</pd
000411d0 f:Keywords>.
000411e0 <xmpRights:
000411f0 WebStatement>
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000411c0 n; immutable</pd
000411d0 f:Keywords>.
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000411f0 WebStatement>htt
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00041220 webrecon.cgi?v1=
00041230 8&ti=1,8&amp
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00041250 oin&Search_C
00041260 ode=TALL&CNT
00041270 =25&PID=smt2
00041280 6T35HVzgQS6nr_EN
00041290 lOgOIjL7h&SE
000412a0 Q=20190522095558
000412b0 &SID=1</xmpR
000412c0 ights:WebStateme
000412d0 nt>. <ph
Explorer 1
Explorer 2
Volume Navigator
Base

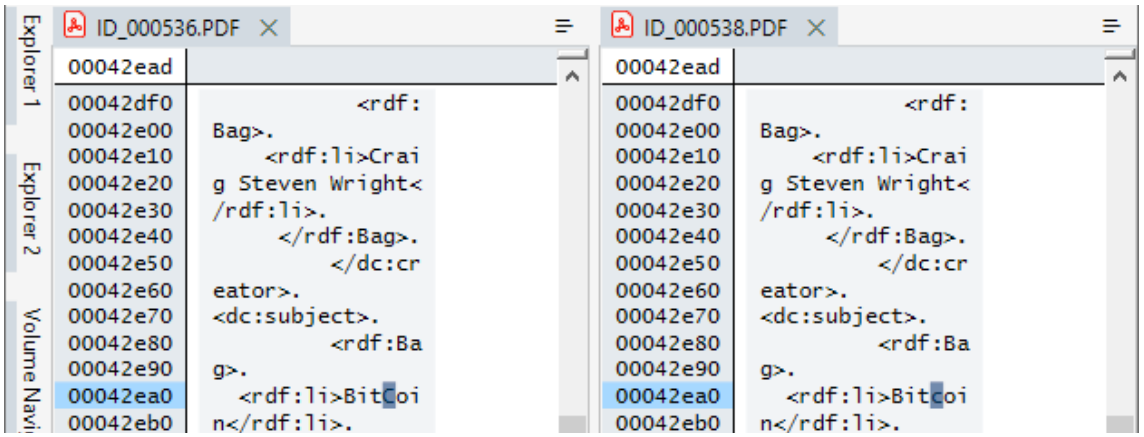

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000412a0 Q=20190522095558
000412b0 &SID=1</xmpR
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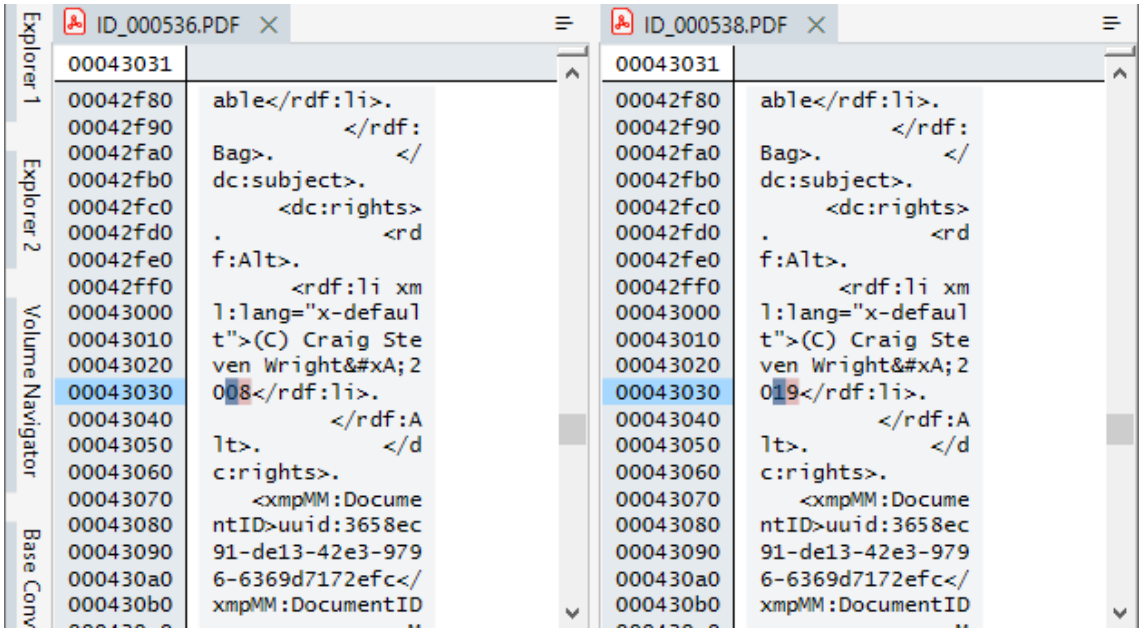

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00041610 f:li>.
00041620 <rdf:li>law
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00041650 >EDI-E-contract<
00041660 /rdf:li>.
00041670 <rdf:li>
ID_000538.PDF
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00041610 f:li>.
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000426e0 lns:x="adobe:ns:
000426f0 meta/" x:xmptk="
00042700 Adobe XMP Core 1
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00042720 6, 2008/10/29-16
00042730 :58:49 ">
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000426b0 ¿" id="W5M0MpCeh
000426c0 iHzreSzNTczkc9d"
000426d0 ?>.<x:xmpmeta xm
000426e0 lns:x="adobe:ns:
000426f0 meta/" x:xmptk="
00042700 Adobe XMP Core 5
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00042730 :58:49 ">
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000428b0 p://purl.org/dc/
000428c0 elements/1.1/".
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00042900 1.0/mm/">.
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00042a50 pdf:Keywords>Bit
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000429d0 mp:MetadataDate>
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00042a50 pdf:Keywords>Bit
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ID_000538.PDF
00042acd
00042a90 n; immutable</pd
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00042b50 6T35HVzgQS6nr_EN
00042b60 lOgOIjL7h&SE
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00042ff0 <rdf:li xm
00043000 l:lang="x-defaul
00043010 t">(C) Craig Ste
00043020 ven Wright
2
00043030 008</rdf:li>.
00043040 </rdf:A
00043050 lt>. </d
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00043070 <xmpMM:Docume
00043080 ntID>uuid:3658ec
00043090 91-de13-42e3-979
000430a0 6-6369d7172efc</
000430b0 xmpMM:DocumentID
ID_000538.PDF
00043031
00042f80 able</rdf:li>.
00042f90 </rdf:
00042fa0 Bag>. </
00042fb0 dc:subject>.
00042fc0 <dc:rights>
00042fd0 . <rd
00042fe0 f:Alt>.
00042ff0 <rdf:li xm
00043000 l:lang="x-defaul
00043010 t">(C) Craig Ste
00043020 ven Wright
2
00043030 019</rdf:li>.
00043040 </rdf:A
00043050 lt>. </d
00043060 c:rights>.
00043070 <xmpMM:Docume
00043080 ntID>uuid:3658ec
00043090 91-de13-42e3-979
000430a0 6-6369d7172efc</
000430b0 xmpMM:DocumentID

155. These changes are in my view consistent with hexadecimal or plain text manipulation of one of the documents, having regard to the fact that each apparent edit occupies the precise byte length of the text it replaces.

{ID_000538} *ID_000538 – actual chronological order*

{ID_000538} {ID_000536} 156. Considering the nature of these changes, it is apparent ID_000538 was generated before ID_000536. I reach this conclusion with particular regard to the comparison between the changes at location 00042ac0 onwards within the file:

{ID_000538}

a. In ID_000538, that location includes a "Rights Web Statement" field containing a link to a "CoCatalog.loc.gov" URL occupying approximately 12 lines as displayed in Hex Edit Neo.

{ID_000536}

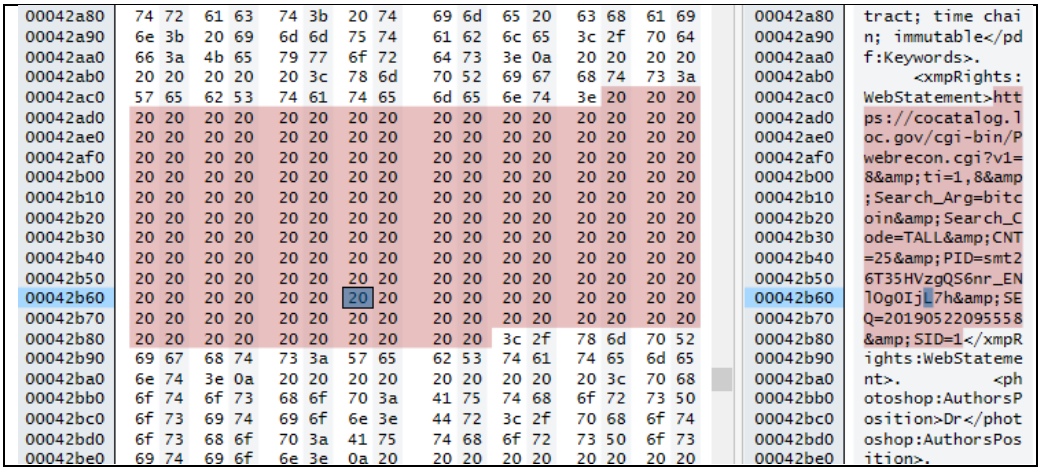
b. By contrast, although ID_000536 contains a "Rights Web Statement" field, the field is blank of human-readable content and merely appears as whitespace.

{ID_000536} {ID_000538}

c. The whitespace in ID_000536 occupies precisely the same amount of space as the CoCatalog.loc.gov URL in ID_000538. It is not plausible that a PDF would be encoded before a URL was generated, not knowing what its content might be; and yet leaving precisely the right amount of space for a URL to later be inserted by hex-editing. Doing so would, in my opinion, also be without purpose, since if a URL was intended to be added later that could be achieved simply by re-generating the PDF at that point in time without the need to engage in an unusual process of hex-editing.

d. I also observe that the whitespace in question has been overwritten not with "Null" characters (as is normally to be expected for vacant areas of files, but rather with text space characters which are encoded in hexidecimal as character number "20". This can be observed in the following screenshot which compares (on the left) the hex values associated with those spaces in ID_000536, and (on the right) the text values of ID_000538.

{ID_000536} {ID_000538}



00042a80	74	72	61	63	74	3b	20	74	69	6d	65	20	63	68	61	69	tract; time chai
00042a90	6e	3b	20	69	6d	6d	75	74	61	62	6c	65	3c	2f	70	64	n; immutable</pd
00042aa0	66	3a	4b	65	79	77	6f	72	64	73	3e	0a	20	20	20	20	f:Keywords>.
00042ab0	20	20	20	20	20	3c	78	6d	70	52	69	67	68	74	73	3a	<xmpRights:
00042ac0	57	65	62	53	74	61	74	65	6d	65	6e	74	3e	20	20	20	WebStatement>htt
00042ad0	20	20	20	20	20	20	20	20	20	20	20	20	20	20	20	20	ps://cocatalog.l
00042ae0	20	20	20	20	20	20	20	20	20	20	20	20	20	20	20	20	oc.gov/cgi-bin/P
00042af0	20	20	20	20	20	20	20	20	20	20	20	20	20	20	20	20	webrecon.cgi?v1=
00042b00	20	20	20	20	20	20	20	20	20	20	20	20	20	20	20	20	8&ti=1,8&amp
00042b10	20	20	20	20	20	20	20	20	20	20	20	20	20	20	20	20	;Search_Arg=bitc
00042b20	20	20	20	20	20	20	20	20	20	20	20	20	20	20	20	20	oin&Search_C
00042b30	20	20	20	20	20	20	20	20	20	20	20	20	20	20	20	20	ode=TALL&CNT
00042b40	20	20	20	20	20	20	20	20	20	20	20	20	20	20	20	20	=25&PID=smt2
00042b50	20	20	20	20	20	20	20	20	20	20	20	20	20	20	20	20	6T35HVzgQS6nr_EN
00042b60	20	20	20	20	20	20	20	20	20	20	20	20	20	20	20	20	lOg0IjL7h&SE
00042b70	20	20	20	20	20	20	20	20	20	20	20	20	20	20	20	20	Q=20190522095558
00042b80	20	20	20	20	20	20	20	20	20	20	3c	2f	78	6d	70	52	&SID=1</xmpR
00042b90	69	67	68	74	73	3a	57	65	62	53	74	61	74	65	6d	65	ights:WebStateme
00042ba0	6e	74	3e	0a	20	20	20	20	20	20	20	20	20	3c	70	68	nt>. <ph
00042bb0	6f	74	6f	73	68	6f	70	3a	41	75	74	68	6f	72	73	50	otoshop:AuthorsP
00042bc0	6f	73	69	74	69	6f	6e	3e	44	72	3c	2f	70	68	6f	74	osition>Dr</phot
00042bd0	6f	73	68	6f	70	3a	41	75	74	68	6f	72	73	50	6f	73	oshop:AuthorsPos
00042be0	69	74	69	6f	6e	3e	0a	20	20	20	20	20	20	20	20	20	ition>.

157. In relation to the link above, I understand that the website is the American Copyright Office Public catalogue and the link relates to a search of the database for the term "bitcoin" seen in the link as "*Search_Arg=bitcoin*".

158. Included in the link there is also a timestamp "SEQ=20190522095558". Under testing of the website conducted in August 2023 I observed a similarly structured timestamp. This was applied with a GMT-04:00 offset (which was during Daylight Saving Time and therefore corresponded to East-Coast US time). Applying this to the timestamp encoded in the link, results in a date and time of 22 May 2019 at 13:55:58 UTC. This suggests that the link was created in May 2019.

{ID_000538} 159. I note that the timestamp embedded in that search URL in relation to ID_000538 is less than 1 minute before the "Last Modified" timestamp which is more plainly visible within the {ID_003732} properties of ID_003732 (13:56:48 UTC on 22 May 2019). This suggests a close association in time for the edits being made to these two documents, consistent with the other observations I have made about the apparent chain of edits to various documents discussed above.

{ID_000538} *ID_000538 – XMP core changes*

160. The individual changes to the XMP Core can be tracked between the two records. As before, this is indicative of changes being made to the actual XMP Core version from 29 October 2018 as if to refer to a different version which appears to date from the identical day, hour, minute and second in 2008, but which did not in fact exist:

{ID_000538}	ID_000538	"Adobe XMP Core 5.6-c016 91.163616, 2018/10/29-16:58:49 "
{ID_000538}	ID_000536	"Adobe XMP Core 1.1-c016 91.132716, 2008/10/29-16:58:49 "

{ID_000538} *ID_000538 – date adjustments*

161. It can be observed how the dates have been adjusted from 2019 to 2008 as in the comparison example below:

{ID_000538}	ID_000538	<xmp:ModifyDate>2019-08-21T19:35:28+01:00</xmp:ModifyDate>
{ID_000536}	ID_000536	<xmp:ModifyDate>2008-08-21T19:35:28+01:00</xmp:ModifyDate>

{ID_000538} *ID_000538 – Name entry edit*

162. The field for "photoshop:CaptionWriter" has been changed as follows:

{ID_000538}	ID_000538	Satoshi Nakamoto
{ID_000536}	ID_000536	Craig S. Wright.

{ID_003732} 163. I note that this metadata field for ID_003732 is recorded as being "Satoshi Nakamoto".

164. It is important to note that when manipulating the hexadecimal content of a PDF file in this manner, it is vital that the bytes offsets remain. It is not possible to increase or reduce the capacity of the files. If even 1 byte is added or removed, the file will be rendered damaged as the internal component addressing will be mismatched.

165. It can be seen that both of these fields are identical in byte capacity. I observe in particular the trailing full stop character "." which is not characteristic of a name entry, and which is more characteristic of choosing a field length to match the replaced characters. (The text in red and green is shown above in a larger font size to make that small character easier to view.) I observe also that the text "Craig S. Wright." is different to the Author name which is stated on the face of the document, "Dr Craig S Wright".

{ID_000538} *ID_000538 – Conclusions*

{ID_000536} {ID_000538} 166. As these are the extent of the differences between ID_000536 and ID_000538, the analysis and findings for ID_000536 in relation to "Touchup_textedit" information within the document apply equally to ID_000538, in that ID_ 000538 includes references to edits seen in ID_003732 that are no longer displayed on the face of ID_000536 or ID_000538.

{ID_003732} {ID_000536} {ID_000538}

167. Considering the various observations, I do not believe that ID_000538 is authentic to its recorded authorship timestamps. The evidence is consistent with it having been created downstream from ID_003732 and having been used in order to create ID_000536 (in particular for the reason set out in in paragraph 163). This would mean that ID_000538 was produced on or after 22 May 2019 and likely in close association with other documents addressed above.

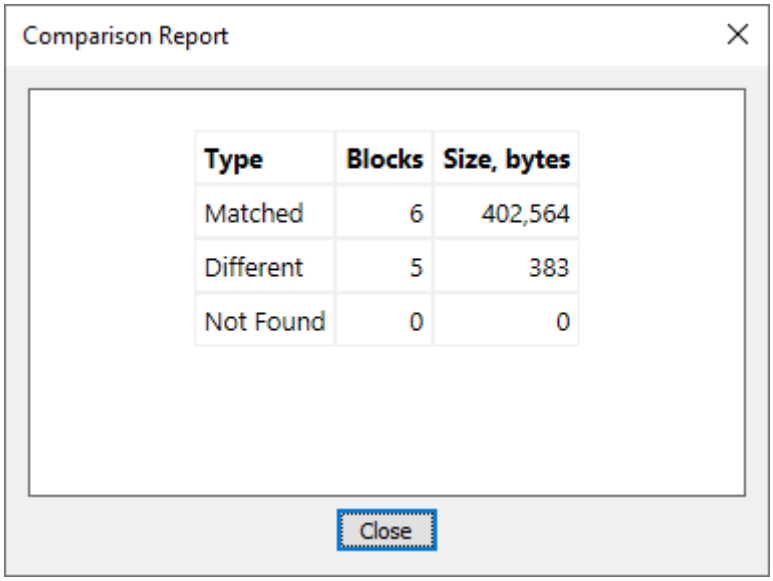
{ID_000540} **ID_000540**

168. In a similar manner to the relationship between ID_000536 and ID_000538, the PDF file ID_000540 is also closely similar to these two documents. I therefore deal with the analysis of this document swiftly: much of my analysis of ID_000536 and ID_000538 also applies to ID_000540.

{ID_000536} {ID_000538} {ID_000540}

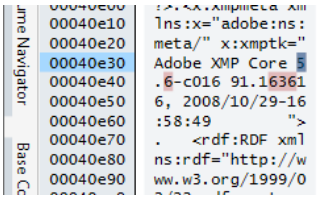
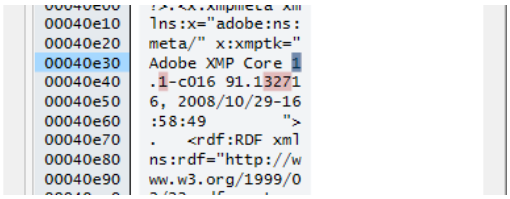
169. I compared ID_000540, first to ID_000536 and then to ID_000538 separately. I found it to be a closer match to ID_000536, with only 5 areas of difference recorded between ID_000540 and ID_000536, as indicated by the Hex Editor Neo comparison screenshot below:

{ID_000540} {ID_000536} {ID_000538}

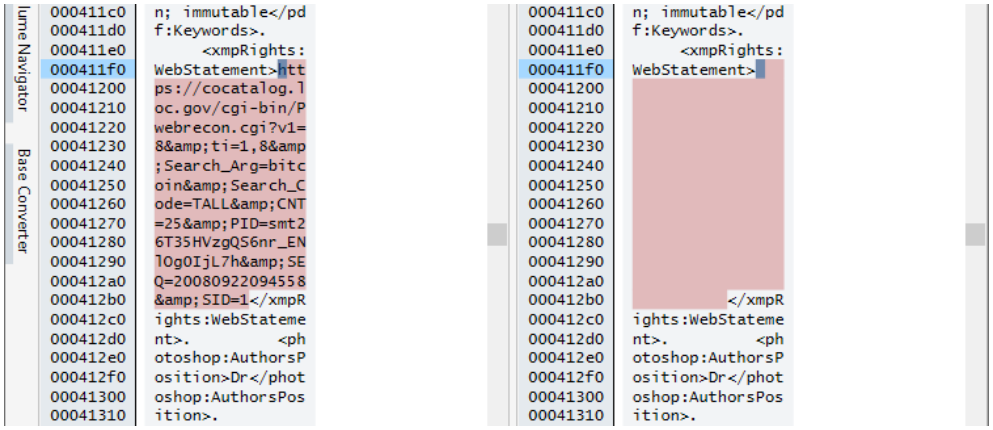
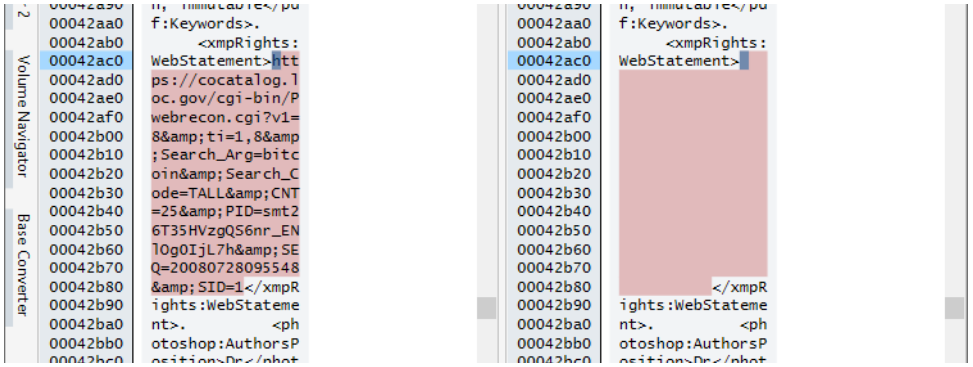


170. I list these differences below:

{ID_005400} {ID_000536}

ID_005400	ID_000536
First three differences: 	

{ID_005400}
{ID_000536}

ID_005400	ID_000536
Fourth difference: 	
Fifth difference: 	

171. The remaining metadata including the timestamp properties otherwise matched those of {ID_000536} ID_000536 including the irregularities discussed above.

{ID_000540} 172. ID_000540 also still included the authorship information from ID_003732 which does not {ID_003732} appear on the face of the document but remains embedded within it as a result of the previous, upstream edit.

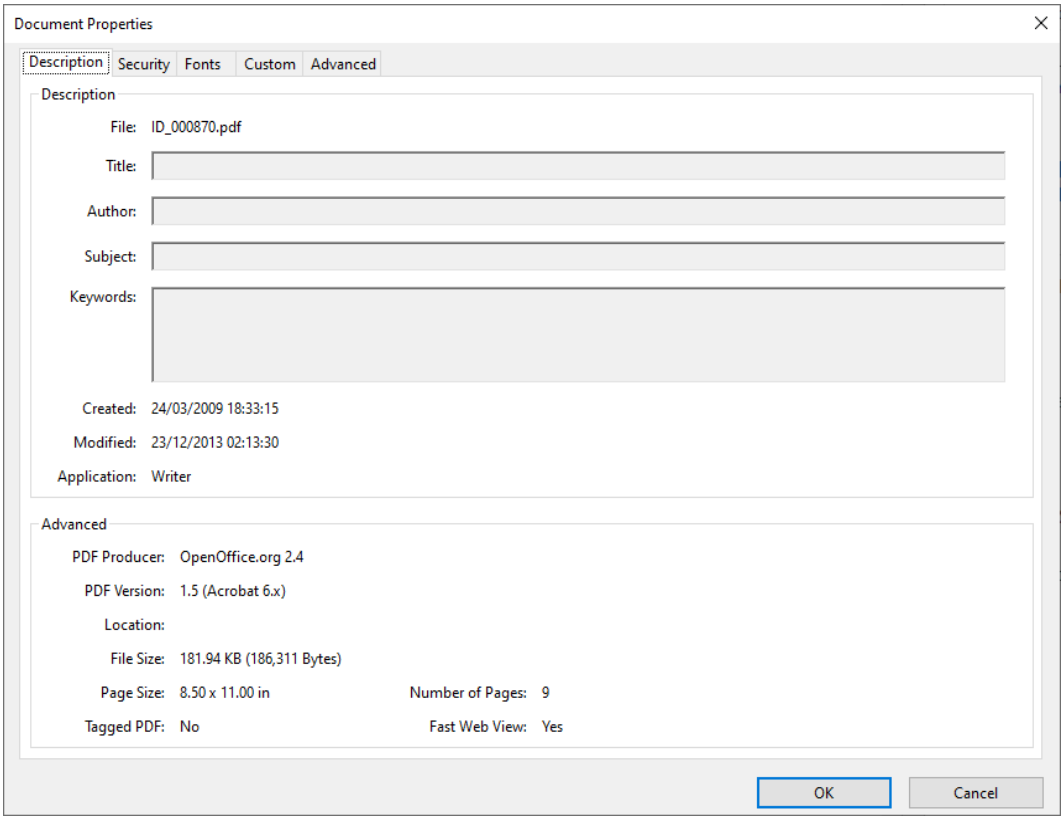
{ID_000540} 173. Considering the various observations, I do not consider that ID_000540 can be authentic to its recorded authorship timestamps.

{ID_000870} **ID_000870**

{ID_000870} 174. ID_000870 is a PDF document that on a visual comparison matches as being content-identical {ID_000865} to the ID_000865 Control Copy including its contact information.

175. While the face of the documents appears to match, the Adobe Acrobat reader properties for {ID_000870} the documents do differ in that ID_000870 has a Modified Timestamp of 23/12/2013 (23 December 2013) at 02:13:30, and it has the PDF Version recorded as “1.5 (Acrobat 6.x)”

{ID_000865} where the ID_000865 Control Copy is recorded as being "1.4 (Acrobat 5.x)". A screenshot of the Adobe Reader Properties tab is shown below listing these properties.



176. {ID_000870} However, although their face-value content is very similar, the files are electronically different, with ID_000870 being structured and encoded very differently. While the file structure of the Control Copies of the BWP is very simple and straightforward, the file {ID_000870} structure of ID_000870 is much more complicated.

177. At the end of both files is a pair of IDs. While the PDF standard does not dictate a rigid approach to the method and manner in which the ID is generated, it does indicate a guideline that requests that it is generated by passing information from the PDF through an algorithm in order to generate a pseudo unique reference identity for the document.

{ID_000870} {ID_000865} 178. I observe that the document IDs in ID_000870 are not encoded in a "trailer" format in the same way as in ID_000865. Rather, they are encoded within a somewhat different formatted plaintext stream as follows, illustrating different surrounding encoding and a lack of structured line breaks unlike those of the two Control Copies (with the two IDs highlighted in green):

```
<</DecodeParms<</Columns 4/Predictor
12>>/Filter/FlateDecode/ID[<CA1B0A44BD542453BEF918FFCD46DC04><804097E
1517B79459264FD99BFD5E291>]/Info 22 0 R/Length 77/Root 24 0 R/Size
23/Type/XRef/W[1 3 0]>>stream
hÞbb &F&Ç÷LŒk□„¸-Œ%¤
c‰¬,‹ÉƒXë€„ ü-HðeÃX\‡□
                                    ˆÅþ$
â.É^bbü§5Äe
0 ã D
endstream
endobj
startxref
116
%%EOF
```

179. The two IDs for each of the two documents are listed below:

{ID_000865}	ID_000865 Control Copy	CA1B0A44BD542453BEF918FFCD46DC04
		CA1B0A44BD542453BEF918FFCD46DC04
{ID_000870}	ID_000870	CA1B0A44BD542453BEF918FFCD46DC04
		804097E1517B79459264FD99BFD5E291

{ID_000865} {D/91}

180. It can be seen for the ID_000865 Control Copy that the two IDs are the same. This is typical of a first generation PDF file that has not been edited or amended and is the same as both Control Copies (and is also as observed in BWP-NB1).

{ID_000870} {ID_000865}

181. For ID_000870, it can be seen that the first ID matches that of the ID_000865 Control Copy. This indicates, in practical terms, that ID_000870 is derived from ID_000865 and is therefore a downstream edit of it.

{ID_000870} {ID_000865}

182. ID_000870 also contains a further XMP metadata data stream not present in Control Copy ID_000865 as per the screenshot below. I have removed line breaks and added highlight to aid review:

```
<?xpacket begin="ï»¿" id="W5M0MpCehiHzreSzNTczkc9d"?>
<x:xmpmeta xmlns:x="adobe:ns:meta/" x:xmptk="Adobe XMP Core 5.2-c001
63.139439, 2010/09/27-13:37:26        ">
   <rdf:RDF xmlns:rdf="http://www.w3.org/1999/02/22-rdf-syntax-ns#">
      <rdf:Description rdf:about=""
            xmlns:xmp="http://ns.adobe.com/xap/1.0/">
         <xmp:CreateDate>2009-03-24T11:33:15-06:00</xmp:CreateDate>
         <xmp:CreatorTool>Writer</xmp:CreatorTool>
         <xmp:ModifyDate>2013-12-23T12:13:30+11:00</xmp:ModifyDate>
         <xmp:MetadataDate>2013-12-23T12:13:30+11:00</xmp:MetadataDate>
      </rdf:Description>
      <rdf:Description rdf:about=""
            xmlns:pdf="http://ns.adobe.com/pdf/1.3/">
         <pdf:Producer>OpenOffice.org 2.4</pdf:Producer>
      </rdf:Description>
      <rdf:Description rdf:about=""
            xmlns:xmpMM="http://ns.adobe.com/xap/1.0/mm/">
```

```
         <xmpMM:DocumentID>uuid:669b7cc8-1347-4165-a691-
89ac97a8bdcc</xmpMM:DocumentID>
         <xmpMM:InstanceID>uuid:9fdd2686-1c4e-4a45-8bf9-
69a66287ceea</xmpMM:InstanceID>
      </rdf:Description>
      <rdf:Description rdf:about=""
            xmlns:dc="http://purl.org/dc/elements/1.1/">
         <dc:format>application/pdf</dc:format>
      </rdf:Description>
   </rdf:RDF>
</x:xmpmeta>
<?xpacket end="w"?>
```

183. From this it can be observed that the Modified timestamp, and the Metadata Date timestamp, have been applied with a +11 hour timezone offset. This is consistent with some time zones in Australia (as well as far eastern Russia, and some islands in Oceania), but is not consistent with any of the time zones in the Control Copies or BWP-NB1 which all have negative UTC offsets (West of GMT). {D/91}

184. In addition, the presence of the "MetadataDate field" is consistent with an edit being made to the properties of the document.

185. I have been unable to determine what change, if any, was made to the document when ID_000870 was produced, as the content is otherwise identical, save for the ID hash as indicated above. As the PDF exhibits very different encoding and file structure, it may simply be that the document was opened and re-saved with different encoding. {ID_000870}

186. In my view, these observations are consistent with ID_000870 having been produced from the ID_000865 Control Copy as a downstream edit of the Control Copy, which may have taken place 2013 (taking the last modified time stamp at face value) or at some other time (if clock manipulation techniques were used). {ID_000870} {ID_000865}

ID_004023 {ID_004023}

187. ID_004023 is a PDF file. It presents at face value as being identical in content to ID_000865, and the observed metadata within the properties dialog of Adobe Acrobat is identical to ID_000865. Were it not for the fact that ID_004023 has a different MD5 hash, I would have considered it to be identical to ID_000865. {ID_004023} {ID_000865} {ID_004023} {ID_000865}

188. However, on a close inspection of the raw (plaintext) data of the file, I observed that it includes the following "Trailer" section at the very end of the file (with colour added for ease of reference).

```
trailer
<</Size 68/Root 66 0 R
/Info 67 0 R
/ID [ <CA1B0A44BD542453BEF918FFCD46DC04>
```

```
<CA1B0A44BD542453BEF918FFCD46DC04> ]
/DocChecksum /6F72EA7514DFAD23FABCC7A550021AF7
>>
startxref
182727
%%EOF
06 2000 1026]/ItalicAngle 0
```

189. This is the only BWP document within the dataset to include a "Trailer" section other than the Control Copies and BWP-NB1 (which I consider to be authentic): no other BWP documents contain this field in this manner. {D/91}

190. In this case however, the presence of a Trailer section is not an indication of authenticity, but in my opinion an indication that the document has been altered. I come to this opinion having regard to the following factors:

a. The final line, highlighted above in red, is not present in the Control Copies.

b. Except for that line, the Trailer section is identical to the ID_000865 Trailer section in every detail. {ID_000865}

c. The line is appended after the %%EOF line. The "EOF" indicates the "End of File", and the addition of content after that is therefore irregular.

d. The line appears to be a truncated copy of a different line in the same file (which is also present in ID_000865, which encodes text formatting, as follows (highlighted in green): {ID_000865}

```
61 0 obj
<</Type/FontDescriptor/FontName/DAAAAA+TimesNewRomanPS-BoldMT
/Flags 6
/FontBBox[-558 -306 2000 1026]/ItalicAngle 0
/Ascent 891
/Descent 216
/CapHeight 1025
/StemV 80
/FontFile2 59 0 R>>
endobj
```

e. As can be seen above, the text appended to the end of the file appears to be a partial copy of a different part of the same file.

f. However, the document ID and Document Checksum are both identical to those of ID_000865. In the ordinary course of creation of a PDF, where these fields are present they ought to be dynamically created at the point of the file being rendered, based on hashed content of the document. The fact that the Checksum in particular is identical {ID_000865}

{ID_000865} to ID_000865 is indicative of the document not being created by exporting from OpenOffice.org 2.4 as indicated in its metadata, but instead having been created by the alteration of a copy of the Control Copy of the BWP.

g. The schedule of properties that accompanied the Disclosure dataset did not include any File timestamp properties for this document. Furthermore it erroneously recorded a date of 21 May 2019 for the metadata modified field. This does not correlate with Adobe Reader properties listed for the document.

191. I cannot determine how this alteration was carried out. It may be as a result of editing the file in plain text and saving it. In view of the fact that the added content is a fragment of other content appearing elsewhere in the document, I cannot discount that it is an artefact of accidental corruption, such as during a download process. Without further explanation, I consider that ID_004023 cannot safely be utilised as a source of information, and that the evidential handling of the document is contaminated.

{ID_004023}

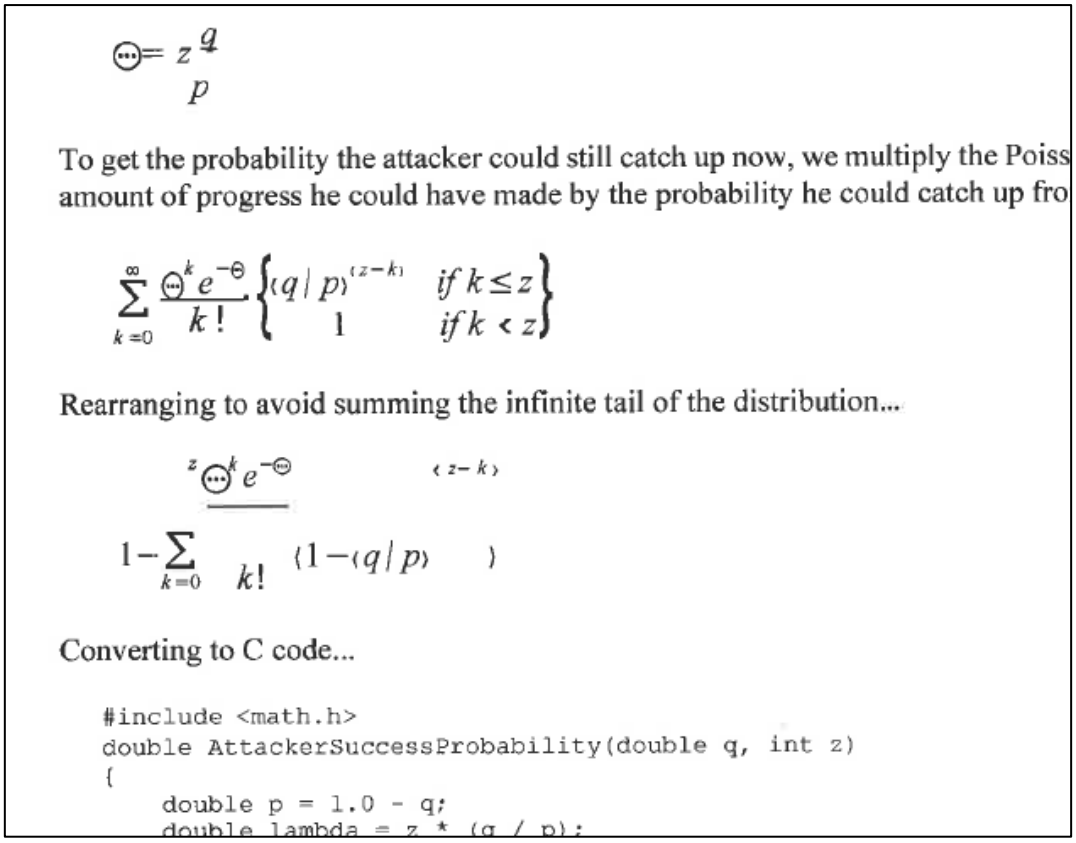
{ID_004010} **ID_004010**

{ID_004010} 192. ID_004010 presents as a PDF scan of a hard copy document and contains artefacts of hard copy scanning including poor horizontal and vertical alignment and physical characteristics such as folded corners where the underlying document is stapled.

193. As with other scanned documents, there is little in the way of metadata relating to its creation to enable a forensic examination of the underlying document. The document contains on the front page the same authorship data as the ID_00865 Control Copy.

{ID_000865}

194. However, I did observe that there are some unusual artefacts in relation to the equations on page 7 of the document, which appear to have been corrupted in a way that has prevented them from properly displaying:



$$\lambda = z\frac{q}{p}$$

To get the probability the attacker could still catch up now, we multiply the Poiss amount of progress he could have made by the probability he could catch up fro

$$\sum_{k=0}^{\infty} \frac{\lambda^k e^{-\lambda}}{k!} \begin{cases} (q/p)^{(z-k)} & \text{if } k \le z \\ 1 & \text{if } k > z \end{cases}$$

Rearranging to avoid summing the infinite tail of the distribution...

$$1 - \sum_{k=0}^{z} \frac{\lambda^k e^{-\lambda}}{k!} \left(1 - (q/p)^{(z-k)}\right)$$

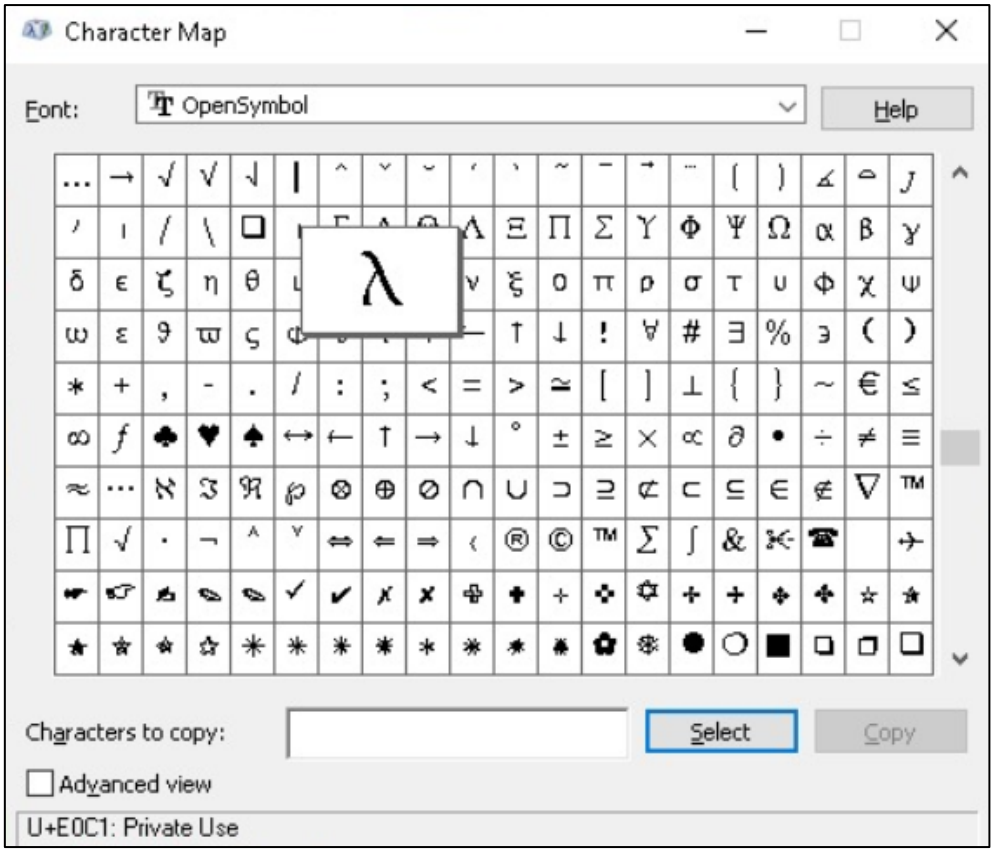
Converting to C code...

```
#include <math.h>
double AttackerSuccessProbability(double q, int z)
{
    double p = 1.0 - q;
    double lambda = z * (q / p);
```

195. As can be seen in the screenshot above, the "lambda" characters within the equations are replaced with a very different character resembling three dots inside a circle.

196. Having observed this, it was my initial view that this was likely an artefact of conversion of the equations from PDF into an editable form, which had not carried over the correct font. I therefore analysed it as follows:

197. I began by observing that the font used to render equations in the Control Copies of the BWP {D/91} (and also in the copy BWP-NB1) was the font Open Symbol, and inspecting the map of that font's characters, using the built-in Windows character map tool. I was able to locate the lambda glyph corresponding to the equations in the Control Copies, as follows:



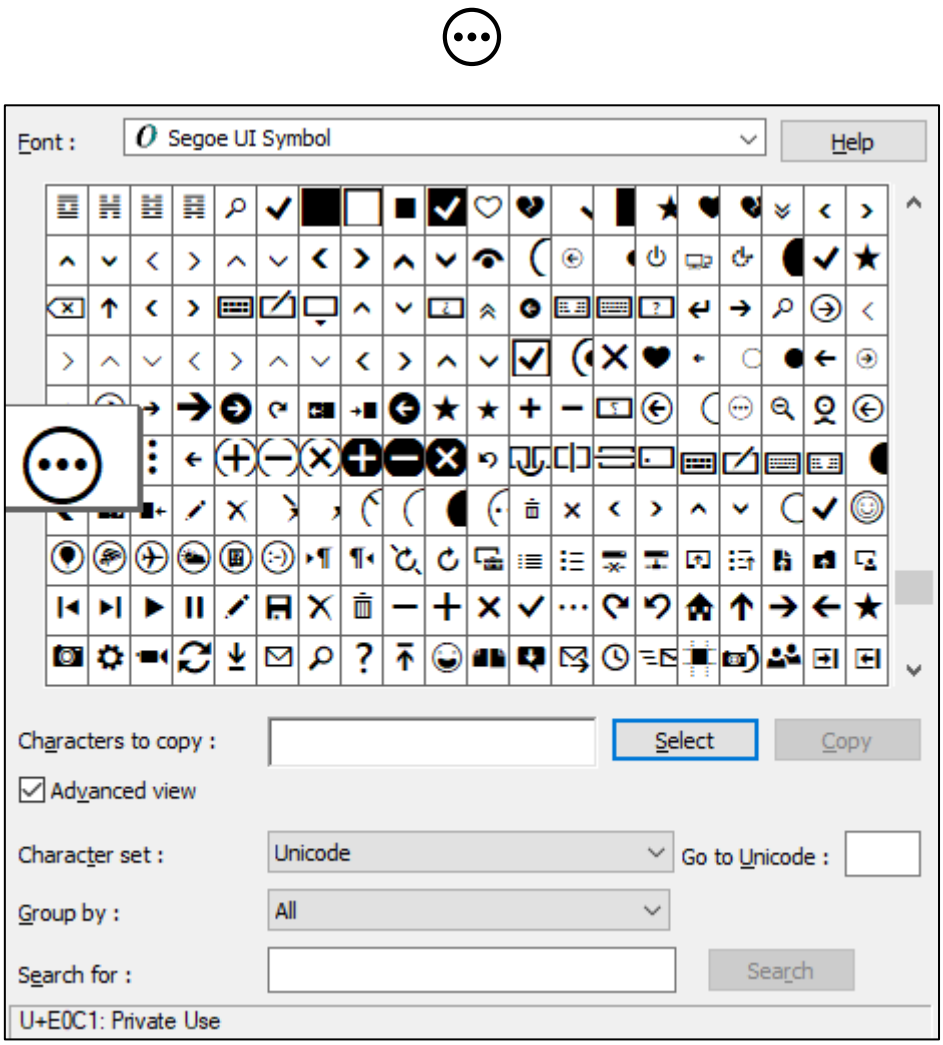
198. Having established the location in the Open Symbol font, which was at location U+E0C1 as can be observed in the bottom part of the screenshot above, this still did not allow me to determine the source of the glyph showing dots in a circle.

199. However, I considered whether the replacement might be due to the mistaken use of a default font during conversion, in circumstances where the computer doing the conversion process did not have the Open Symbol font on their computer, such that a different symbol font had been substituted instead.

200. I therefore refreshed my memory of the default fonts of various iterations of Windows, and consulted the website https://granneman.com/webdev/coding/css/fonts-and-formatting/default-fonts. As listed there, the default font used in Windows XP was called Tahoma, and the default font used in versions of Windows after that (including Windows Vista, Windows 7 and 8, and Windows 10) was Segoe UI.

201. I investigated the glyph map of the Tahoma font and found that it did not contain a corresponding location to U+E0C1. I then investigated the character map of the Segoe UI font and found that it also did not have a location listed in the character map for U+E0C1. However, I also observed that the glyph location U+E0C1 was listed in the character map named “Segoe UI Symbol” (which I take to be a subset of the same font as displayed in the

{ID_004010} character map utility) and that did indeed contain the symbol observed in ID_004010, which can also be entered into this Report appendix by copying from the character map and pasting as follows: □, and again below using size 20 font for clarity:



202. As I have shown in various screenshots towards the beginning of this Appendix in which the Open Symbol font was shown highlighted among the list of embedded fonts within the BWP Control Copies, the PDF files in question contain the font information necessary to ensure {ID_004010} they display appropriately. Therefore, had the document of which ID_004010 is a scan been created directly from PDF, it should not have resulted in the glyph replacement irregularity seen above, since the font information would have been filled in from the embedded font.

{ID_004010} 203. It is therefore my conclusion that the document of which ID_004010 is a scan was created by conversion of a PDF file in an imperfect manner which did not retain the font information, by the use of a computer which did not have the relevant font (Open Symbol) installed, and which substituted its own default symbol font (Segoe UI) in place of Open Symbol.

204. It is possible for fonts to receive updates through the course of their lifetime and so I could not initially say when or on which operating system this took place, other than to say it is likely to have been done on a later version of Windows than Windows XP. However, I checked as follows:

a. On a clean installation of Windows 7, I examined the character map for the version of Segoe UI and Segoe UI Symbol included within that operating system. It did not have any character corresponding to location E0C1.

b. I then examined Segoe UI and Segoe UI symbol as presented on a version of Windows 7 which was not a clean installation, but had received updates until 26 March 2020 (which was the date that Windows 7 ceased to be supported). There was no character corresponding to location E0C1.

c. I then examined the font as it appeared in a Windows 8 build, which was the next version of Windows after Windows 7.[9] I observed that the Segoe UI and Segoe UI Symbol fonts did contain the E0C1 circle-with-dots character.

d. The version of Windows 8 I used was not a clean (non-updated) installation but had received some updates. The version of Segoe UI included was dated with a copyright notice to 2013, which is the year after Windows 8 was released. It is possible that earlier versions of Segoe UI shipped with Windows 8 (which was released in August 2012) may also have had the character in question. Since I was not able to check the period between August 2012 and the end of 2013, I proceed on the assumption that the character was included from 2012.

e. Summarising overall how these observations translate into dates: The character was not in the version of Segoe UI which was released in October 2009 with Windows 7. The character first appears to have been included with Windows 8, which was released in August 2012.

f. It is therefore my conclusion that the font character could not have been included in {ID_004010} ID_004010 before August 2012 and that ID_004010 therefore dates from after that date.

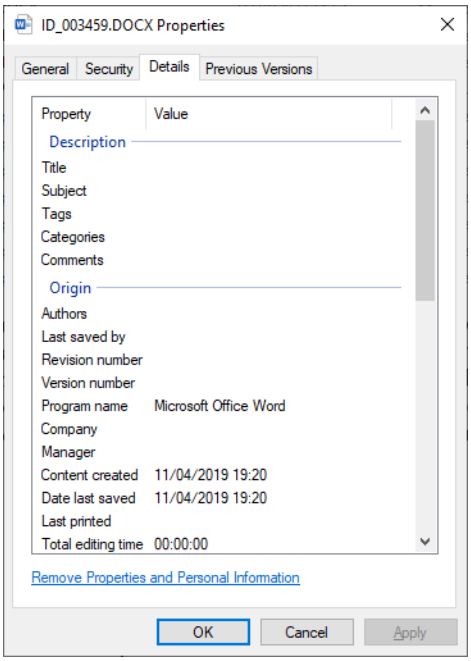
[9] This is not always obvious from the numbering alone – for example, Windows ”9” was skipped, and some versions of Windows had names like XP instead of well-known numbers.

DOC and DOCX files

205. Having reviewed the PDF versions of BWP-related documents, I turn to the remaining DOC and DOCX files.

{ID_003459} **ID_003459**

{ID_003459} 206. ID_003459 is a DOCX file, that has recorded metadata timestamps showing the document being Created and Last Modified on 11/04/2019 (11 April 2019) at 19.20 (expressed in BST) as per the screenshot below:

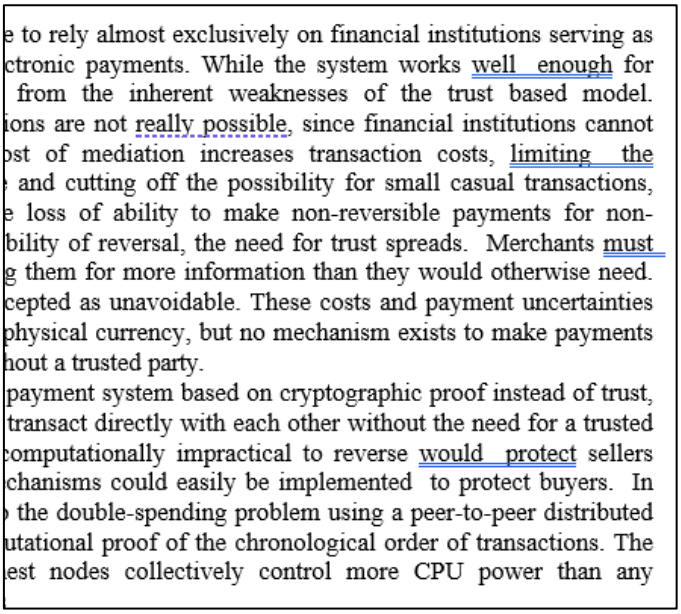


207. The document is the only document within the disclosure dataset whose authorship content appears in the form:

Dr Craig S Wright
craigswright@acm.org

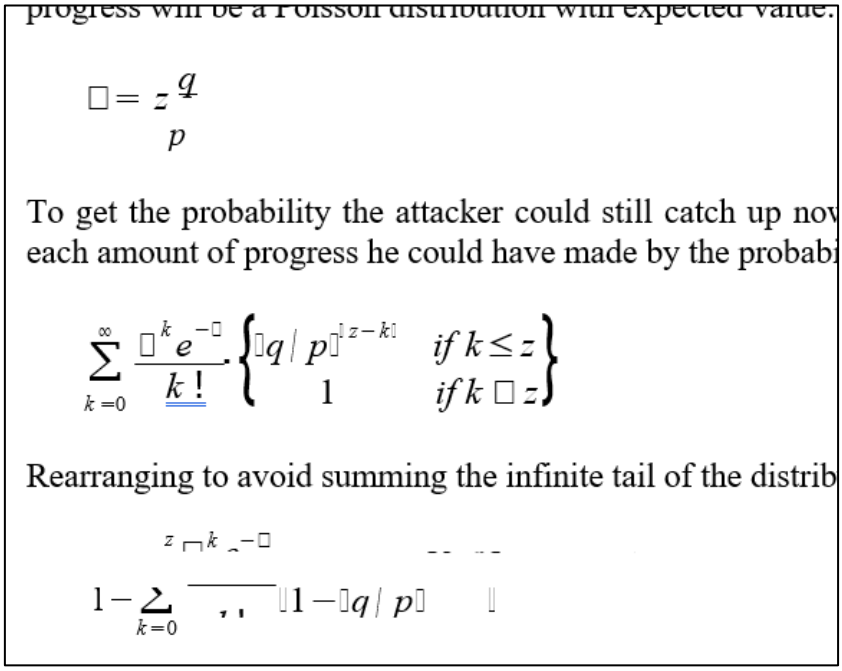
208. The document does not appear to have been created by typing into MS Word, but rather presents as may be expected of a conversion into DOCX format from a PDF file. I note in particular there are many artefacts typical of such a conversion present in the file, including:

a. On its face, the content presents with many double-spaces between words, as can be seen in the following partial screenshot in which the use of double spaces is flagged by MS Word as a grammar error with a blue double-underline in four sections:



e to rely almost exclusively on financial institutions serving as
ctronic payments. While the system works well enough for
from the inherent weaknesses of the trust based model.
ions are not really possible, since financial institutions cannot
ost of mediation increases transaction costs, limiting the
and cutting off the possibility for small casual transactions,
e loss of ability to make non-reversible payments for non-
bility of reversal, the need for trust spreads. Merchants must
g them for more information than they would otherwise need.
cepted as unavoidable. These costs and payment uncertainties
physical currency, but no mechanism exists to make payments
hout a trusted party.
payment system based on cryptographic proof instead of trust,
transact directly with each other without the need for a trusted
computationally impractical to reverse would protect sellers
chanisms could easily be implemented to protect buyers. In
the double-spending problem using a peer-to-peer distributed
utational proof of the chronological order of transactions. The
est nodes collectively control more CPU power than any

b. The equations within the document do not appear to have been generated with equation editing tools or by typing, but also by imperfect conversion, as can be seen by the corrupted nature of them shown in the following screenshot. It can be observed that the lambda characters, present in the equations present in the Control Copies, have been encoded with replacement empty rectangle characters, indicative of the use of a font to render a document in circumstances when the font does not possess a suitable glyph to display the character in question:



progress will be a Poisson distribution with expected value:

$$\square = z\frac{q}{p}$$

To get the probability the attacker could still catch up nov
each amount of progress he could have made by the probabi

$$\sum_{k=0}^{\infty} \frac{\square^{k} e^{-\square}}{k!} \cdot \begin{cases} \square q/p \square^{\square z-k\square} & \text{if } k \le z \\ 1 & \text{if } k \square z \end{cases}$$

Rearranging to avoid summing the infinite tail of the distrib

$$1-\sum_{k=0}^{z \square k _^{-\square}} \overline{\quad\quad} \square 1-\square q/p \square \quad \|$$

c. When viewed within the internal structure of the file, in many places the text is not arranged in cohesive sections but is interspersed with many apparently redundant formatting tags. This can be seen in several areas of the embedded text in the XML of the document but is most easily visible in the Abstract section, which presents as follows (viewed in Notepad++, a text editor which displays XML code markup with colour coding to aid review):

w:val="18"/></w:rPr><w:t>**Abstract.** </w:t></w:r><w:r><w:rPr><w:w w:val="105"/><w:sz w:val="18"/></w:rPr><w:t>**A purely peer-to-peer version of electronic cash would allow online payments to be sent directly from one party to another without going through a financial institution. Digital signatures provide part of the solution, but the main benefits are lost if a trusted third party is still required to prevent double-spending.** </w:t></w:r><w:r><w:rPr><w:spacing w:val="-8"/><w:w w:val="105"/><w:sz w:val="18"/></w:rPr><w:t>**We**</w:t></w:r><w:r><w:rPr><w:spacing w:val="-11"/><w:w w:val="105"/><w:sz w:val="18"/></w:rPr><w:t> </w:t></w:r><w:r><w:rPr><w:w w:val="105"/><w:sz w:val="18"/></w:rPr><w:t>**propose**</w:t></w:r><w:r><w:rPr><w:spacing w:val="-12"/><w:w w:val="105"/><w:sz w:val="18"/></w:rPr><w:t>
</w:t></w:r><w:r><w:rPr><w:w w:val="105"/><w:sz w:val="18"/></w:rPr><w:t>**a**
</w:t></w:r><w:r><w:rPr><w:spacing w:val="-11"/><w:w w:val="105"/><w:sz w:val="18"/>
</w:rPr><w:t> </w:t></w:r><w:r><w:rPr><w:w w:val="105"/><w:sz w:val="18"/>
</w:rPr><w:t>**solution**</w:t></w:r><w:r><w:rPr><w:spacing w:val="-10"/><w:w w:val="105"/><w:sz w:val="18"/></w:rPr><w:t> </w:t></w:r><w:r><w:rPr><w:w w:val="105"/><w:sz w:val="18"/></w:rPr><w:t>**to**</w:t></w:r><w:r><w:rPr><w:spacing w:val="-12"/><w:w w:val="105"/><w:sz w:val="18"/></w:rPr><w:t> </w:t></w:r><w:r><w:rPr><w:w w:val="105"/><w:sz w:val="18"/></w:rPr><w:t>**the**</w:t></w:r><w:r><w:rPr><w:spacing w:val="-12"/><w:w w:val="105"/><w:sz w:val="18"/></w:rPr><w:t> </w:t></w:r><w:r><w:rPr><w:w w:val="105"/><w:sz w:val="18"/></w:rPr><w:t>**double-spending**
</w:t></w:r><w:r><w:rPr><w:spacing w:val="-12"/><w:w w:val="105"/><w:sz w:val="18"/>
</w:rPr><w:t> </w:t></w:r><w:r><w:rPr><w:w w:val="105"/><w:sz w:val="18"/>
</w:rPr><w:t>**problem**</w:t></w:r><w:r><w:rPr><w:spacing w:val="-12"/><w:w w:val="105"/><w:sz w:val="18"/></w:rPr><w:t> </w:t></w:r><w:r><w:rPr><w:w w:val="105"/><w:sz w:val="18"/></w:rPr><w:t>**using**</w:t></w:r><w:r><w:rPr><w:spacing w:val="-12"/><w:w w:val="105"/><w:sz w:val="18"/></w:rPr><w:t> </w:t></w:r><w:r><w:rPr><w:w w:val="105"/><w:sz w:val="18"/></w:rPr><w:t>**a**</w:t></w:r><w:r><w:rPr><w:spacing w:val="-10"/><w:w w:val="105"/><w:sz w:val="18"/></w:rPr><w:t> </w:t></w:r><w:r><w:rPr><w:w w:val="105"/><w:sz w:val="18"/></w:rPr><w:t>**peer-to-peer**</w:t></w:r><w:r><w:rPr><w:spacing w:val="-12"/><w:w w:val="105"/><w:sz w:val="18"/></w:rPr><w:t>
</w:t></w:r><w:r><w:rPr><w:w w:val="105"/><w:sz w:val="18"/></w:rPr><w:t>**network.**
The network timestamps transactions by hashing them into an ongoing chain of hash-based</w:t></w:r><w:r><w:rPr><w:spacing w:val="-9"/><w:w w:val="105"/><w:sz w:val="18"/></w:rPr><w:t> </w:t></w:r><w:r><w:rPr><w:w w:val="105"/><w:sz w:val="18"/></w:rPr><w:t>**proof-of-work**,</w:t></w:r><w:r><w:rPr><w:spacing w:val="-8"/><w:w w:val="105"/><w:sz w:val="18"/></w:rPr><w:t> </w:t></w:r><w:r><w:rPr><w:w w:val="105"/><w:sz w:val="18"/></w:rPr><w:t>**forming**</w:t></w:r><w:r><w:rPr><w:spacing w:val="-9"/>

209. In view of these characteristics indicative of conversion from PDF, and the internal metadata indicating the document to be from 2019, it is my view that this document was created by conversion from PDF in April 2019. In view of the fact that it is the only document in the dataset which bears the contact details in the form shown above, it is not possible to determine from which specific PDF file ID_003459 was created. However, I observe that it is likely to be created either from ID_000865 or from a document that was created from ID_000865, based on the following observation:

{ID_003459}
{ID_000865}

{ID_003459}

a. Within the compound doc file of ID_003459 is a short XML stream labelled "custom.xml". The content of custom.xml presents as follows:

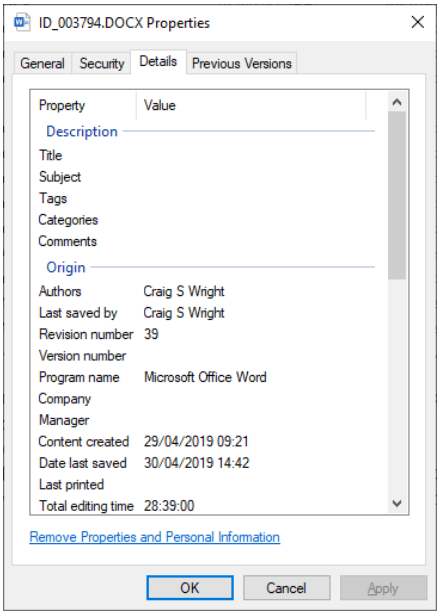
```
<?xml version="1.0" encoding="UTF-8" standalone="yes"?>
<Properties xmlns=
"http://schemas.openxmlformats.org/officeDocument/2006/custom-properties"
xmlns:vt=
"http://schemas.openxmlformats.org/officeDocument/2006/docPropsVTypes"
><property fmtid="{D5CDD505-2E9C-101B-9397-08002B2CF9AE}" pid="2" name=
"Created"><vt:filetime>2009-03-24T00:00:00Z
</vt:filetime></property><property fmtid=
"{D5CDD505-2E9C-101B-9397-08002B2CF9AE}" pid="3" name="Creator"><vt:lpwstr>
Writer</vt:lpwstr></property><property fmtid=
"{D5CDD505-2E9C-101B-9397-08002B2CF9AE}" pid="4" name="LastSaved"
><vt:filetime>2019-04-11T00:00:00Z</vt:filetime></property></Properties>
```

b. As can be observed in the screenshot above, the final XML tag lists the "filetime" as being last saved on 11 April 2019. However, the "Created" filetime tag in the middle of that content lists the created date as 24 March 2009, which corresponds to the created date of ID_000865.

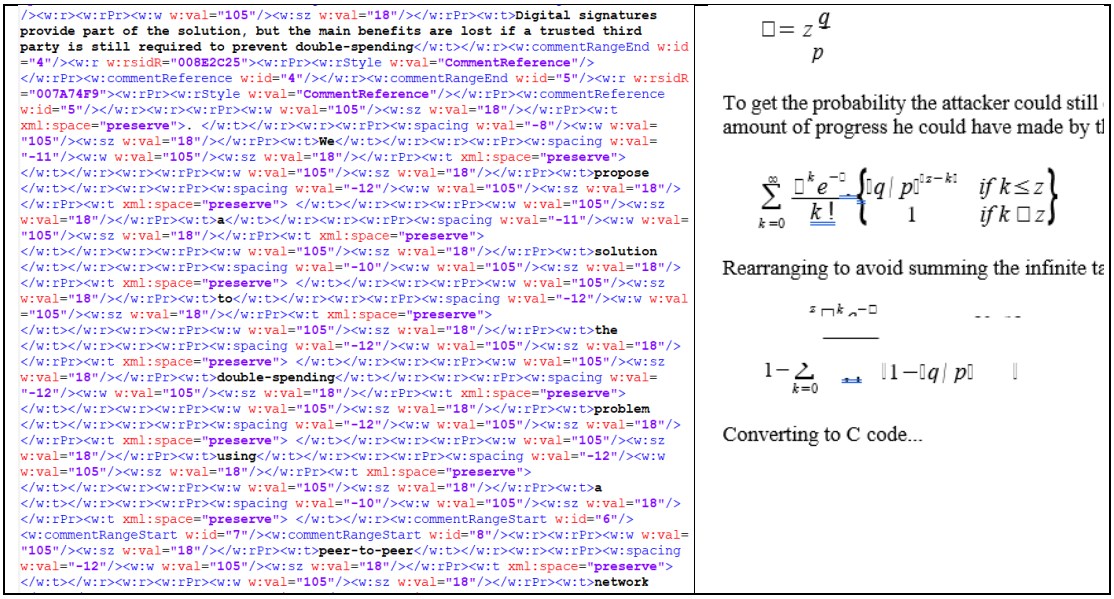
{ID_000865}

{ID_003794} **ID_003794**

{ID_003794} 210. ID_003794 is a DOCX file, listed as being Created on 29/04/2019 (29 April 2019) at 09.21 and Last Modified just over a day later on 30/04/2019 at 14:42 (both expressed in BST) as shown in the screenshot below:



{ID_003459} 211. {ID_003794} Much of my analysis of ID_003459 also applies to ID_003794, including the artefacts of conversion such as double spaces, text encoded interspersed with redundant formatting characters, and the presence of corrupted equations:



212. The document presents on its face with a large number of markup comments added to the document, in the apparent form of a discussion between two comment authors "Craig S Wright" and "jerry.d.chan@bittoku.co.jp". It is possible to print the comments to a single document by performing a "print" operation and selecting the option of printing "Markup" only. **Exhibit PM3.8** contains a printout of the comments of ID_003794 generated in this way.

{H/28}
{ID_003794}

{ID_003459} 213. Similarly to my conclusion above in respect of ID_003459, it is my view that this document was created by conversion from PDF in April 2019.

{ID_000510} **ID_000510**

ID_000510- overview

{ID_000510} 214. ID_000510 is a simple MS Word .DOC document. There are 5 electronic duplicates by MD5 hash of this document in the disclosure dataset. The timestamp and metadata properties for these documents are listed in the table below:

Metadata field	**ID_000510**	**ID_000511, ID_000512, ID_000513, ID_000514**	**ID_004024**
Original File name	Bitcoin.doc	Bitcoin.doc	
type / Extension	DOC	DOC	doc
OS Created - Date and Time	21/12/2008 21:55:00	16/11/2008 16:18:00	
OS Last Modified - Date and Time	21/12/2008 21:55:00	16/11/2008 16:18:00	
OS Last Accessed - Date and Time	21/12/2008 21:55:00	16/11/2008 16:18:00	
Annotation Authors	Unknown	Unknown	Unknown
Author	Craig S Wright	Craig S Wright	Craig S Wright
Last Author	Lynn Wright	Lynn Wright	Lynn Wright
Created	14/11/2008 21:06:00	14/11/2008 21:06:00	14/11/2008 21:06:00
Last Saved	16/11/2008 05:18:00	16/11/2008 05:18:00	16/11/2008 05:18:00

Difference between Internal Created and Internal Last Modified	1 days 8hrs12mins	1 days 8hrs12mins	1 days 8hrs12mins
Difference between Internal Created and Internal Last Modified as minutes	1932	1932	1932
Revision Number	2	2	2
Edit Time OLE as minutes	1931	1931	1931
Difference between Internal timestamp difference and Edit Time	1	1	1
% Edit Time of difference	100	100	100
AppName	Microsoft Office Word	Microsoft Office Word	Microsoft Office Word
Application Version OLE / readable	11.9999	11.9999	11.9999
Company	University of Newcastle	University of Newcastle	University of Newcastle
Title	Bitcoin	Bitcoin	Bitcoin
Subject	A Peer to Peer electronic cash system	A Peer to Peer electronic cash system	A Peer to Peer electronic cash system
Template	Normal.dot	Normal.dot	Normal.dot
Edit Time OLE Full	32 hours, 11 minutes, 0 seconds	32 hours, 11 minutes, 0 seconds	32 hours, 11 minutes, 0 seconds
Hyperlink Base	http://www.newcastle.edu.au	http://www.newcastle.edu.au	http://www.newcastle.edu.au

215. I observe that the document is recorded as being last written by "Lynn Wright", and I refer to Appendix PM24 for a discussion of the Lynn Wright documents.

216. An anomaly identified in several of these documents is that they have recorded Edit Times that match or are near 100% of the time difference between the recorded metadata Created and Last Saved timestamps. This behaviour is observed for ID_000510 where the Edit Time is just one minute less than the time difference between Created and Last Written. Noting also that the Edit Time of this document is 1,931 minutes (over 32 hours), I consider it to be anomalously long.

{ID_000510} *ID_000510- content review*

{ID_000865} 217. The document presents content-wise as very similar to ID_000865, a Control Copy BWP. There are, however, some notable differences which I discuss below.

218. My first observation is that it does not contain any of the authorship information as seen in the majority of the BWP documents. It presents as follows:

Bitcoin: A Peer-to-Peer Electronic Cash System

Abstract. A purely peer-to-peer version of electronic cash would allow online payments to be sent directly from one party to another without going through a financial institution. Digital signatures provide part of the solution, but the main benefits are lost if a trusted third party is still required to prevent double-spending. We propose a solution to the double-spending problem using a peer-to-peer network. The network timestamps transactions by hashing them into an ongoing chain of hash-based proof-of-work, forming a record that cannot be changed without redoing the proof-of-work. The longest chain not only serves as proof of the sequence of events witnessed, but proof that it came from the largest pool of CPU power. As long as a majority of CPU power is controlled by nodes that are not cooperating to attack the network, they'll generate the longest chain and outpace attackers. The network itself requires minimal structure. Messages are broadcast on a best effort basis, and nodes can leave and rejoin the network at will, accepting the longest proof-of-work chain as proof of what happened while they were gone.

1. Introduction

Commerce on the Internet has come to rely almost exclusively on financial institutions serving as trusted third parties to process electronic payments. While the system works well enough for most transactions, it still suffers from the inherent weaknesses of the trust based model. Completely non-reversible transactions are not really possible, since financial institutions cannot avoid mediating disputes. The cost of mediation increases transaction costs, limiting the minimum practical transaction size and cutting off the possibility for small casual transactions, and there is a broader cost in the loss of ability to make non-reversible payments for non- reversible services. With the possibility of reversal, the need for trust spreads. Merchants must be wary of their customers, hassling them for more information than they would otherwise need. A certain percentage of fraud is accepted as unavoidable. These costs and payment uncertainties can be avoided in person by using physical currency, but no mechanism exists to make payments over a communications channel without a trusted party.

What is needed is an electronic payment system based on cryptographic proof instead of trust, allowing any two willing parties to transact directly with each other without the need for a trusted third party. Transactions that are computationally impractical to reverse would protect sellers from fraud, and routine escrow mechanisms could easily be implemented to protect buyers. In this paper, we propose a solution to the double-spending problem using a peer-to-peer distributed timestamp server to generate computational

219. I observe that the heading is also not centred but is left aligned, and continuation paragraphs do not start with an indent unlike other documents referred to in this Appendix.

220. {ID_000510} {ID_003459} {ID_003794} There are a number of double spaces between words throughout the text. This is characteristic of a document that has been created by converting the source material from a PDF file. As ID_000510 is a .DOCfile it is not possible to extract the formatting out of the XML code in the same manner as ID_003459 and ID_003794 which are .DOCX files.

221. {ID_000510} {ID_000865} As can be observed, ID_000510 also has no diagrams or complicated equations. There are however whitespaces where these are to be found in ID_000865.

222. {ID_000865} {ID_000510} The other textual differences between ID_000865 and ID_000510 are described in the tables below:

{ID_000865} {ID_000510}

ID_000865	ID_000510

Section 7 page 4	Section 7 page 4
"…transactions are hashed in a Merkle Tree [7][2][5],…"	"…transactions are hashed in a Binary Tree [7][2][5],…"

{ID_000865}
{ID_000510}

ID_000865	ID_000510
Section 8 page 5	Section 8 page 4
"…and obtain the Merkle branch linking the transaction to the block it's timestamped in."	"…, and obtain the branch of the binary tree [7] linking the transaction to the block it's timestamped in."

{ID_000865}
{ID_000510}

ID_000865	ID_000510
Section 11 page 8	Section 11 page 6
q=0.1 z=0 P=1.0000000 z=1 P=0.2045873 z=2 P=0.0509779 z=3 P=0.0131722 z=4 P=0.0034552 z=5 P=0.0009137 z=6 P=0.0002428 z=7 P=0.0000647 z=8 P=0.0000173 z=9 P=0.0000046 z=10 P=0.0000012	q=0.1 z=0 P=1.000000 z=1 P=0.204587 z=2 P=0.050978 z=3 P=0.013172 z=4 P=0.003455 z=5 P=0.000914 z=6 P=0.000243 z=7 P=0.000065 z=8 P=0.000017 z=9 P=0.0000046 z=10 P=0.0000012

{ID_000865}
{ID_000510}

ID_000865	ID_000510
References page 9	References page 7
[6] A. Back, "Hashcash - a denial of service counter-measure," http://www.hashcash.org/papers/hashcash.pdf, 2002. [7] R.C. Merkle, "Protocols for public key cryptosystems," In *Proc. 1980 Symposium on Security and Privacy*, IEEE Computer Society, pages 122-133, April 1980.	[6] A. Back, "Hashcash - a denial of service counter-measure," [7] R.C. Merkle, "Protocols for public key cryptosystems," In *Proc. 1980 Symposium on Security and Privacy*, IEEE Computer Society, pages 122-133, April 1980.

{ID_000865}
{ID_000226}
{ID_000226}
{ID_000510}

223. Firstly, I observe that in respect of these four differences, the content for ID_000865 is consistent with that found in ID_000226, being the other Control Copy BWP. ID_000226 has a created timestamp of 03/10/2008, being approximately 5 weeks before the creation of ID_000510 being 14/11/2008. This means that the differences identified in ID_000510 conflict with the content of the Control Copy BWP documents from before and after the recorded creation of ID_000510: taking the chronology at face value, the changes would have had to be made after ID_000226 and then reverted back at the point of creating ID_000856. The chronology also does not fit in with BWP-NB1.

{ID_000226}
{ID_000856}
{D/91}

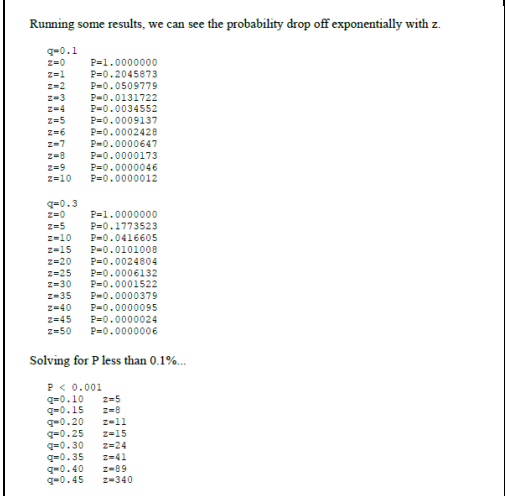
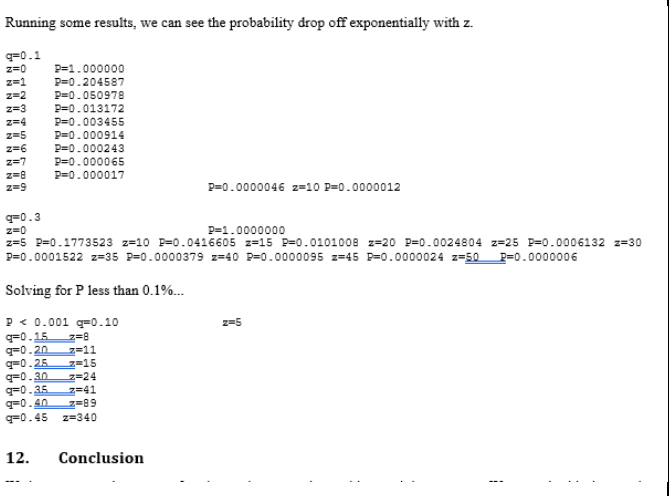
224. Considering the first two of these differences, I believe that it is relevant to point out that item [7] from the references section of both documents is "*[7] R.C. Merkle, "Protocols for public key cryptosystems," In Proc. 1980 Symposium on Security and Privacy, IEEE Computer Society, pages 122-133, April 1980.*". While it is a topic that is outside of my expertise, I note that the two differences relate to the author's name, "Merkle". I do not draw any opinion from this connection beyond noting it.

225. For the third difference, I note it relates to the decimal point to which the numbers are enumerated, with ID_000865 rounding at 7 decimal points and ID_000510 rounding at 6 decimal points. It is a simple fact that numbers can be round to a lower (shorter) decimal place, but not extended without having to recalculate from source, and that is intuitively consistent with moving from the more precise source to the less precise source and not the other way around.

{ID_000865} {ID_000510}

226. For the third difference, I observe that the formatting for the entries from z=9 appears to have been corrupted. A larger pair of screenshots, shown below, demonstrate this more clearly.:

{ID_000865} {ID_000510}

ID_000865	ID_000510
	

227. This loss of structure and formatting is not typical of standard user drafting behaviour but is consistent with the type of corruption resulting from a document conversion process such as from PDF to MS Word.

228. The fourth difference relates simply to the absence of the web link for HashCash. As noted above, it is present in both control Copy BWP documents that pre and post- date the recorded creation of ID_000510 and in BWP-NB1.

{ID_000510} {D/91}

229. Considering these points, I do not believe that ID_000510 can be accepted as authentic to its purported creation in November 2008. I believe that it has been produced downstream of ID_000865.

{ID_000510} {ID_000865}

BWP Documents with insufficient data for forensic analysis.

230. Certain BWP related documents do not present a good target for forensic analysis. I address these in summary in the following section.

{ID_000515} *ID_000515*

{ID_000515} 231. Document ID_000515 presents as a ".TIF" file, with monochrome (black and white) encoding. TIFF (abbreviated as "TIF" in three-letter extension) stands for Tag Image File Format, and is an image file format initially commonly associated with Fax transmissions and older scans of documents. While later versions of TIF format files can encode more complex information, ID_000515 is a simple monochrome, relatively low resolution document.

{ID_000515}

{ID_000515} 232. Although often associated with scans and hard copy files, ID_000515 presents as a digital conversion, which I take from the fact that it exhibits perfect horizontal and vertical alignment and presents very cleanly without the artefacts usually associated with hard-copy scanning. Monochrome TIFF format files are not in my experience commonly preferred for digital conversions, over more widely-used file formats like PDF.

233. TIFF files do not characteristically carry much metadata to enable forensic analysis, and this {ID_000515} is true of ID_000515. I note that there is an embedded OLE metadata stream within the document encoding a timestamp relating to 16 November 2008, but I am unable to examine that in more detail in view of the lightweight `nature of the TIF format, save to observe that the provided date does not appear to correspond to any metadata date provided in the {ID_000515} disclosure load file other than ID_000510 and the duplicates thereof. I discuss ID_000510 {ID_000510} above in this report., The content of the document superficially appears to match that of {ID_000536} ID_000536 in terms of its content, including the textual content and fact that they both display contact details in the form of Style 4 listed towards the beginning of this report. I have {ID_000536} determined above that ID_000536 is not authentic to its purported creation time for a number of independent reasons and to be consistent with being authored with the use of clock manipulation techniques. In view of the limitations of the format of this document, and the {ID_000536} possibility that it is a copy of ID_000536 converted into the TIF format, it is my view that the embedded OLE metadata stream in the TIF file cannot be relied upon as indicative of its true date of authorship.

{ID_004007} *ID_004007 (un-named)*

{ID_004007} 234. Document ID_004007 presents as a scan of a printed copy of the ID_000865 Control Copy {ID_000865} BWP. The metadata properties report that the scan was produced on 06/09/2019 at 17:00:39 (expressed as BST) using a Canon MFP device recorded as "Canon iR-ADV C5560 PDF" again with a UTC offset of UTC+1. No original filename has been provided.

{ID_004007} 235. From a visual comparison between ID_004007 against the ID_000865 Control Copy I did not {ID_000865} identify any differences other than differences caused by the conversion via printed copy to scanned PDF.

236. Being a scan of a hard copy document dating from 2019, there is no information within the document that relates to the original document from which the printout was produced which would enable me to conduct any deeper forensic analysis.

{ID_003857} *ID_003857*

{ID_003857} {ID_004007} 237. Document ID_003857 is also a scan of a hard copy printout. It is dated as being produced a few months later than ID_004007, on 10/02/2020 (10 February 2020) at 13:50:16 (expressed as BST) and is recorded as being scanned with the same Canon device as ID_004007. The content appears to be consistent with ID_000865 and being a scan, it does not present any additional information to enable forensic analysis save for one indication below.

{ID_000865}

238. I record this observation for completeness although it has not assisted me to form any conclusion.

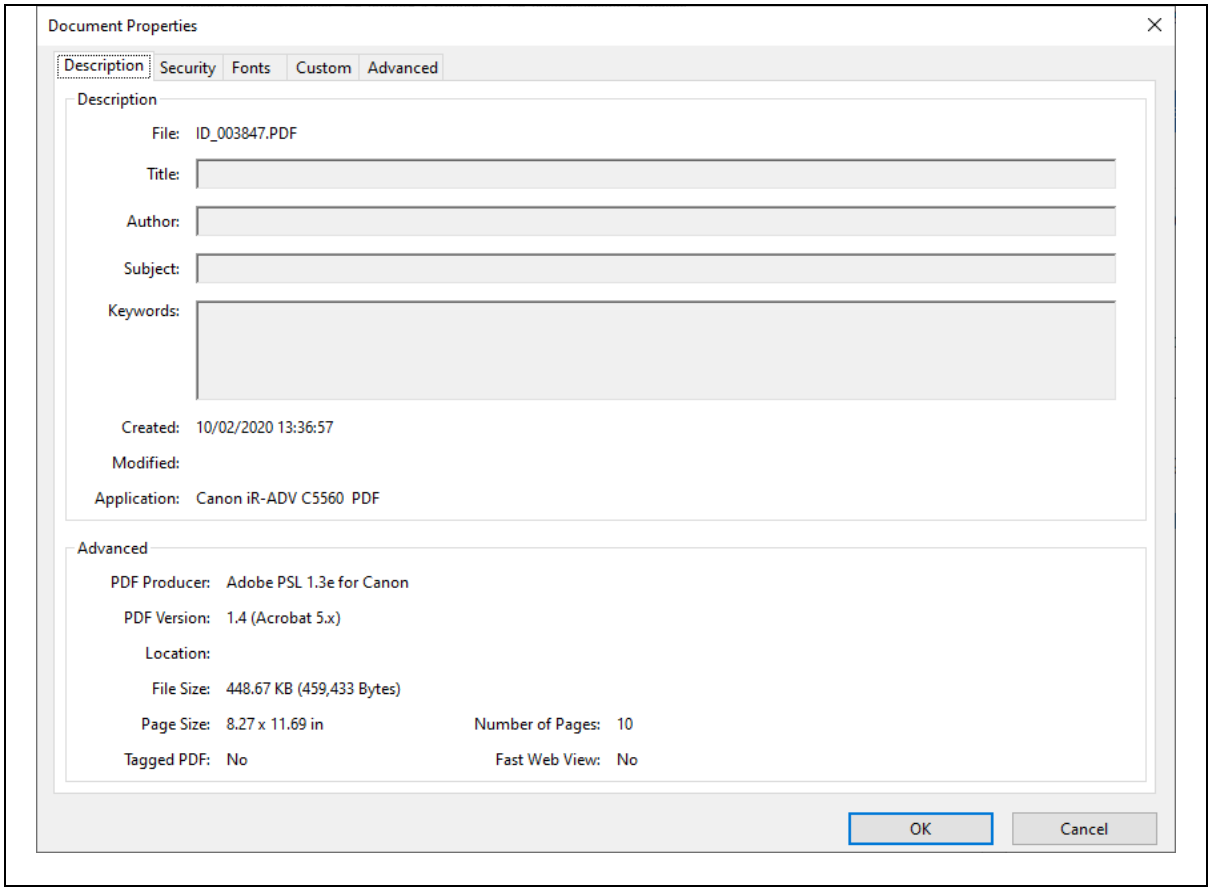
{ID_003857} 239. Within the metadata of ID_003857 there is a metadata stream which indicates that the scanner was configured with a +1 hour time zone offset as per the screenshot below. I have added highlight to aid review.

```
…
                    <xap:CreateDate>2020-02-10T13:50:16+01:00</xap:CreateDate>
                    <xap:CreatorTool>Canon iR-ADV C5560  PDF</xap:CreatorTool>
…
```

240. This is consistent with the document being produced in a place with a time zone of UTC+1 such as Central European Time. It is not consistent with British Summer Time, which would not have been active in February 2020. Otherwise, as a scanned document there is no metadata within the document that would relate to the electronic document on which the printout was based. The analysis is limited to the scanned copy.

{ID_003847} *ID_003847*

241. The document ID_003847 is a scan of a hard copy printout of a version of the BWP and is a Reliance Document. The Adobe Acrobat Reader properties indicate that the scan was produced on 10/02/2020 at 13:36:57 (expressed as BST) as per the screenshot below. The scan was produced using a Canon iR-ADV c5560 Multifunction scanner/printer device.



{ID_003847} {ID_000536} {ID_000358} {ID_000840} 242. From a visual comparison, the content of ID_003847 is consistent with ID_000536 (and thereby ID_000358 and ID_000840), from which I infer that ID_003847 is a scanned copy of a printout of one of those documents. I have formed the opinion that those documents cannot be relied upon as authentic to their purported creation dates for the reasons set out above.

{ID_003847} {ID_003857} 243. Within the metadata of ID_003847 there is a metadata stream which indicates that the scanner was configured with a +1 hour time zone offset as per ID_003857 mentioned above. Otherwise, as a scanned document there is no metadata within the document that would relate to the source document from which the scan was generated.

{ID_003747} *ID_003747*

{ID_003747} 244. Document ID_003747 is an email which attaches two BWP-related documents. The email dates from 2019 and contains attachment properties relating to 2019, so it does not appear to be directly relevant to the authorship of the BWP itself. The two documents attached to it are hash-identical to other documents in the dataset, respectively: ID_003732 and ID_003459, which I have addressed separately.

{ID_003732} {ID_003459}

{ID_004011} *ID_004011*

{ID_004011} 245. Document ID_004011 presents as a scan of a hard copy document and is a Reliance Document. Adobe Acrobat Reader properties indicate that the scan was produced on 10/9/2019 (10 September 2019) at 13:45:05 (expressed as BST). The scan was produced using a Canon iR-ADV c5560 Multifunction scanner/printer device configured to a UTC+1 time zone. It is marked up and stained, but as with other such documents, there is little metadata for further digital forensic analysis.